Thesis Dissertation / Soutenance De These

EVALUATION OF MEDICAL IMAGING DEVICES USING SYNTHETICALLY MODIFIED CLINICAL IMAGES AND 3D PRINTED PHANTOMS



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e Motion Tech

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Industrial Collaboration

eMotion Tech

GE Healthcare

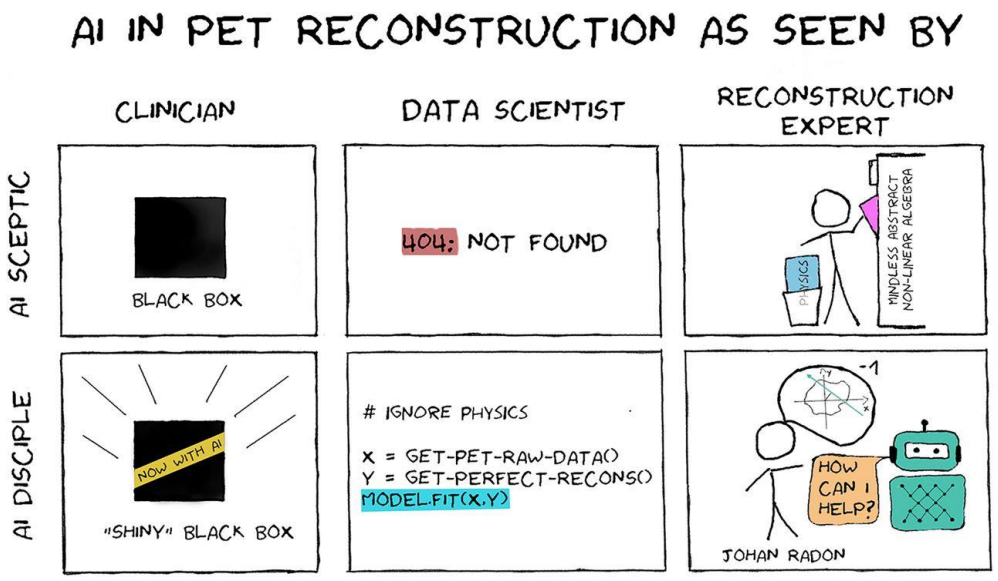
Buc · Simulation Software
Industrial Collaboration

GE HealthCare

Thesis Context

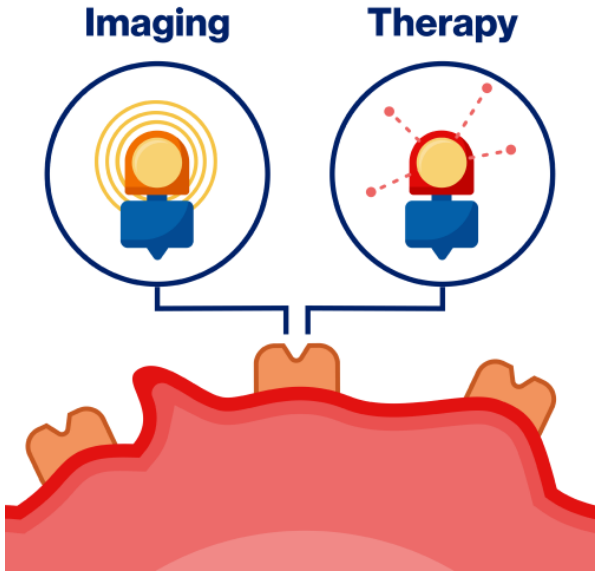
Two parallel revolutions in Nuclear Medicine

AI



Reconstruction Reporting Analyzing

Theranostics



Select Treat Monitor

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https://doi.org/10.2967/jnumed.121.262303 ; https://doi.org/10.2967/jnumed.126.272362; Novartis RLT academy

How do we know that scanner systems are giving us trustworthy numbers?



Treatment Decisions

Ex: 30% drop in uptake - change the treatment plan.



Clinical Studies

Unvalidated systems = unreliable trial outcomes.



Physician Confidence

Trust across scanners and institutions.



AI & Radiomics

Models developed using scanner data

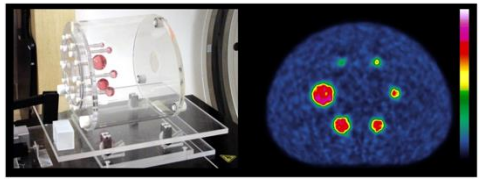
All of these require a benchmarking of system response against a ground truth

Complementary validation methods

Each approach provides a different form of reference information

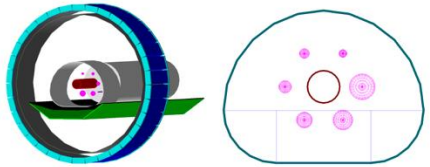
Physical Phantoms

Real acquisition and reconstruction response



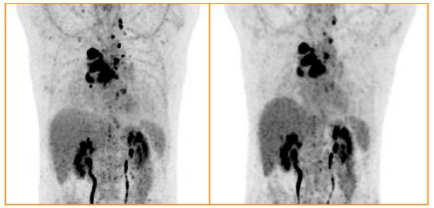
Simulation

Known input with flexible, repeatable conditions



Clinical Study

Real anatomy and radiotracer biodistribution



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Shared designs and PET metrics can connect physical measurement, simulation and clinical context.

https://doi.org/10.2967/jnmt.107.040782; https://doi.org/10.1186/s40658-023-00610-2

Custom / 3D Printed Phantoms in Nuclear Medicine

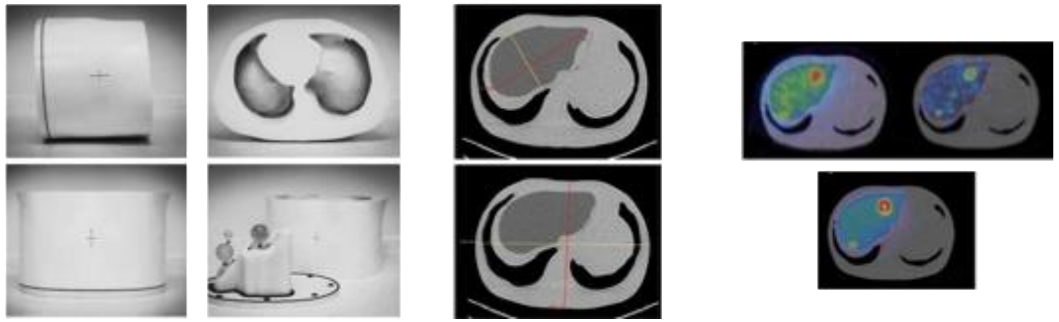
ANTHROPOMORPHIC

Looks like / Feels like / Behaves like

Morphological

Shape & Structure

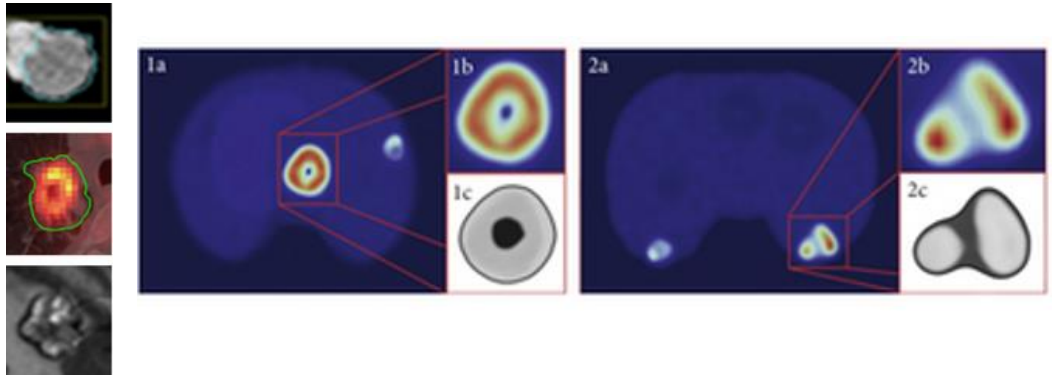
Realistic Geometry



Voxel-Level

Internal Pattern or Distribution

Realistic Voxel information



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https://doi.org/10.1186/S40658-016-0151-6;

Custom / 3D Printed Phantoms in Nuclear Medicine

ANTHROPOMORPHIC

Looks like / Feels like / Behaves like

WHAT IT ADDS

Known 3D geometry

CAD + region masks

Designed uptake patterns

Controlled internal structure

Real scanner response

Acquisition to reconstruction

WHAT TO VERIFY

Print fidelity

CAD ≠ fabricated shape

Realized activity

Theoretical value might not be accessible for all designs

Preparation burden

No standardizes protocol

Simulation Methods in PET/CT

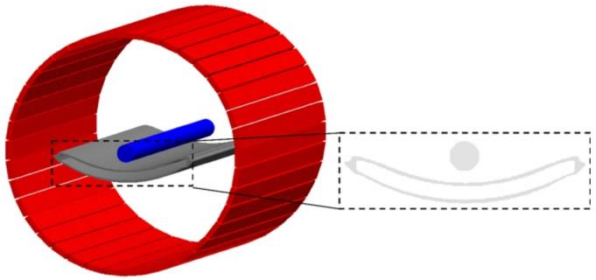
SIMULATIONS

Model of reality but not reality itself

Monte Carlo

Photon Transport Physics

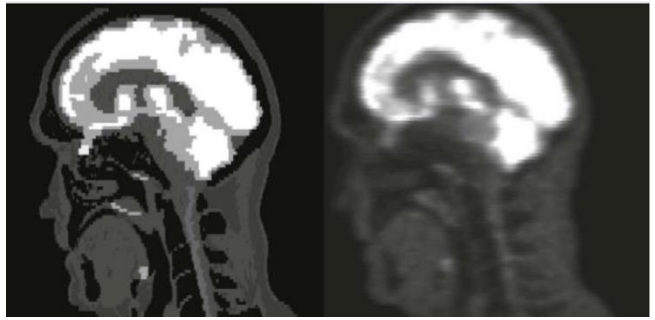
Accurate Physics of Detection



Analytical Sim

Activity Maps to reconstructed image

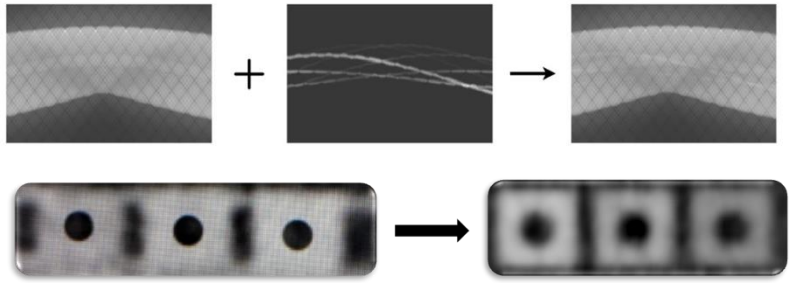
Fast simulators with approximations



Data Insertion

Sinogram / Image based data addition

Vendor specific sinogram / reconstruction



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Pommranz, C.M. et al https://doi.org/10.1186/s40658-025-00738-3 ; Beatrice Berthon et al 10.1016/j.ejmp.2015.07.139 ;Quentin Maronnier et al https://doi.org/10.1186/s40658-022-00496-6

Simulation Methods in PET/CT

SIMULATIONS

Model of reality but not reality itself

WHAT IT CONTROLS

Prescribed target
Geometry + activity

Isolated variables
Size · contrast · counts

Measured realistic background
Phantom or patient data

Repeatable tests
Same lesion, new conditions

WHAT NEEDS VALIDATION

Model-defined truth
Input ≠ physical lesion

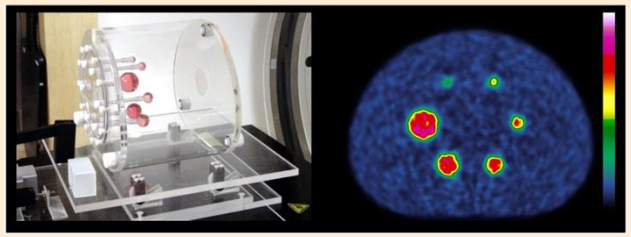
Scanner approximation
projection + corrections

Noise may differ based on fidelity
Spatial texture after reconstruction

Transfer is conditional
Scanner · tracer · task

Gap in validation methods

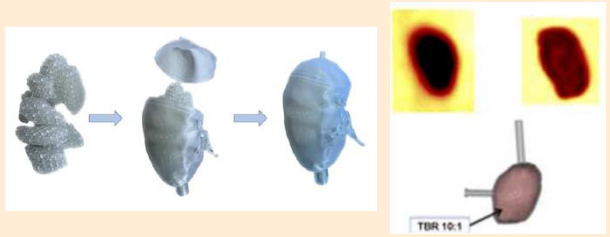
Standard Phantoms



Simple walled spheres · Uniform distribution ✗

Controllable· Standardized. Analyzable ✓

Custom Phantoms



Realistic shapes· Uniform distribution ✓

Standardized in design / material ✗

Real Clinical Data



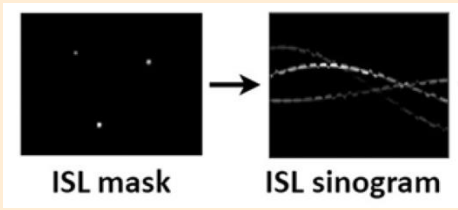
Irregular · Heterogeneous · Complex 3D morphology ✓

✓ **Model Validated**

✗

✓ **Model Applied**

Synthetic Lesion Insertion



Vendor Specific & Analytical · Sinogram based · Extendable to clinical data easily ✓

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https://doi.org/10.1186/s40658-024-00613-7

Motivation, tools and objectives

Problem

How can realism be introduced in controlled PET phantoms and simulations while retaining an interpretable reference for both

Tools

1. **3D printing**
   Physical Targets
2. **DUETTO / CASSOULET**
   Synthetic Lesion Insertion
3. **MATLAB / 3D Slicer**
   Data Extraction and Analysis

01

Uniform wall-less targets

02

Structured multi uptake

03

Patient derived morphology

This Work

Design Evaluation of Porous 3D printed Grids: Experiment + Simulation

Pilot study · multi-contrast encoding · single centre

This Work

Dual-Centre Transferability of Multi contrast Phantom : Experiment

Same phantom · two independent institutions · consistent results

This Work

Heterogeneous Uptake in Single Encoded grid: Experiment + Simulation

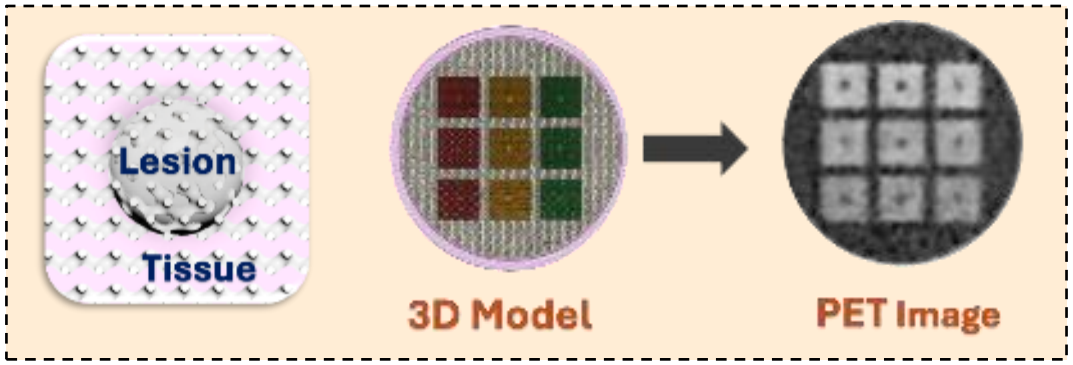
First order and second order metrics . Dual tracer

This Work

Hybrid Clinical implementation workflow for Anthropomorphic Lesions

8 clinical shapes · hybrid framework · proof of concept validated

Porous Phantoms for Voxel Emulation

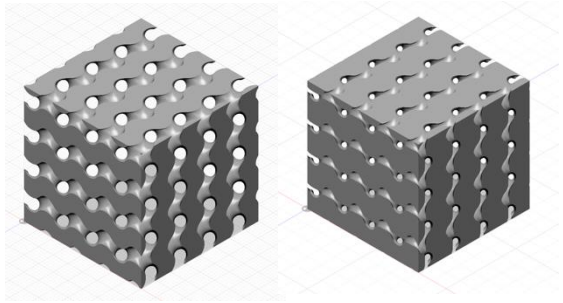


$$AC_{tissue\ /\ background} = \varphi \times AC_{lesion\ /\ solution} \quad (1)$$

Porosity (φ) controls contrast

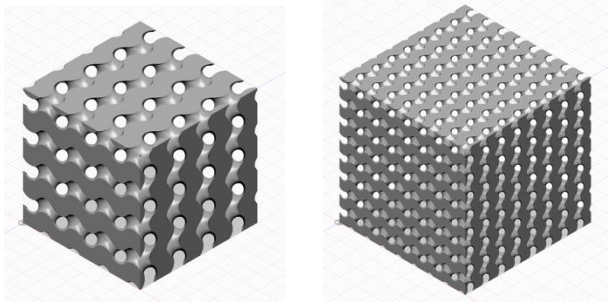
Solidity (S)	Porosity (ϕ)	Nominal TBR
50%	0.50	2.00
60%	0.40	2.50
70%	0.30	3.33
75%	0.25	4.00

Solidity (S)



Controls Contrast

Pore Distance (D_P)



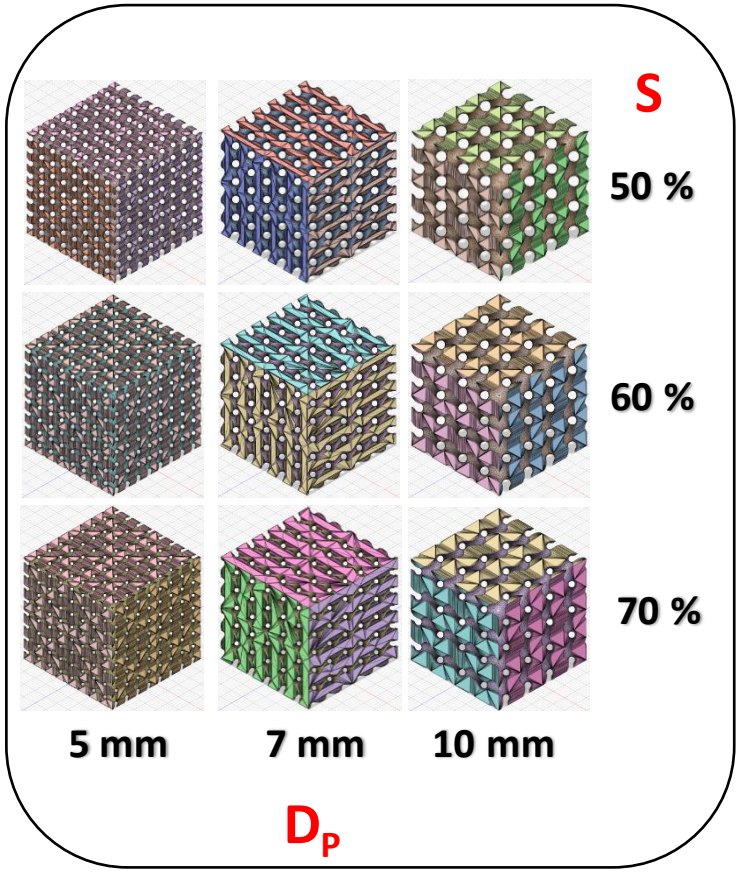
Controls Resolution / Diffusion

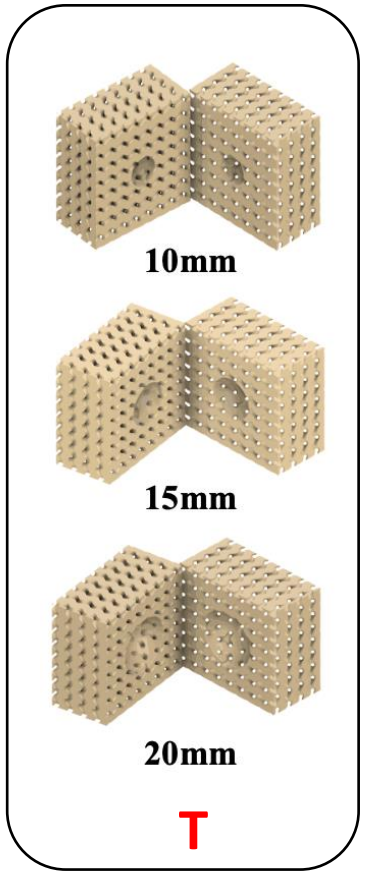
$$\textbf{Target to Background Ratio } (\mathbf{TBR_{insert}}) = \frac{\mathbf{100}}{\mathbf{100} - \mathbf{S}\ (\%)} \quad (2)$$

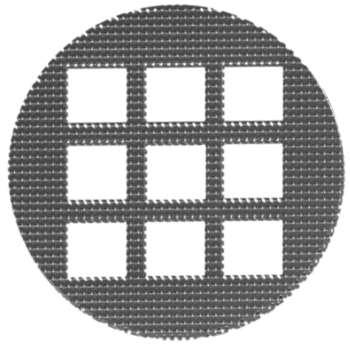
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Phantom Design

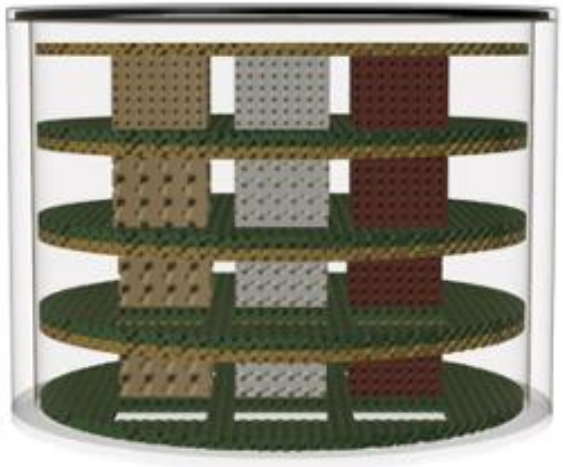
Each insert is 40 mm cube and identified in a S-D_P-T notation (Ex: 50-5-20)



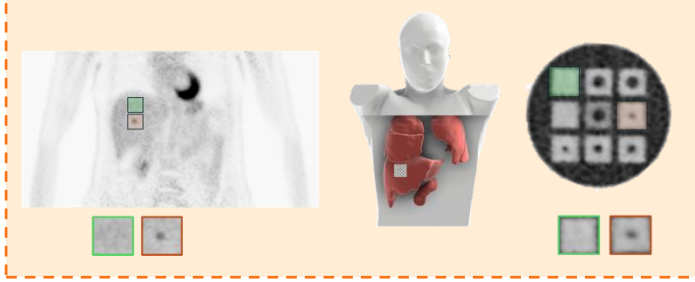




3 x 3 layout
S50-D5



Multi contrast Phantom with
36 inserts in a jazasck



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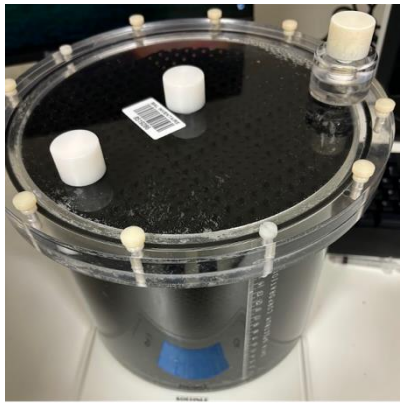
Do porous inserts retain their quantitative response across repeated preparations and acquisitions?

METHODS : Based on 36 insert phantom design

- Acquisitions performed on **TOF - 5R Discovery MI PET / CT** system with **[^{18}F] FDG** as radiotracer
- 27 hot targets + 9 homogeneous cubes (**TBR : 2.00 ; 2.50 ; 3.33**)
- **TOF-BSREM** (Q clear – BPL) reconstruction with clinical reconstruction parameters used at IUCT Oncopole
- PET reconstruction on each day : **256 × 256 x 89** with reconstruction $\boldsymbol{\beta = 550}$

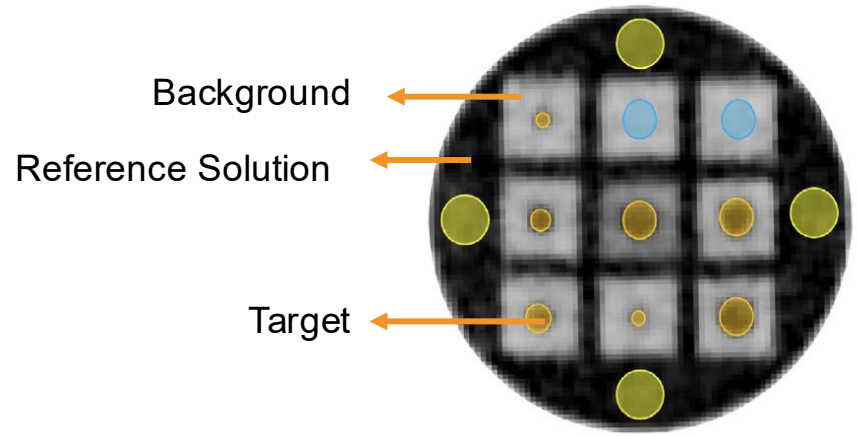
ANALYSIS

- Visual comparison of CT and PET images
- Manual VOI (target diameter) in PETvCAR
- 4 Semi-quantitative metrics : **1 CT + 3 PET**
- Statistical tests and numerical equivalence tests (10% bounds)



8 kg Total phantom weight

Inserts Printed in Acrylonitrile Butadiène Styrène (ABS)



Methods : Quantitative Metrics & Statistical Tests

Friedman + Kendall W
(Significance p < 0.05)
$N_{samples}$: 36

Wilcoxon + Holm
(Significance p < 0.05)
$N_{samples}$: 36

MAPE;
GMR with 90% CI ;
TOST ±10% / ±15%

HU

Hounsfield Units

Directly extracted from the CT at 2.5 mm resolution

SUV

Standardized Uptake Value

$$SUV_{mean/max} = \frac{AC_{VOI,\,mean/max}\left(\frac{kBq}{ml}\right) * M_{Phantom}\,(g)}{A_{inj}\,(kBq)} \quad (5)$$

TRI / RC

Target Representation Index
(Recovery Coefficient)

$$TRI_{mean/max} = \frac{SUV_{target;\,mean\,/max}}{SUV_{reference\ solution;\,mean}} \quad (6)$$

DC

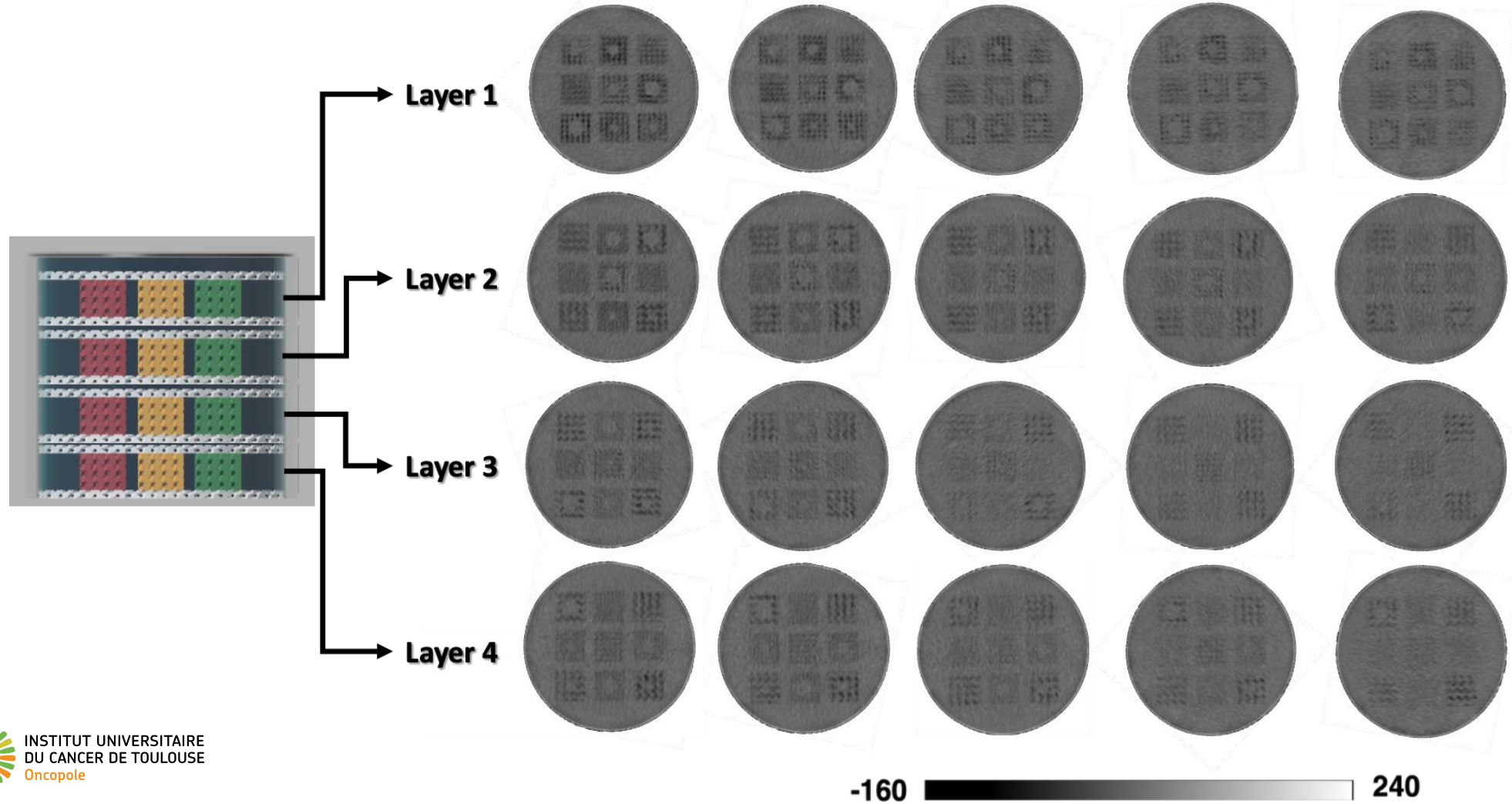
Dilution Coefficient
(For Homogeneous Cubes)

$$DC_{mean/max} = \frac{SUV_{background;\,mean\,/max}}{SUV_{reference\ solution;\,mean}} \quad (7)$$

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Results : CT (2.5mm) for different days

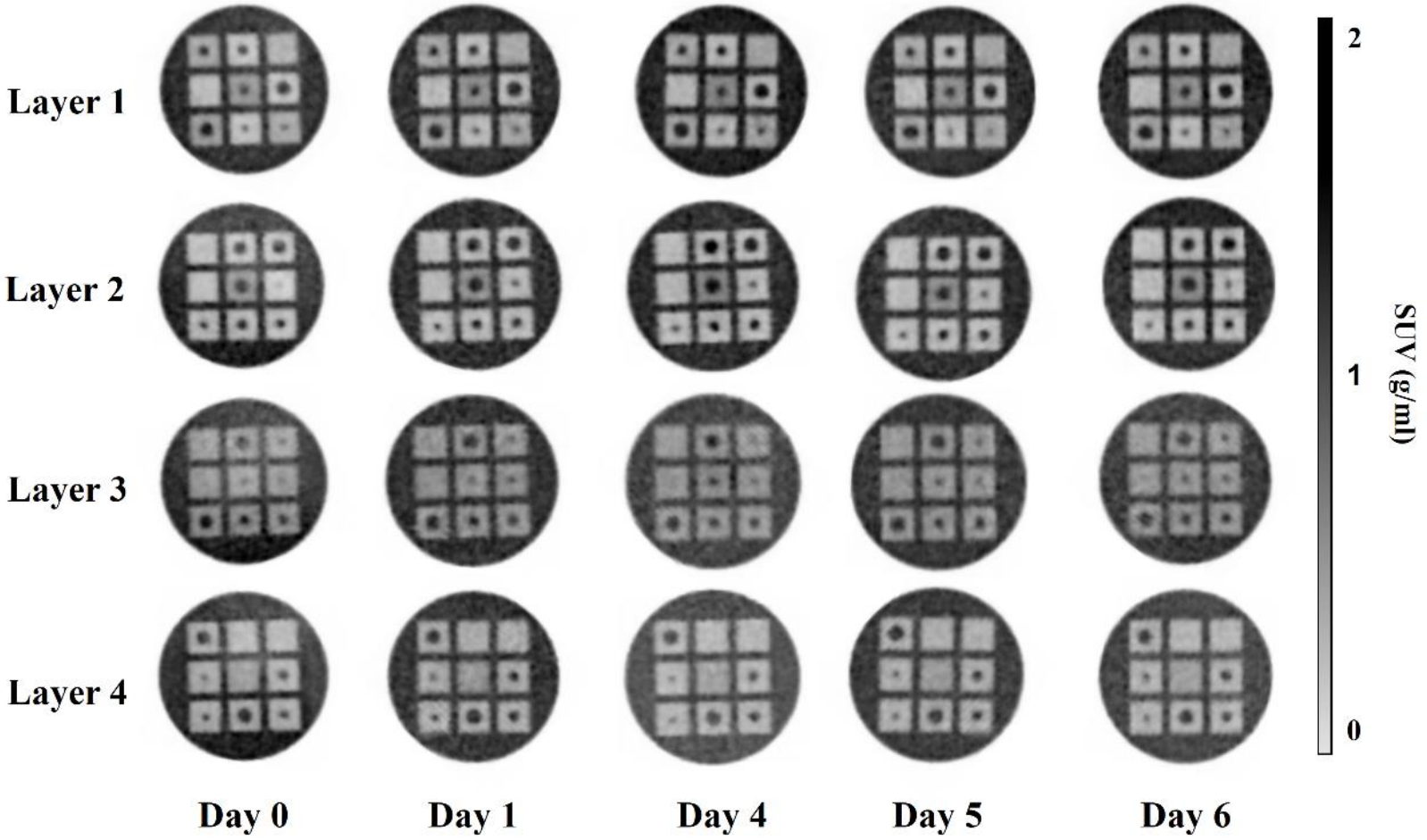
- Visible differences of attenuation across days (progressive improvement towards water)

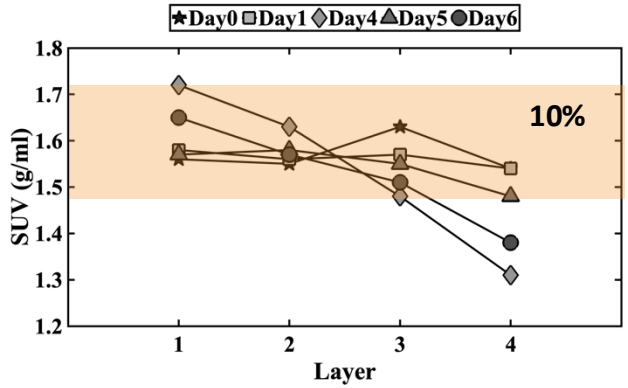


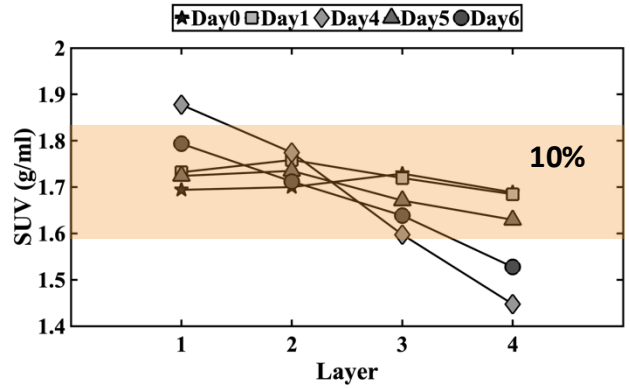
HU drift
+1.5–2.7 HU/day

D_p : 5 mm have the best attenuation by day 6

Results : PET for different days



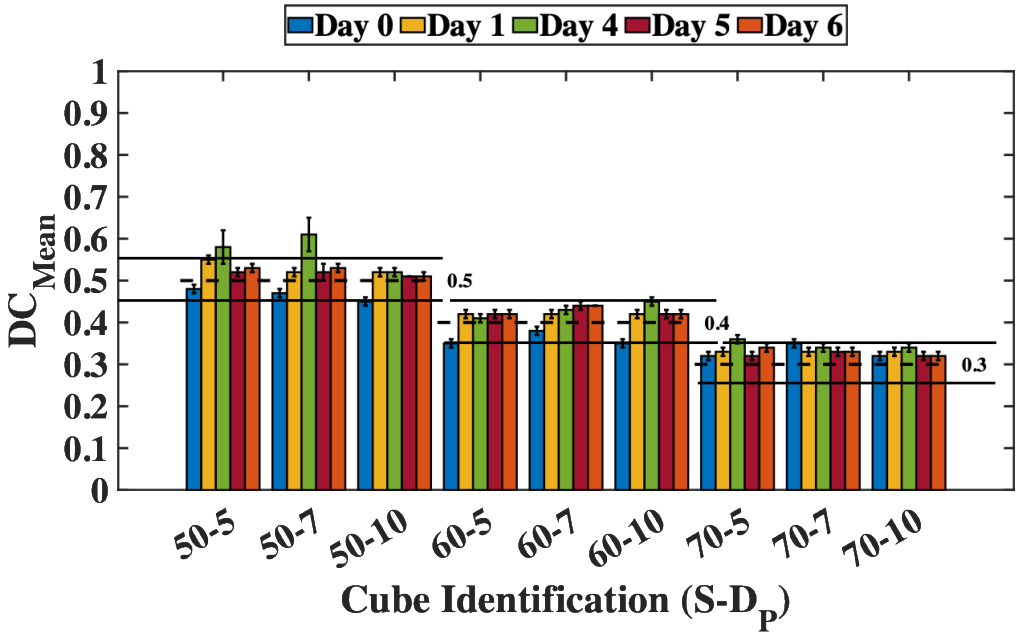


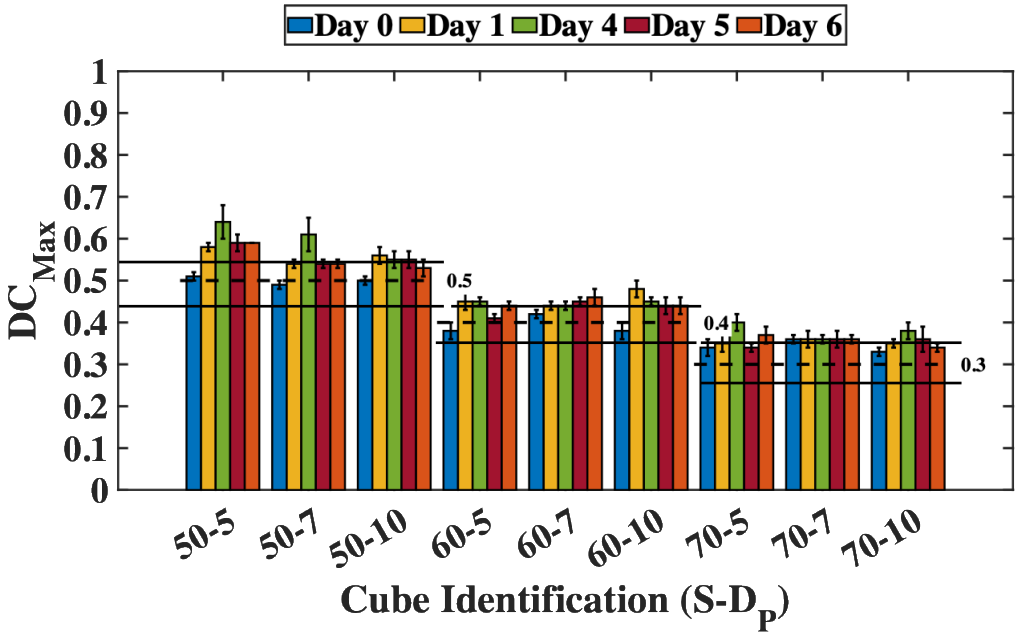


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- **Day 4 & Day 6 are visually different across layers and fall out of the prescribed 10% margin**

Results : Dilution Coefficient – PET for different days





50% solidity
DC 0.60–0.49 vs 0.50

60% solidity
DC 0.35–0.48 vs 0.40

70% solidity
DC 0.32–0.40 vs 0.30

- **All DC values fluctuate around theoretical but generally fall in 10% margin – Due to PET noise**
- **Finding similar to Jessen et al : An intercept (bias) in a linear fit of volume fraction vs contrast**

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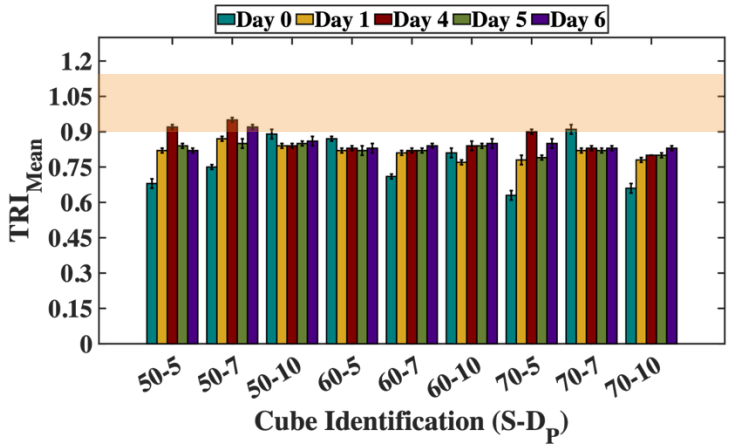
https://doi.org/10.1186/s40658-024-00613-7

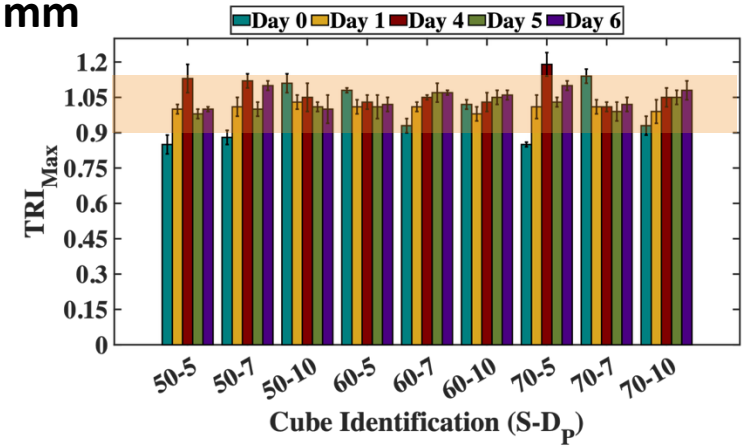
TRI for different days

- ✓ No clear quantitative / qualitative trend as function of S or D_P
- ✓ Reduction in TRI values based on T i.e. PVE as function of T

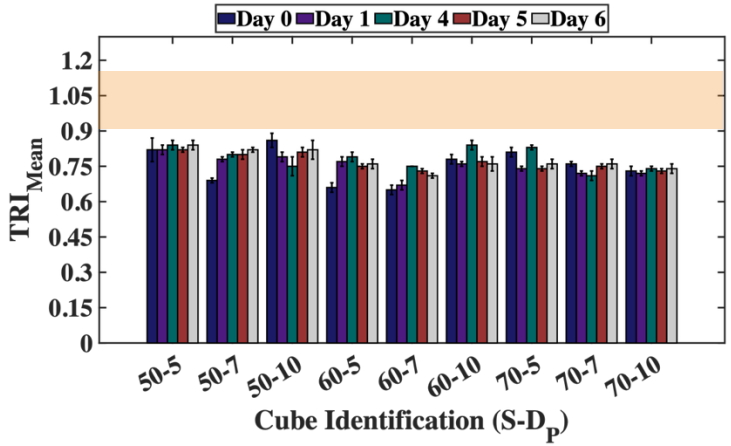
1 ± 0.1

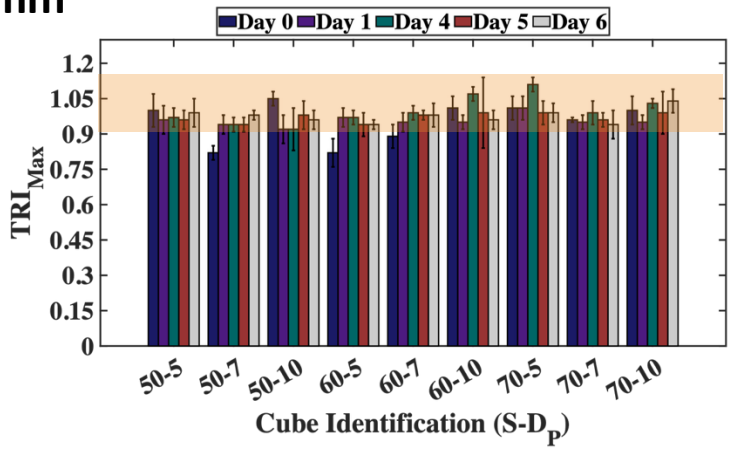
20 mm



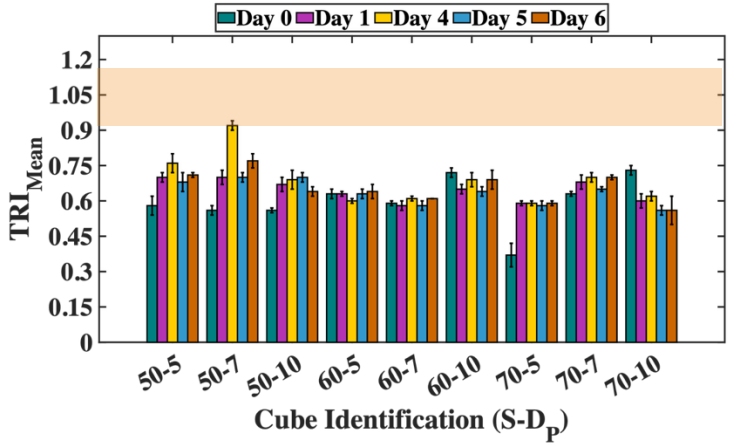


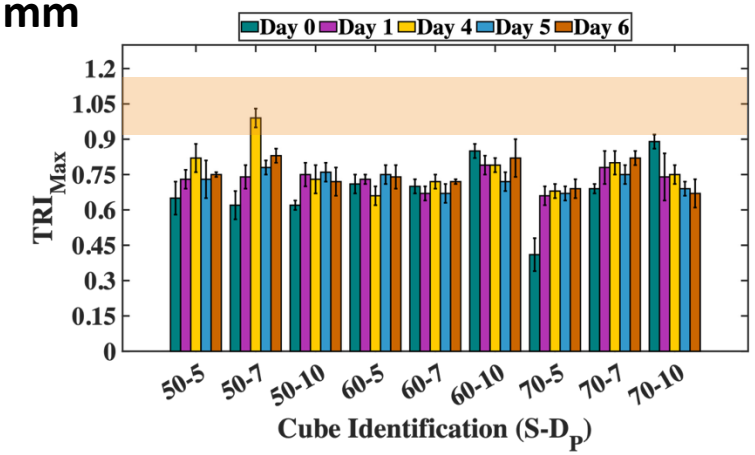
15 mm





10 mm





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Results : Statistical Tests on Metrics

- Global Friedman tests show significance only for DC_{max}
- D4 (For SUV_{mean} & SUV_{max}) & D6 (For SUV_{mean}) led to significance in pairwise tests

Statistical Test	Key Results (Significance – 0.05)
Friedman Test	TRI_{max}: p-value = 0.67 TRI_{Mean}: p-value = 0.93 DC_{max}: p-value = 0.03 DC_{Mean}: p-value = 0.07

Mean SUV — Holm-adjusted p

	D0	D1	D4	D5	D6
D0	—	0.0998	0.0063 **	0.0786	0.0096 **
D1	0.0998	—	0.0001 ***	0.3185	0.0018 **
D4	0.0063 **	0.0001 ***	—	0.0056 **	0.0998
D5	0.0786	0.3185	0.0056 **	—	0.0017 **
D6	0.0096 **	0.0018 **	0.0998	0.0017 **	—

Max SUV — Holm-adjusted p

	D0	D1	D4	D5	D6
D0	—	0.1750	0.0190 *	0.1600	0.0600
D1	0.1750	—	0.0007 ***	0.4270	0.0620
D4	0.0190 *	0.0007 ***	—	0.0056 **	0.1750
D5	0.1600	0.4270	0.0056 **	—	0.1060
D6	0.0600	0.0620	0.1750	0.1060	—

* p<0.05 ** p<0.01 *** p<0.001

Pairwise Wilcoxon with Holm correction

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Discussion

Designed Contrast remained visible, while preparation protocol contributed to the largest day-to-day changes

LIMITATIONS

- Preparation protocol must be modified to improve repeatability
- Material standardization will play a key role especially for CT
- Manual VOI extraction can induce errors
- Calculation used full phantom weight for SUV computation

RECOMMENDATIONS

- Use fillable weight / volume (4.5 – 4.6 L for the 36-cube design) for SUV computation
- Verify material attenuation maps used for PET AC

Presented at

SFPM 2024

Conception et évaluation d'un modèle imprimé en 3D pour une préparation facilitée et robuste de fantômes radioactifs en médecine nucléaire : application à l'évaluation de la détection et de la quantification des systèmes TEP/TDM

Quentin Maronnier*[1], Sai Kiran Kumar Nalla[2], and Olivier Caselles[1]

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Résumé

Introduction : Les études fantômes sont nécessaires à la calibration et l'évaluation des performances des systèmes d'imagerie. Cependant, en médecine nucléaire, les études expérimentales peuvent être affectées par l'introduction d'incertitudes liées à la manipulation de substance radioactive et la préparation du fantôme, comme la répétabilité ou la reproductibilité. Dans cette étude, nous testons un modèle expérimental créé par fabrication additive sur un système de Tomographie par Emission de Positons (TEP) couplée à un tomodensitomètre (TDM). Ce modèle repose sur des objets imprimés qui sont immergés dans un fantôme radioactif, visant à réduire les incertitudes associées aux expérimentations.
Matériel et méthodes : 36 cubes, de 4 cm de côté, ont été imprimés et forment des maillages perméables aux liquides. Parmi ces cubes, 9 sont totalement homogènes, tandis que les 27 autres renferment en leur centre une cible sphérique vide de tailles variables (1, 1,5 et 2 cm). De plus, parmi l'ensemble des cubes, 12 ont été imprimés avec une proportion de remplissage à 50%, 12 à 60%, et les 12 derniers à 70%. Ces taux de remplissage correspondent respectivement à des contrastes avec les cibles sphériques remplies de 2, 2,5 et 3,3. Pour chaque niveau de remplissage, nous explorons plusieurs tailles de maillage (10, 7 et 5 mm) afin de sélectionner la meilleure configuration assurant une perméabilité optimale tout en minimisant les bulles d'air dans le maillage. Les cubes sont répartis sur 4 niveaux dans un fantôme homogène, avec 9 cubes par niveau. Nous réalisons 5 séries de mesures (réparties sur 5 jours) au cours desquelles nous injectons dans le fantôme une activité équivalente de Fluor-18. Chaque série comprend 5 acquisitions consécutives. Nous avons évalué la robustesse du modèle proposé en mesurant et en comparant les Unités Hounsfield (UH) sur les images TDM ainsi que la quantification obtenue à partir des données TEP.

Résultats : D'après les images TDM, des bulles d'air sont présentes en faible quantité et disparaissent progressivement. Cette observation rejoint la tendance des valeurs d'UH reportées dans les cubes homogènes qui augmentent au fil des séries et se rapprochent de 0.

*Intervenant

sciencesconf.org:sfpm-js2024:551264

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SCIENTIFIC PAPER

Enhancing PET/CT target assessment with porous 3D printed grids: a pilot study

Sai Kiran Kumar Nalla[1] · Quentin Maronnier[2] · Tala Palchan-Hazan[3] · John A. Kennedy[3,4] · Olivier Caselles[2,5]

Received: 30 July 2025 / Accepted: 3 December 2025
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Abstract

Phantom experiments are widely used for standardisation in positron emission tomography (PET), but current practices do not necessarily reflect clinical reality and require meticulous phantom preparation for repeatability. 3D printing can reduce these limitations by optimizing preparatory methods and improving phantom features. This work proposes employing 3D-printed porous grids as an alternative mechanism to emulate targets with contrast. Acrylonitrile butadiene styrene (ABS) cubic grids (4 cm/side) with varying design characteristics and targets were printed. Grids were immersed in a [^{18}F] FDG solution with soap within a conventional phantom. Five consecutive acquisitions were repeated on five different days (Day 0, 1, 4–6) using a Discovery MI PET/CT. Target representation index (TRI) (analogous to recovery coefficient) and dilution coefficient (DC) were the metrics used for the analysis. Friedman test was utilized for statistical inference across days. PET images resulted in clear demarcation of various contrast regions produced by the dilution grid. Quantitative metrics showed consistent results across trials, confirming robustness. Dilution coefficient achieved (mean±std. dev.) were 0.55±0.05, 0.41±0.06, and 0.33±0.03 versus 0.5, 0.4 and 0.3 (theoretical), falling within 10% threshold. Observed $TRI_{max,\ mean}$ were in range of 0.4–1.2. Correlation across days was strong for $TRI_{max,\ mean}$ (p-values≥0.67) but the DC_{max} (p-values ~ 0.03) values denoted minor bias in generated contrast due to noise. 3D-printed grids offer a reliable, reproducible alternative for PET/CT assessment. 27 hot targets with varying contrasts and size were produced with a single tracer administration and the metrics stayed stable across different acquisitions.

Keywords Radionuclide imaging · Additive manufacturing · Fused deposition modeling · Quantitative PET/CT · Nuclear medicine · Rapid prototyping · Molecular imaging

Abbreviations

ABS	Acrylonitrile butadiene styrene
AC	Activity concentration in kBq/ml
CNR	Contrast to noise ratio
DC	Dilution coefficient
DMI	Discovery MI
D_p	Distance between pores
FDM	Fused deposition modelling
FDG	Fluorodeoxyglucose
FOV	Field of view
HU	Hounsfield units
IEC	International Electrotechnical Commission
NEMA	National Electrical Manufacturing Association
NM	Nuclear medicine
PET/CT	Positron emission tomography/computed tomography
PLA	Polylactic acid

The data supporting the findings of this study are available from the authors upon reasonable request.

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Published online: 24 December 2025

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Does synthetic lesion insertion reproduce the measured response of uniform printed targets?

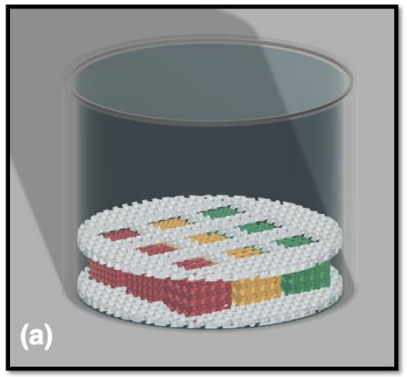
METHODS : Based on 1 layer in 36 insert design

- 1 Acquisition (3 frames) performed on TOF - 5R DMI system
- [^{18}F] FDG (**58.7** MBq ; **15** mins decay)
- **TOF-BSREM** (Q clear – BPL) reconstruction with $\boldsymbol{\beta = 550}$

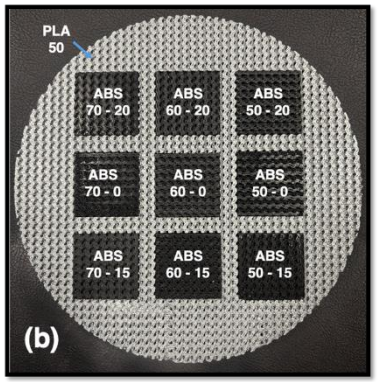
Only D_p: 5 mm inserts used

ANALYSIS

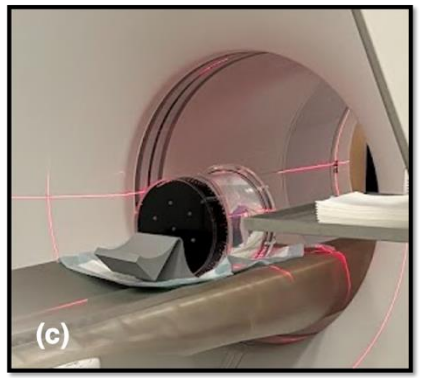
- Visual comparison of CT and PET images
- Manual VOI (target diameter) in PETvCAR
- 2 Semi-quantitative metrics : 1 CT + 1 PET
- Mean absolute percentage error (10% bounds)



Envisioned Setup

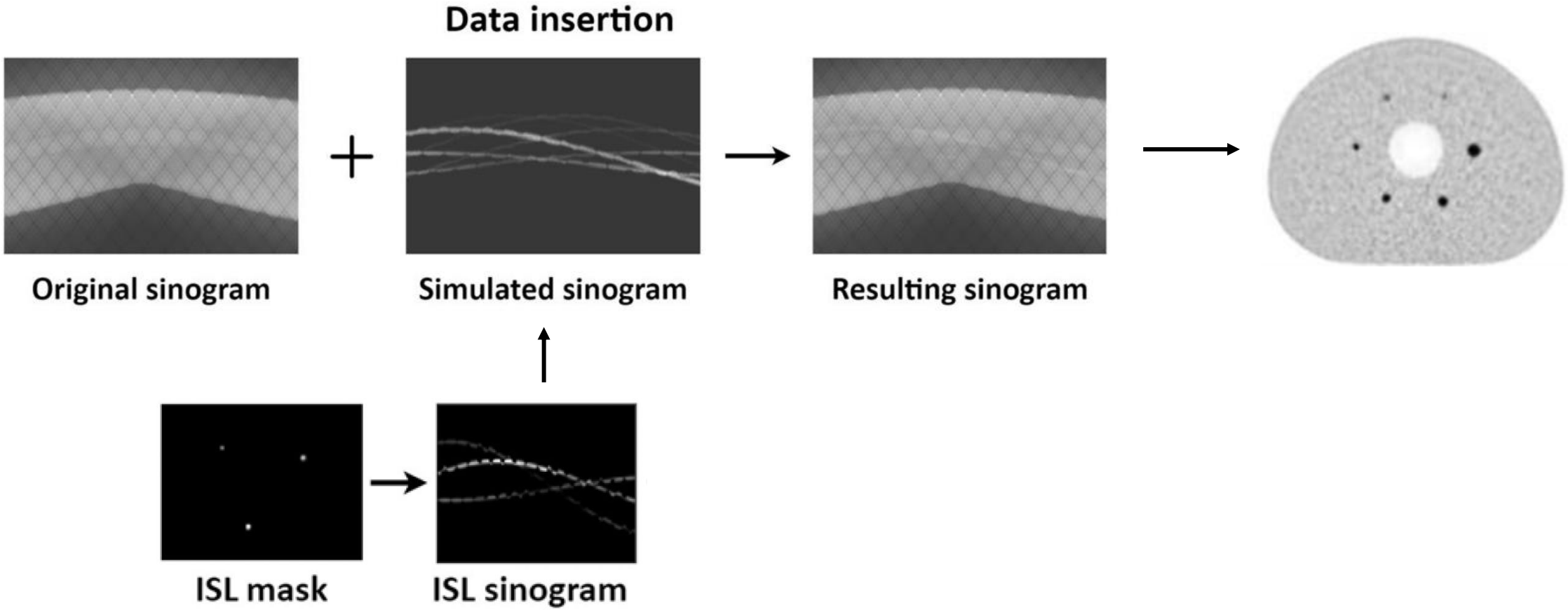


Cube Arrangement



Setup on Scanner

DUETTO – Synthetic Lesion Insertion



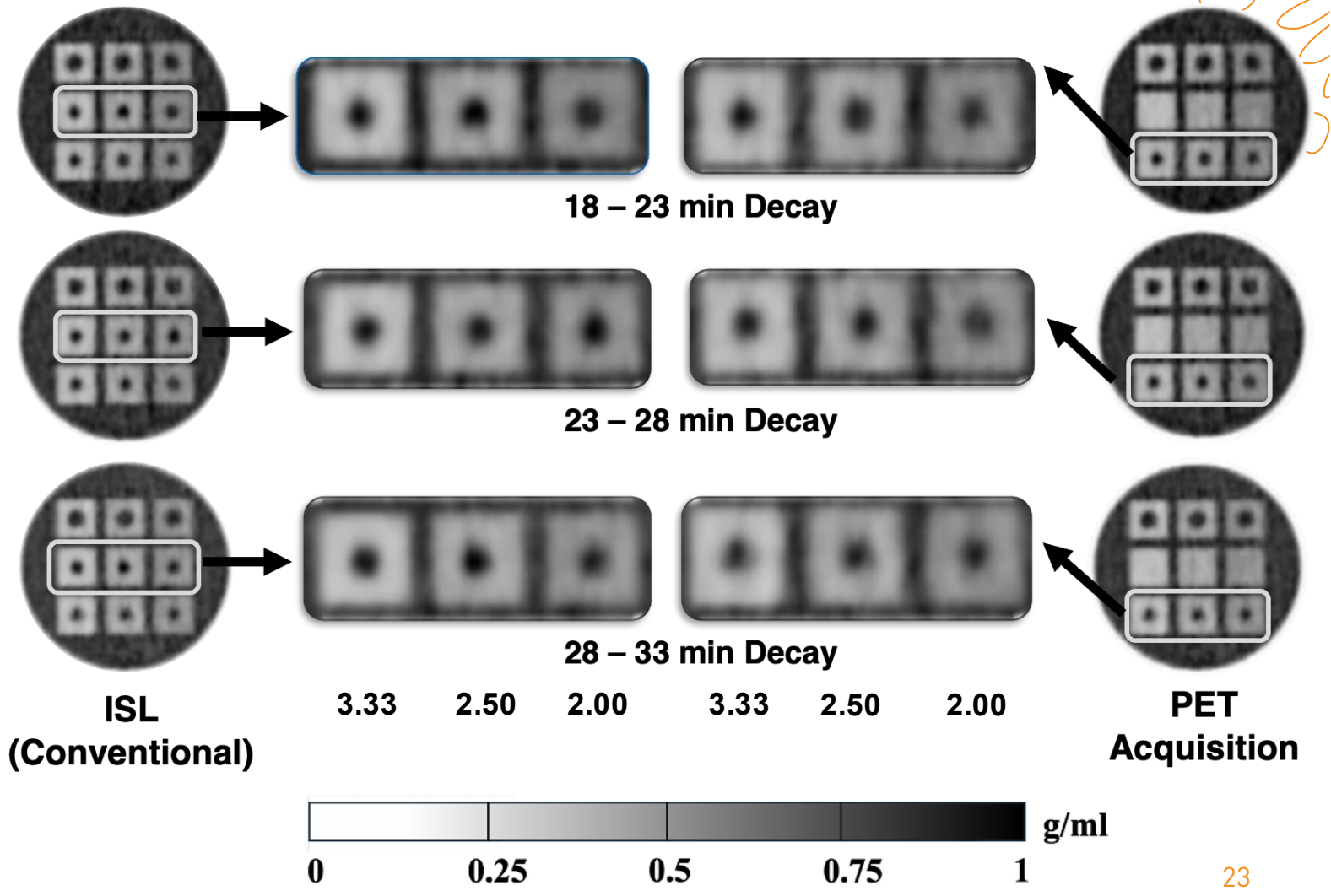
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Synthetic vs Acquired Image for 15 mm targets

S	Exp	Sim	Abs. diff. (%)
50%	0.82	0.82	0.0
60%	0.80	0.83	3.8
70%	0.75	0.77	2.7

S	Exp	Sim	Abs. diff. (%)
50%	1.03	1.16	12.6
60%	1.09	1.20	10.1
70%	1.10	1.18	7.3

- No major noticeable differences visually



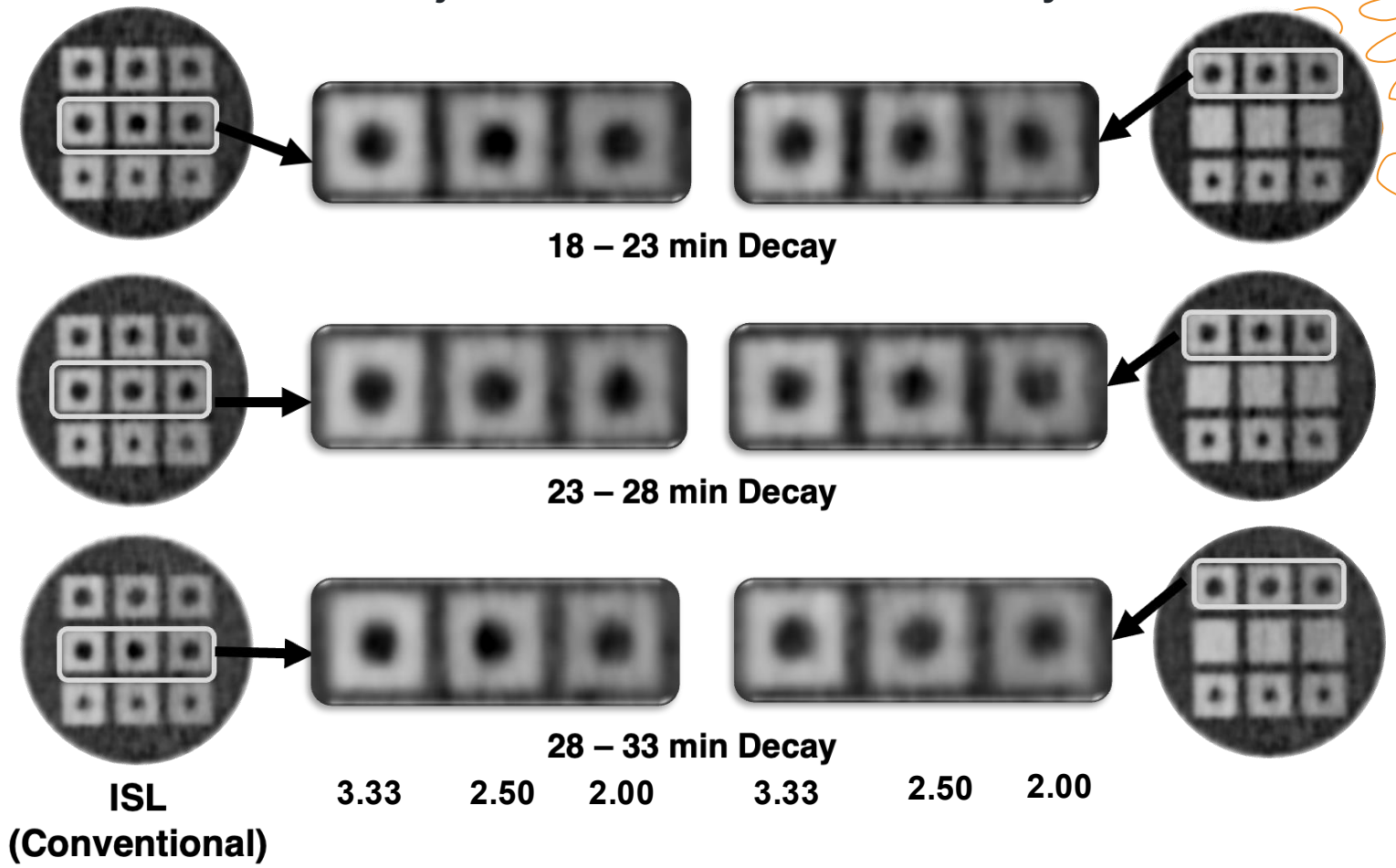
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Synthetic vs Acquired Image for 20 mm targets

S	Exp	Sim	Abs. diff. (%)
50%	0.87	0.82	5.7
60%	0.85	0.82	3.5
70%	0.81	0.82	1.2

S	Exp	Sim	Abs. diff. (%)
50%	1.10	1.23	11.8
60%	1.15	1.32	14.8
70%	1.12	1.23	9.8

- No major noticeable differences visually



Discussion

Uniform insertion reproduced the mean PET contrast response

LIMITATIONS

- Peak values remained systematically biased
- Only two target diameters were assessed

RECOMMENDATIONS

- Calibration of inserts (designed vs measured contrast)
- Higher contrasts (Ex: 10:1) needs to be tested

Accepted at

ECR 2025

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EPOS

Fillable 3D-Printed Phantom Grids: Enhancing Accuracy in PET/CT Quality Control

Poster No.: C-28068

Congress: ECR25

Authorblock: S. K. K. Nalla[1], Q. Maronnier[1], O. Caselles[1], J. Kennedy[2], T. Palchan Hazan[2]; [1]Toulouse/FR, [2]Haifa/IL

Disclosures:
Sai Kiran Kumar Nalla: Nothing to disclose
Quentin Maronnier: Nothing to disclose
Olivier Caselles: Nothing to disclose
John Kennedy: Nothing to disclose
Tala Palchan Hazan: Nothing to disclose

Keywords: Hybrid Imaging, Molecular imaging, Nuclear medicine, PET-CT, Molecular imaging, Quality assurance

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Please note: Links to movies, ppt slideshows and any other multimedia files are not available in the pdf version of presentations.

This Work

Design Evaluation of Porous 3D printed Grids: Experiment + Simulation

Pilot study · multi-contrast encoding · single centre

This Work

Dual-Centre Transferability of Multi contrast Phantom : Experiment

Same phantom · two independent institutions · consistent results

This Work

Heterogeneous Uptake in Single Encoded grid: Experiment + Simulation

First order and second order metrics . Dual tracer

This Work

Hybrid Clinical implementation workflow for Anthropomorphic Lesions

8 clinical shapes · hybrid framework · proof of concept validated

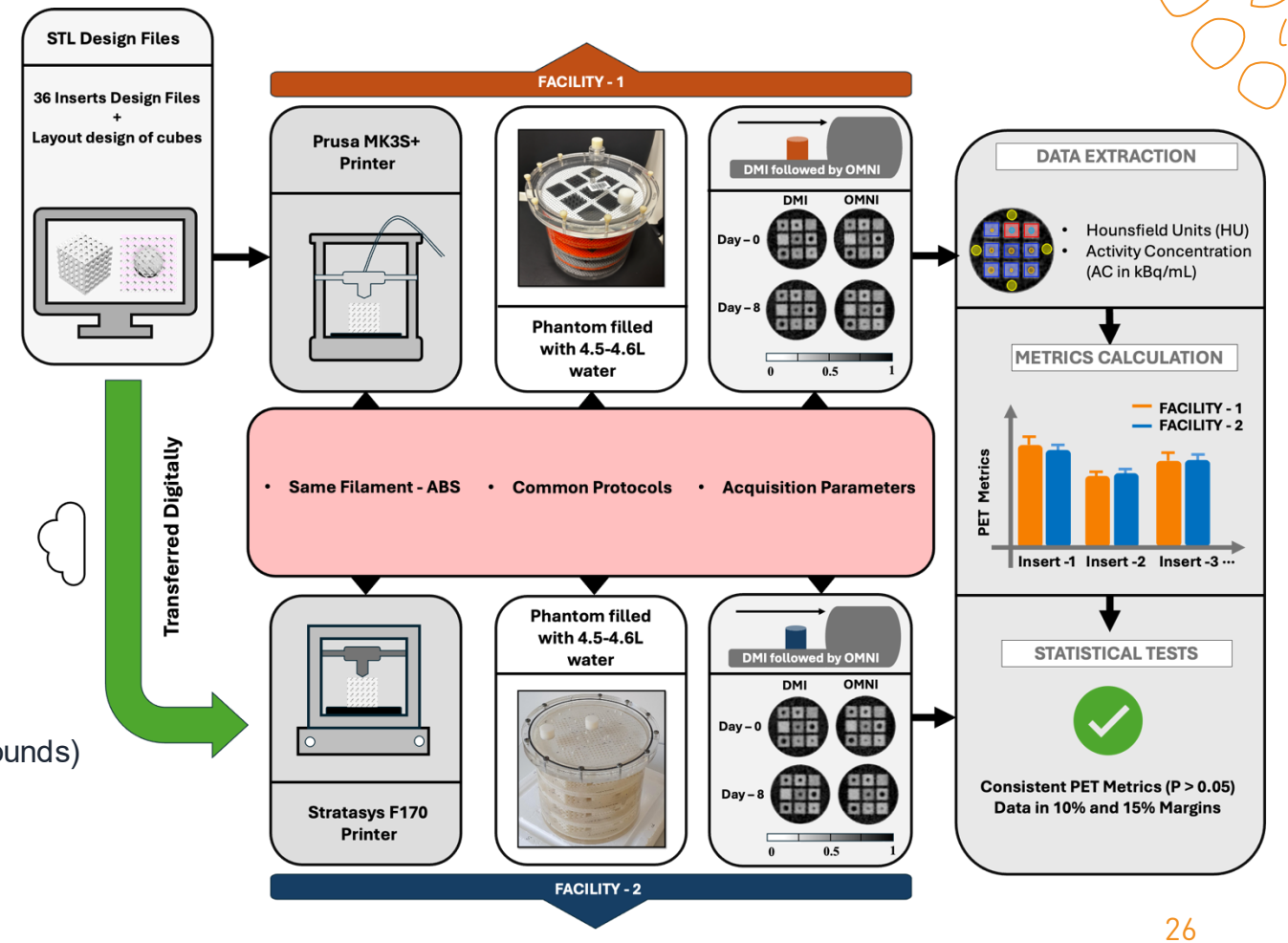
Do inserts produced at different facilities from common design files and protocols produce similar quantitative response?

METHODS : Based on 36 insert design

- Acquisitions performed on Day 0 and Day 8
- **TOF - 5R DMI** (5 x 300s) & **Non TOF - OMNI legend** (5 x 420s)
- [^{18}F] FDG (**matched Decay corrected activity**) – EARL preparation
- **BSREM** reconstruction with $\mathbf{TOF, \beta = 550}$ (DMI) and $\mathbf{\beta = 300}$ (OMNI)
- Matrix size : **256 × 256 x 89 (DMI) & 256 × 256 x 153 (OMNI)**

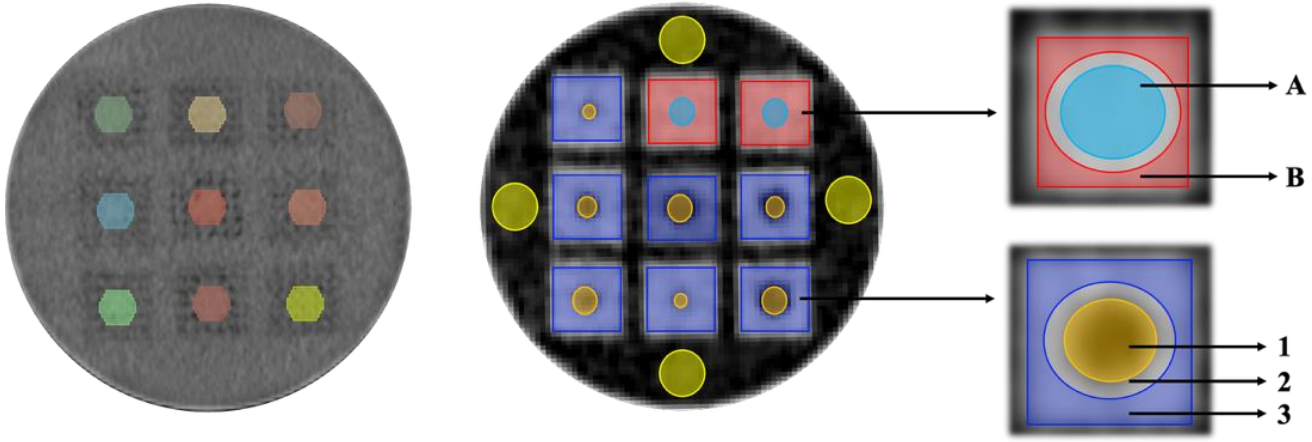
ANALYSIS

- Visual comparison of CT and PET images
- Semi-automatic ROI (target diameter) in MATLAB
- 4 Semi-quantitative metrics : 1 CT + 3 PET
- MAPE, global & pairwise statistical tests, metric equivalence (10% bounds)



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Methods : Data Extraction & Metrics



Modified Protocol

For injection, we use EARL protocol of removing around 50 mL solution and adding it back

KEY Change

SUV calculation uses the fillable weight (4.5 kgs) instead of total weight (8 kgs)

- HU, SUV, RC, DC are the metrics calculated (frame - day averaging approach)
- Additionally, new metric TBR also calculated for target cubes
- Same 3 step statistical test hierarchy as previous experimental study

TBR

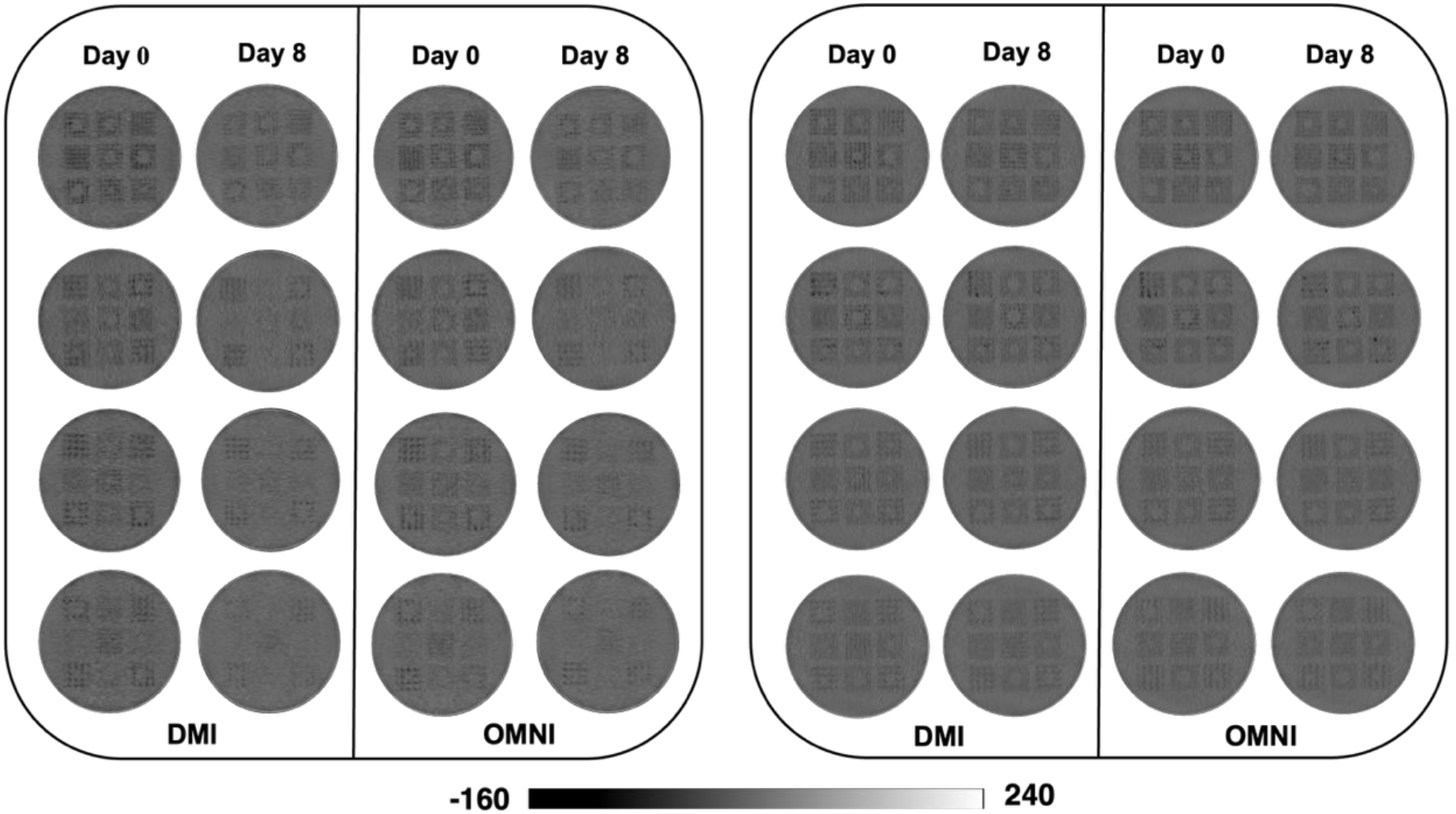
Target Background Ratio
(Only for Target Cubes)

$$\mathrm{TBR}_{\mathrm{mean/max}} = \frac{\mathrm{SUV}_{\mathrm{target;\ mean\ /max}}}{\mathrm{SUV}_{\mathrm{background;\ mean}}} \quad (8)$$

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CT at 2.5mm for cross facility

- Visible differences of attenuation across facilities
- D8 is much better than D0

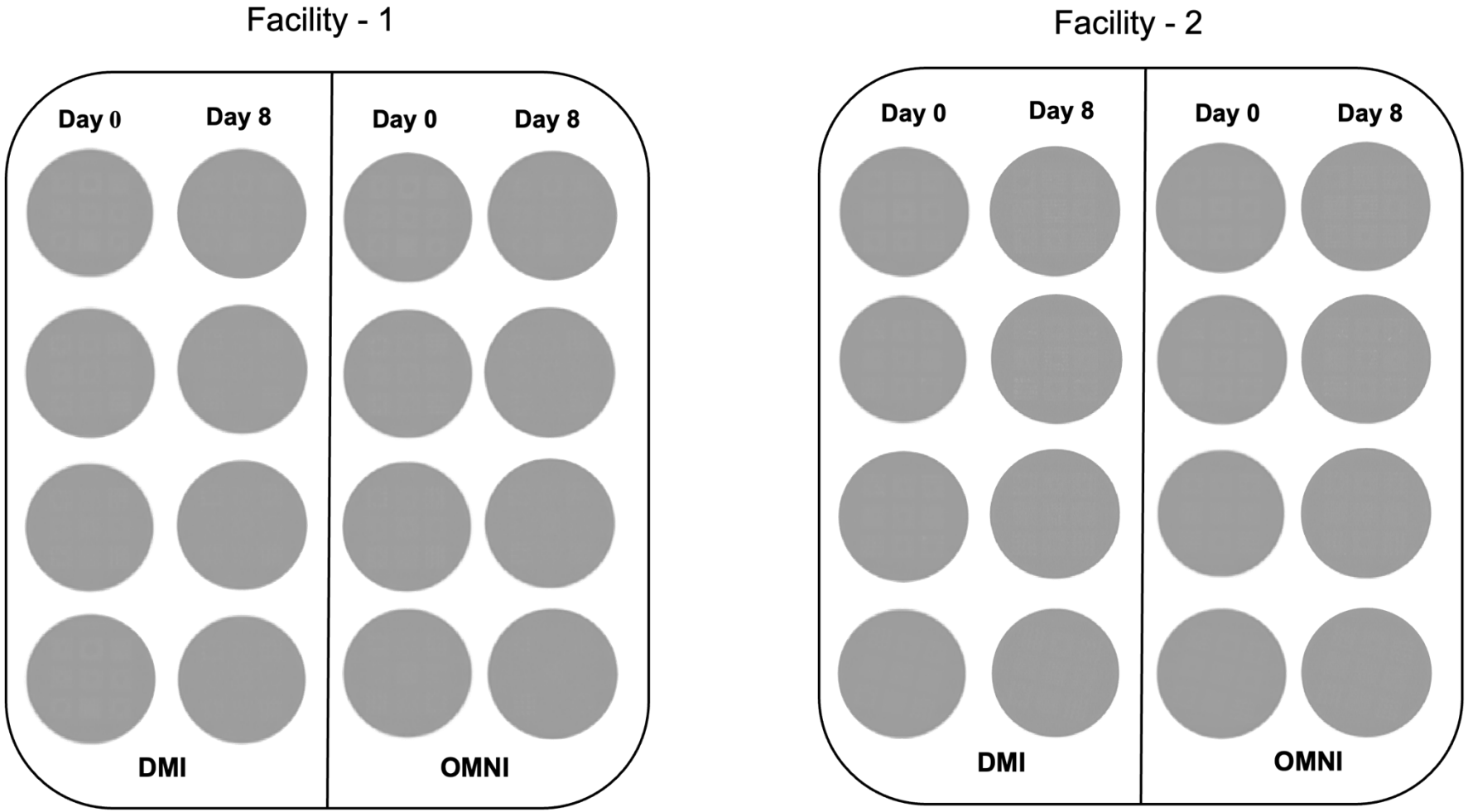


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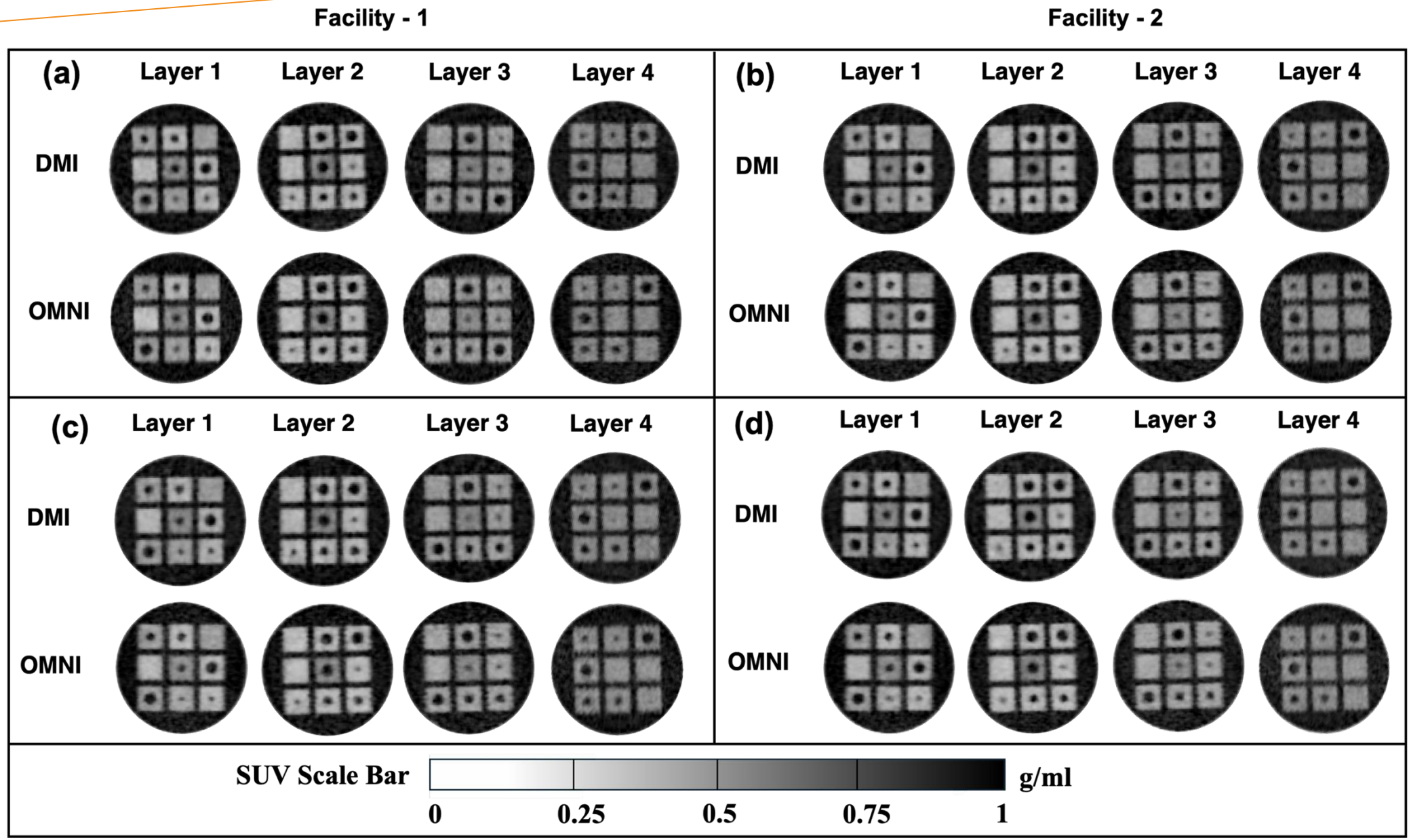
CT attenuation map used for PET correction

- No differences in attenuation contribution for PET correction across facilities

Note
Maps calculated retrospectively based on vendor specific bilinear model



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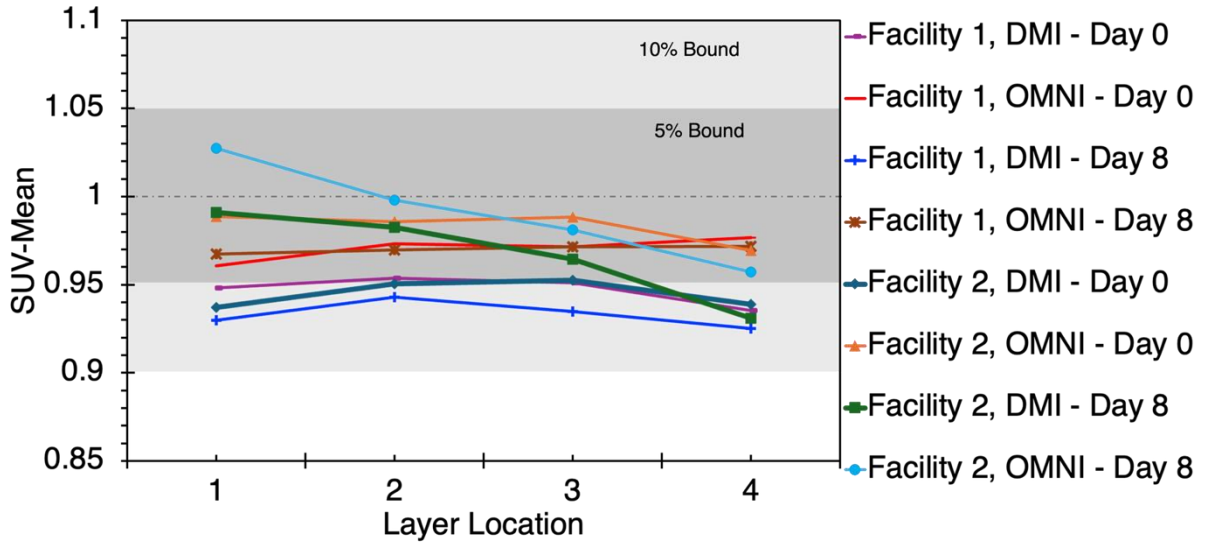


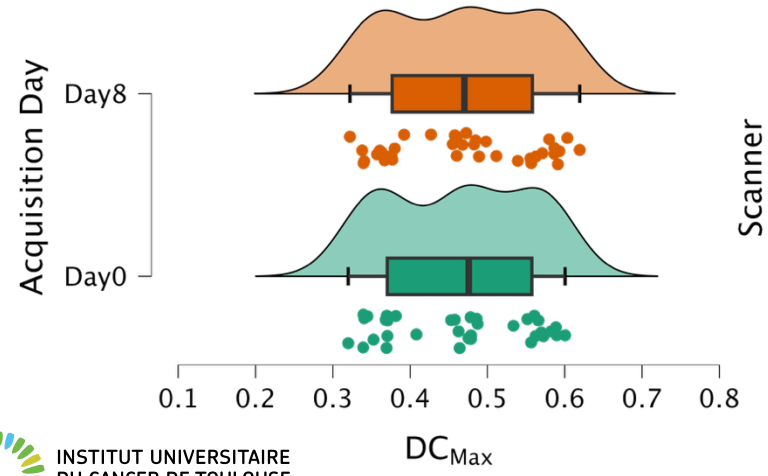
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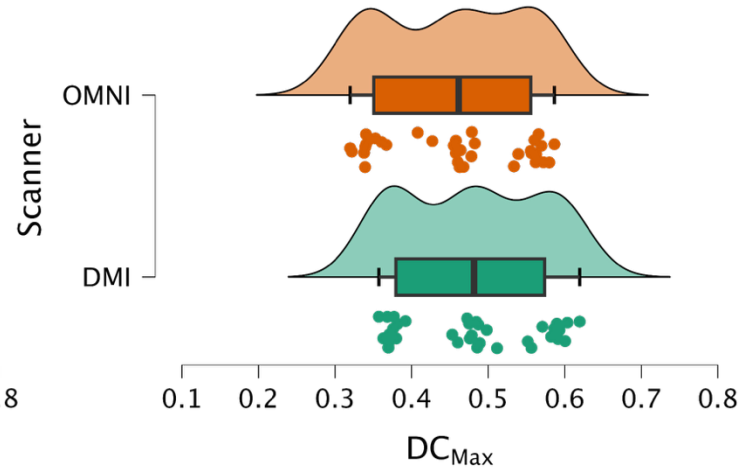
Central PET slice for each layer

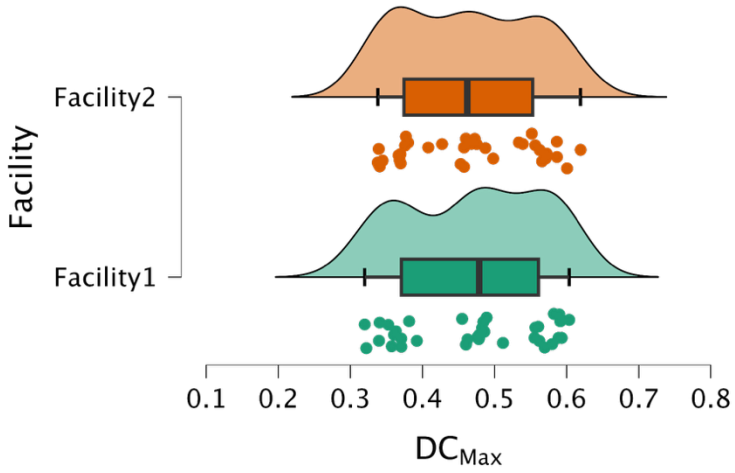
Layer variability and Dilution Coefficient

✓ **Homogeneous mixing confirmed**
(All comparisons within 10% bounds)









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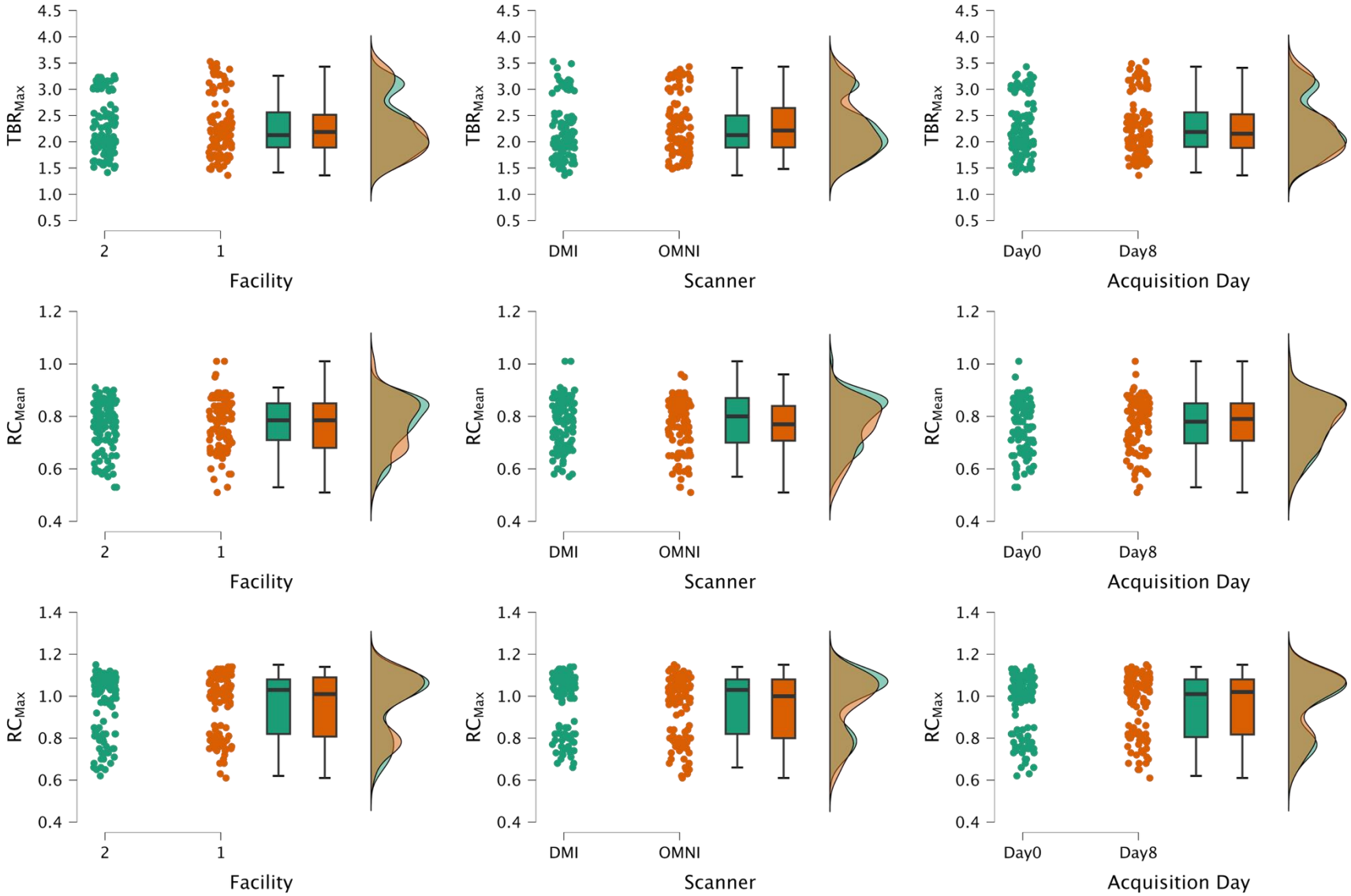
✓ **3 different clusters of data** x **Centered on theoretical values (0.3, 0.4, 0.5)**

PET Metrics distribution for Target cubes

Note

Data assorted parametrically for each column

(Ex: Facility 1 vs Facility 2 corresponds to combined OMNI + DMI and Day 0 + Day 8 of each facility)



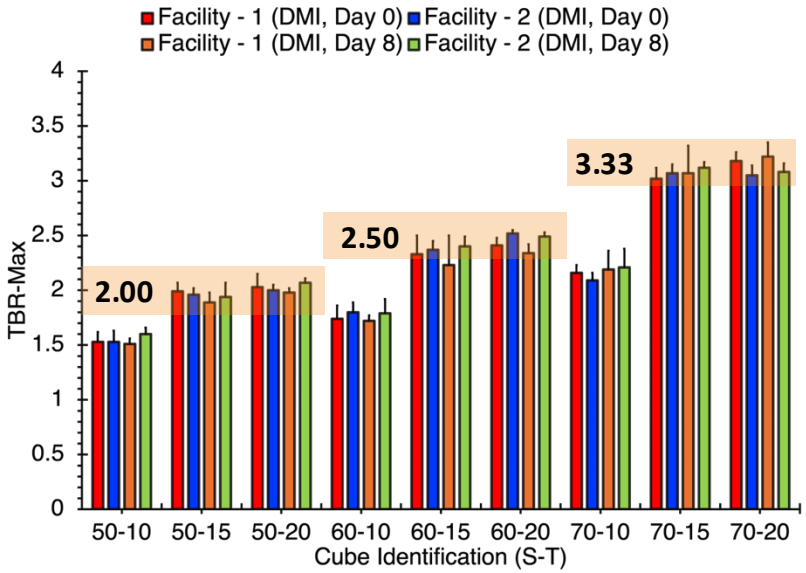
✓ **Data Distribution is similar across metrics**

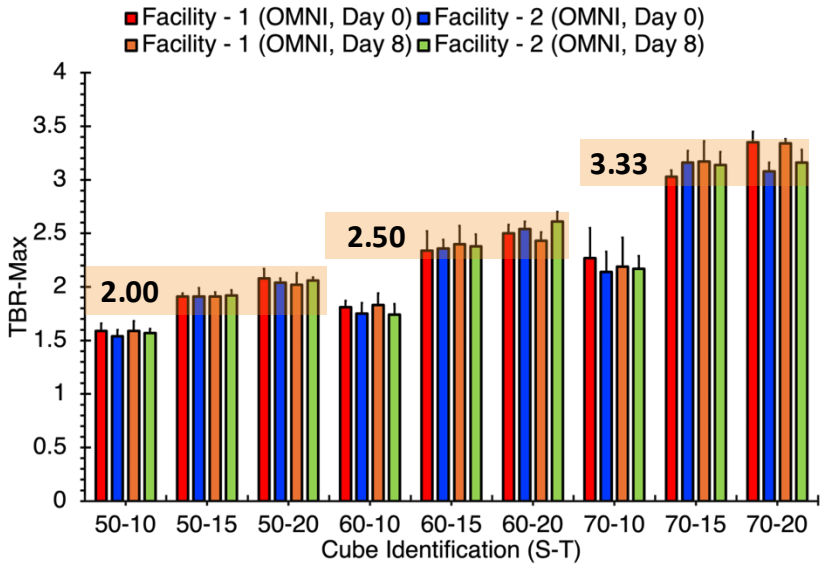
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Data Extraction & Metrics

Consistent across facilities

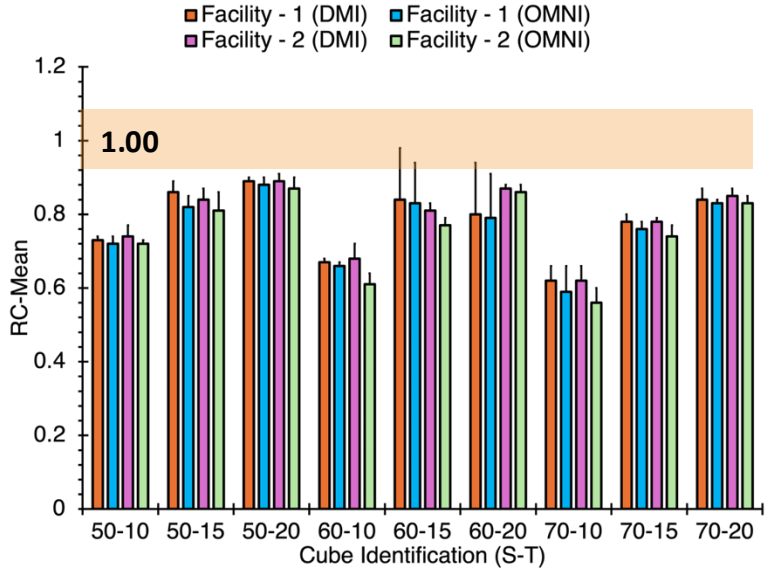
- ✓ 15 mm and 20 mm fall within 10% of TBR_{insert}
- ✓ All 10 mm values are outside 10% bounds

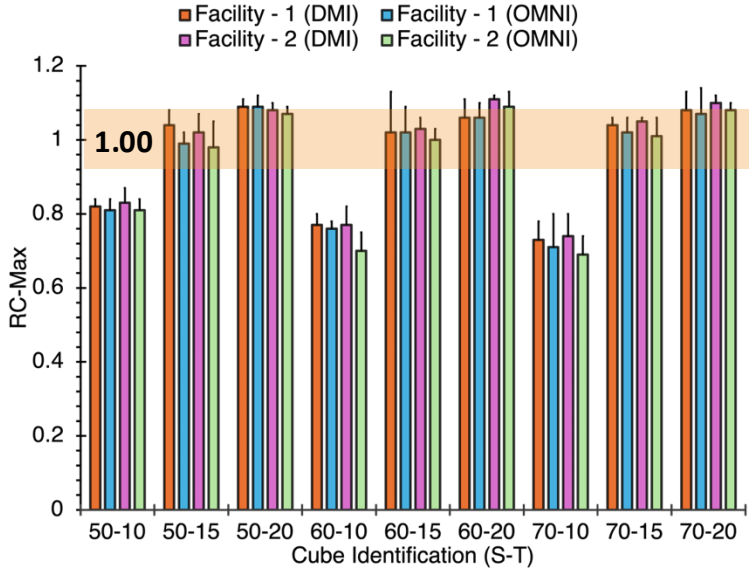




Consistent across facilities

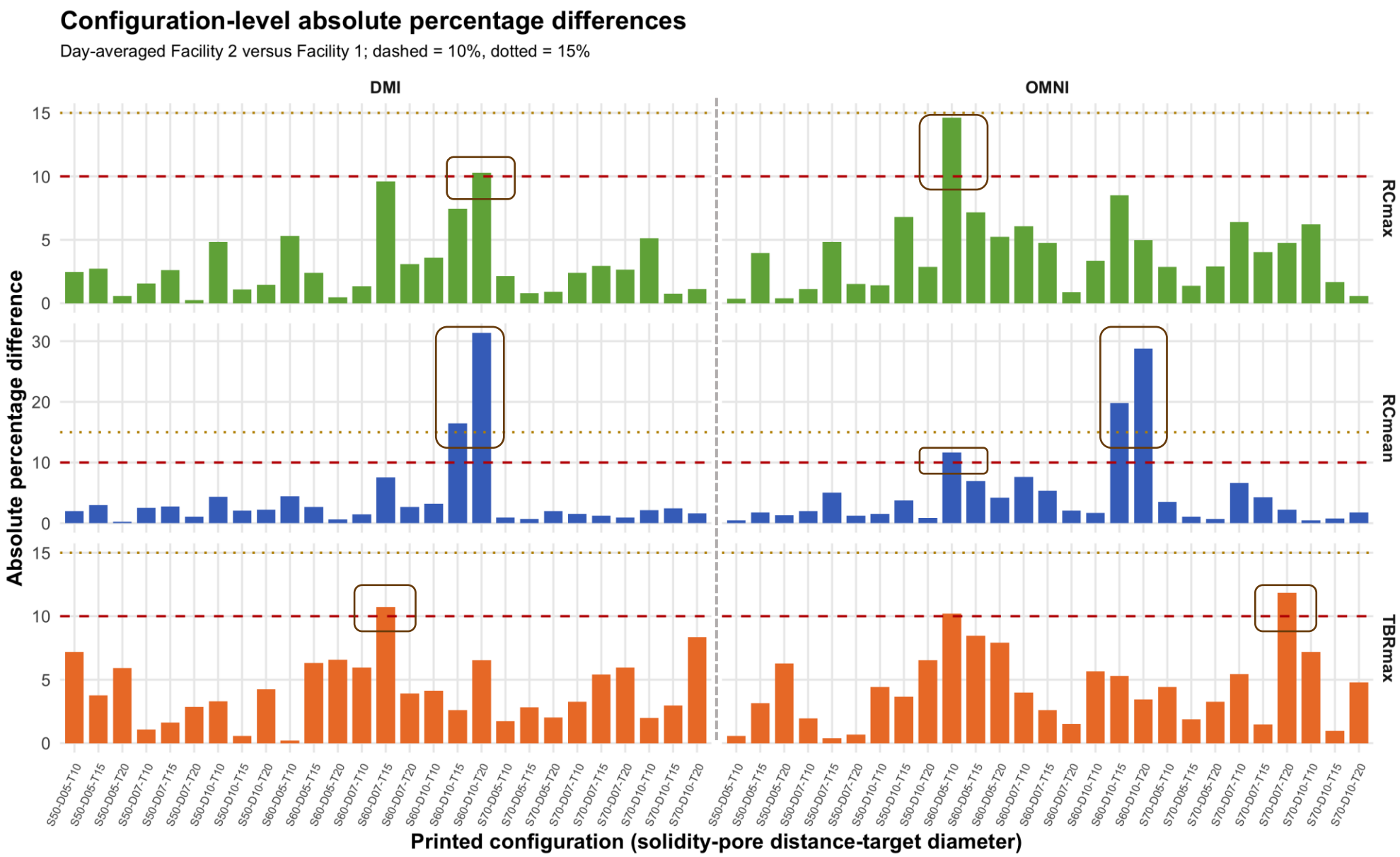
- ✓ 15 mm and 20 mm closer to 1 for RC_{max}
- ✓ No RC_{mean} in 10% bounds





10% bounds around theoretical / expected value

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Above 10%

2 / 54

3 / 54

2 / 54

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MAPE for RC and TBR of Target Cubes

Discussion

Average phantom response met ±10% equivalence bounds at two facilities

LIMITATIONS

- Only two facilities and GE scanner platforms were tested
- Material standardization will play a key role especially for CT
- Nominal ROIs did not measure printed geometry errors

RECOMMENDATIONS

- Repeat at more facilities and across scanner vendors
- Standardise printing, materials and imaging; include a NEMA reference study

Under Review at

Physica Medica

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Physica Medica

From Digital File to Transferable Phantom: A Dual-Centre Evaluation of Multi-contrast 3D Printed Phantom for PET Quantification

--Manuscript Draft--

Manuscript Number:	EJMP-D-26-00590
Article Type:	Technical note
Section/Category:	
Keywords:	PET/CT, 3D Printing, Dual-Centre

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	Quentin Maronnier, PhD
	Tala Palchan-Hazan
	Olivier Caselles, PhD
	John A. Kennedy, PhD

Abstract:	Background Three-dimensional printing enables cost-effective, customizable PET phantoms, but the reproducibility of locally manufactured designs across facilities remains underexplored. This study evaluated whether a shared multi-contrast phantom design could reproduce consistent design-defined PET image features after independent fabrication at two centres. Methods Thirty-six porous cubic inserts (40 mm), including homogeneous inserts and inserts containing 10, 15, or 20 mm spherical targets, were fabricated independently at two facilities from a common digital file. The inserts were placed in a Jaszczak phantom and scanned on 5R-Discovery MI and OMNI Legend PET/CT systems on Day 0 and Day 8. CT Hounsfield units, estimated attenuation coefficients at 511 keV, target-to-background ratio (TBR), and recovery coefficient (RC) were evaluated. TBR and RC were interpreted as relative image-derived reproducibility metrics. Results CT images showed minor inter-facility differences, while estimated attenuation coefficients remained within 5% of the reference water value. PET target visibility was preserved across facilities, scanners, and acquisition days. Mean TBRmax values were 1.56 ± 0.03, 2.35 ± 0.05, and 3.10 ± 0.05 for expected design ratios of 2.0, 2.5, and 3.33. RCmax and RCmean ranged from 0.69–1.11 and 0.56–0.89. No statistically significant differences were detected for TBRmax, RCmean, or RCmax. Bland–Altman mean biases were below 10%, although limits of agreement exceeded ±10% in some comparisons. Conclusion The findings support the feasibility of digitally transferring porous phantom designs and reproducing their design-defined PET image features after independent local manufacture. Broader multicentre and multivendor validation is required before use in harmonisation studies.

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This Work

Design Evaluation of Porous 3D printed Grids: Experiment + Simulation

Pilot study · multi-contrast encoding · single centre

This Work

Dual-Centre Transferability of Multi contrast Phantom : Experiment

Same phantom · two independent institutions · consistent results

This Work

Heterogeneous Uptake in Single Encoded grid: Experiment + Simulation

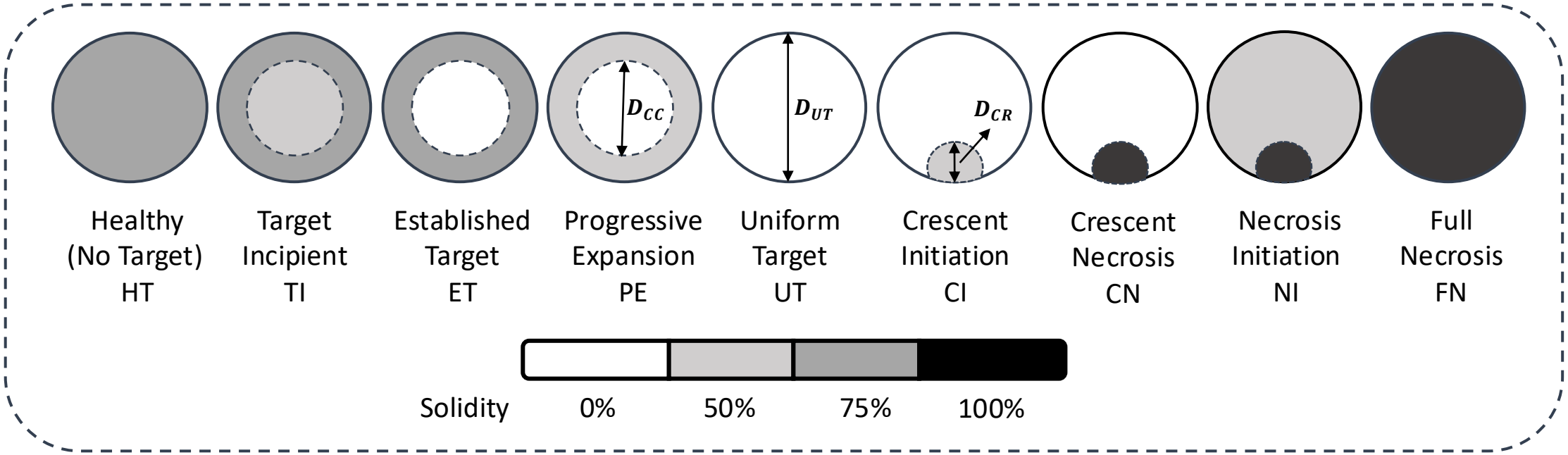
First order and second order metrics . Dual tracer

This Work

Hybrid Clinical implementation workflow for Anthropomorphic Lesions

8 clinical shapes · hybrid framework · proof of concept validated

Design of Multi-uptake Targets



- **Engineered sequence of inserts with nested regions**
- **D_{UT} tested are 17 , 22 and 28 mm respectively**
- **D_{CC} are 10, 12 and 17 mm for corresponding D_{UT}**

$TBR_{insert,\ theoretical}$ = 4.00 (S : 75%)

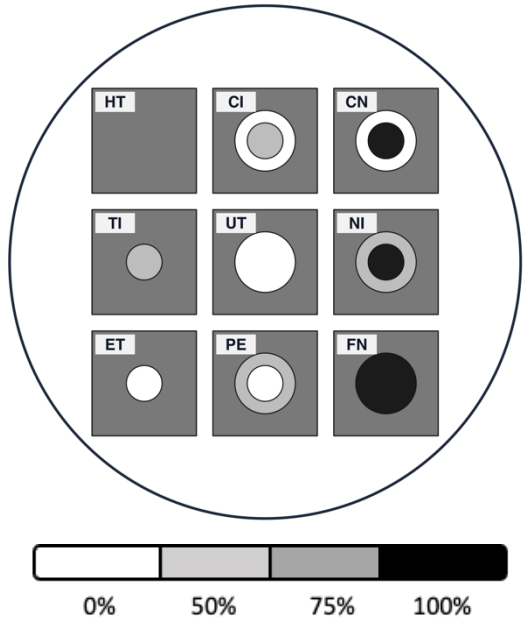
Heterogeneity is produced as a single printed object for each insert

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Can controlled 3D printed multi-uptake inserts be distinguished and ranked consistently with [^{18}F]FDG and [^{68}Ga]PSMA?

METHODS : Based on 27 inserts with 9 pattern design in each layer

- 3 diameters (17, 22, 28 mm) & TBR_{insert} for most active regions – 4:1
- Acquisitions performed on OMNI legend PET/CT system
- ^{18}F FDG and ^{68}Ga PSMA on separate days, at 50 and 30 MBq scan levels
- **BSREM** reconstruction with $\boldsymbol{\beta = 650}$ (^{18}F) and $\boldsymbol{\beta = 750}$ (^{68}Ga)



ANALYSIS

- Visual comparison of CT and PET images
- VOI data extraction and 2 PET metrics – SUV , GI
- Exploratory 60 panel radiomic metric extraction (14 first order + 46 second order)
- MAPE, global & pairwise statistical tests, rank preservation and correlation

Data Extraction and Preliminary Analysis

Reconstructed PET

FDG or Ga-68; native image grid

→

Place spherical VOI

Insert centroid; Ø = 17, 22 or 28 mm

→

Extract voxels

SUV values $x_1 \ldots x_n$ inside each VOI

→

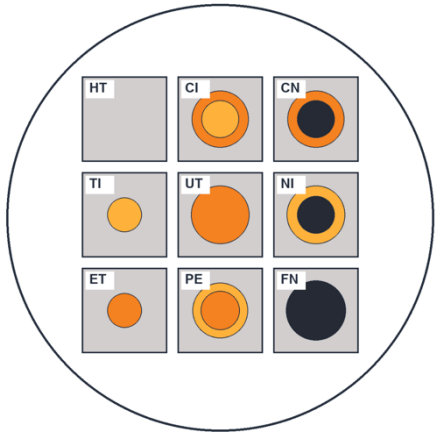
Primary metrics

SUV, TBR, Gini

Gini Index

Target Background Ratio
(Only for Target Cubes)

$$GI = \frac{2\left(\Sigma_i\, i \cdot x_i\right)}{n\left(\Sigma_i\, x_i\right)} - \frac{n+1}{n} \quad (8)$$



THEORETICAL APPROXIMATION FOR GI

Nominal region geometry → voxel activities from solution-accessible fraction

0% → 1 | 50% → 0.5 | 75% → 0.25 | 100% → 0

Voxelise a 100% VOI on the native grid; background shell = 0.25.

Sort assigned activities → same Gini formula; no blur, noise or PVE.

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Radiomic features: extraction and exploratory evaluation

1 VOI + normalise

Sphere = 110% target diameter

Mean reference activity

2 Prepare image

2.07 mm · linear interpolation

Fixed 0.05 bins

3 Extract + average

60 features per frame

3 frames → condition mean

60 features across four families

Intensity / histogram

14 features

12 first-order + 2 histogram



GLCM

9 features

Neighbouring grey-level pairs



GLRLM + GLSZM

32 features

16 runs + 16 zones



NGTDM

5 features

Local grey-tone differences



Three complementary evaluations



PATTERN EFFECT

Friedman

Nine insert configurations



RANK PRESERVATION

Spearman + Holm

F ↔ Ga ranks · rank correlation



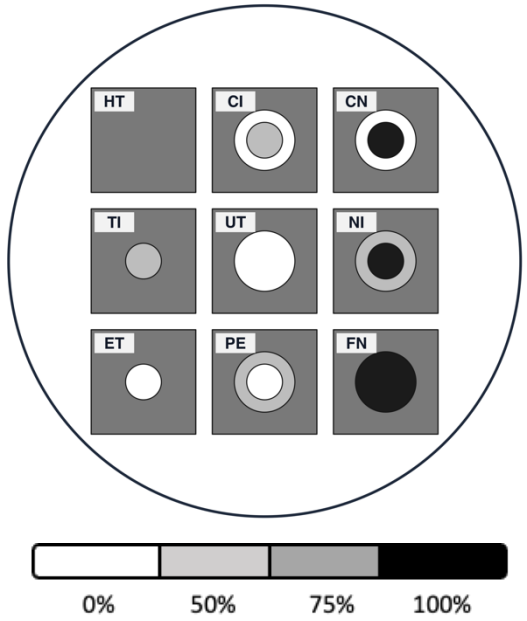
ABSOLUTE VALUES

Absolute % difference

Cross-tracer magnitudes (descriptive)

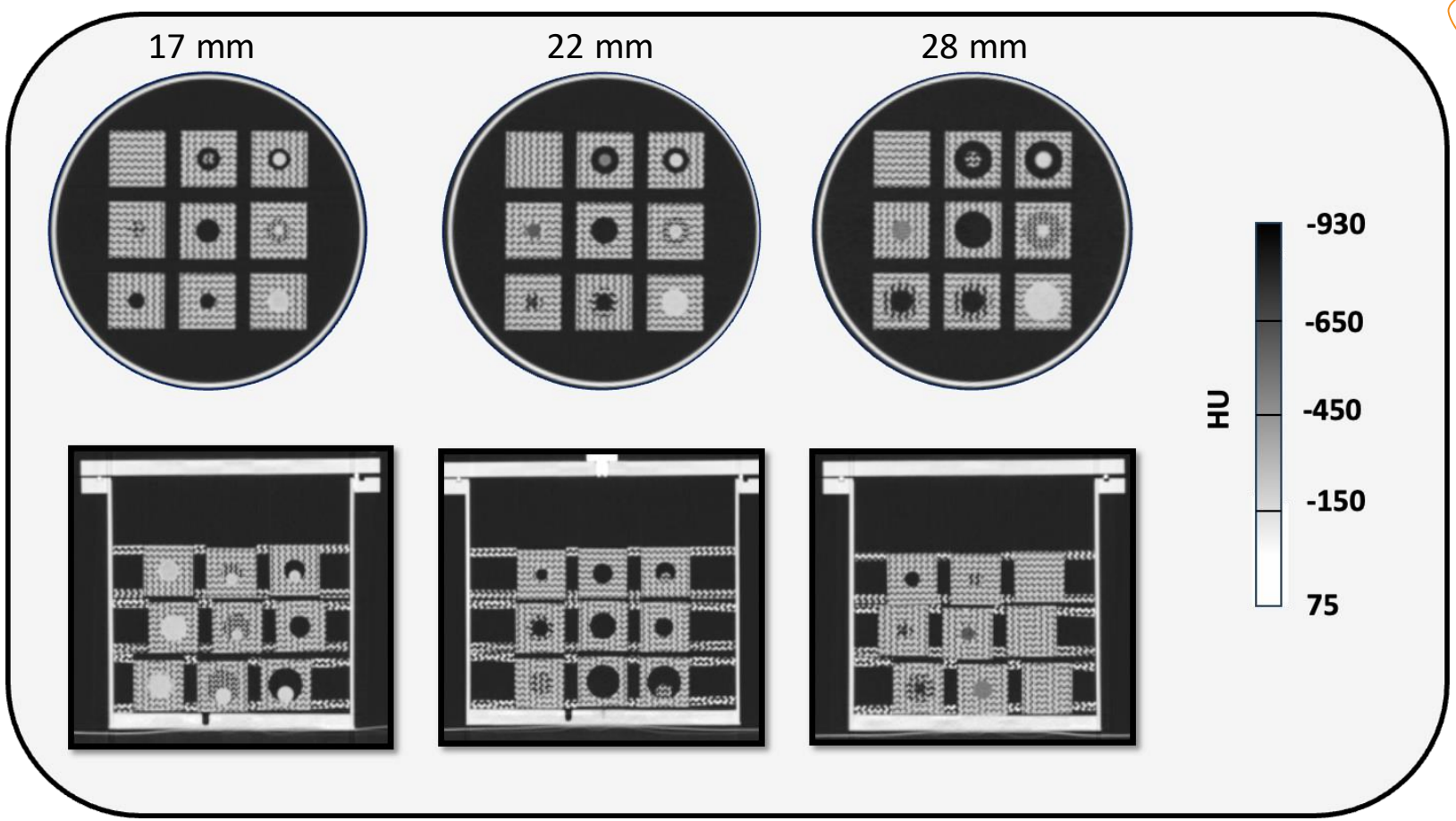
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CT at 1.5mm resolution in AIR

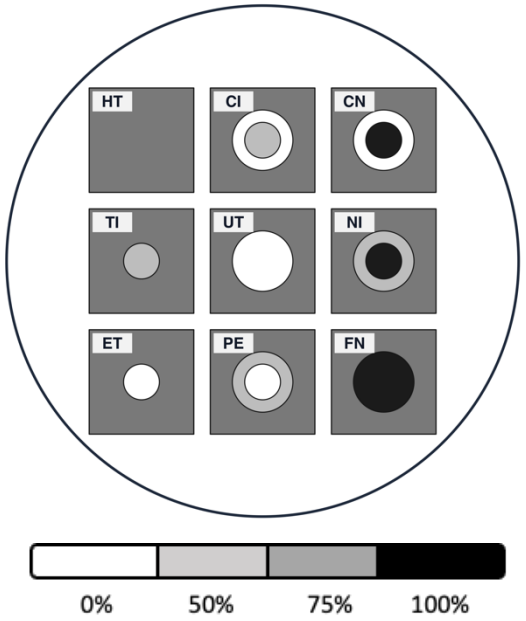


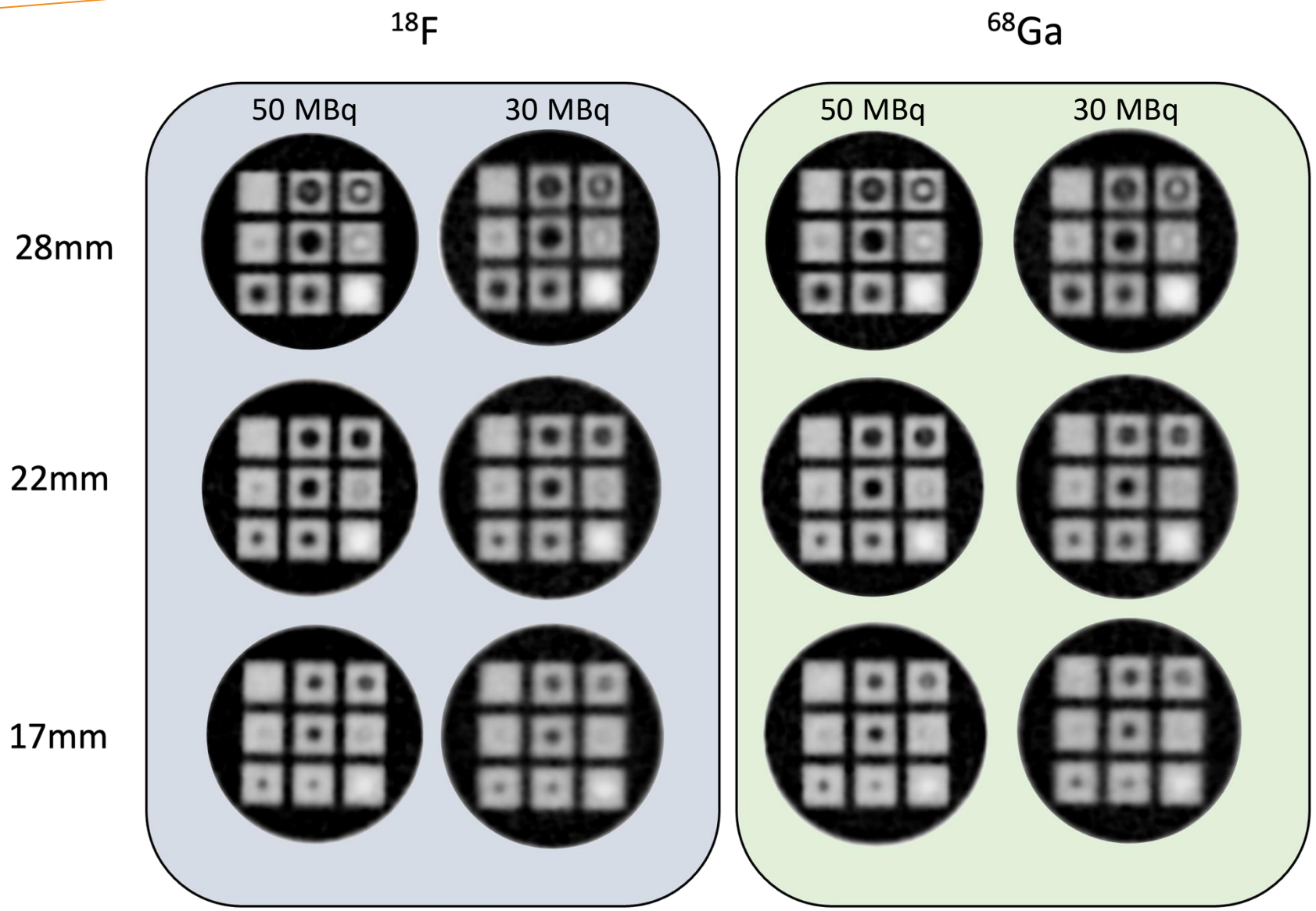
Theoretical Illustration

✓ **All insert regions are successfully printed**



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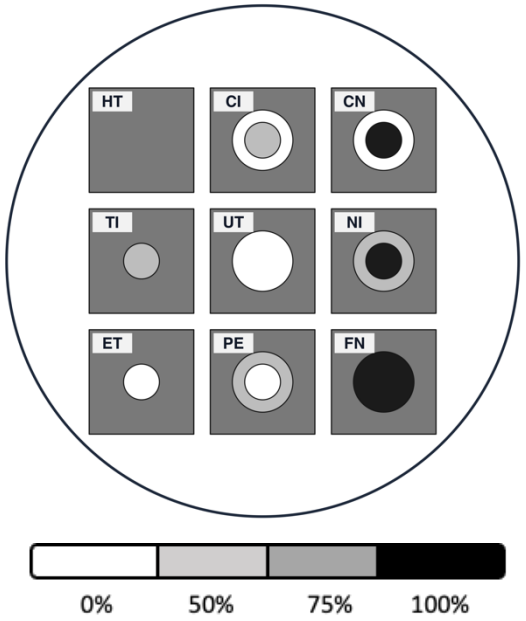


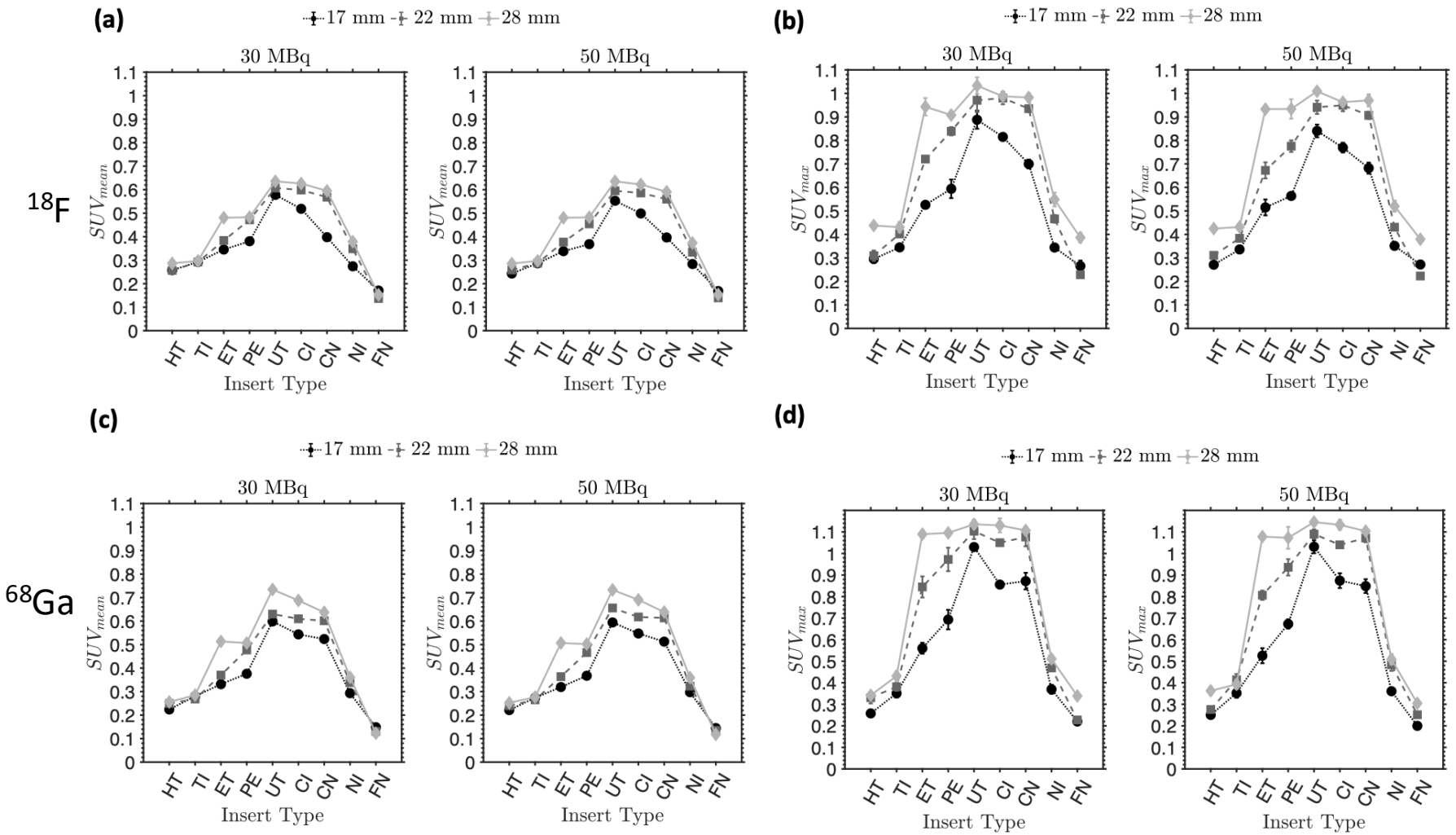


PET Images



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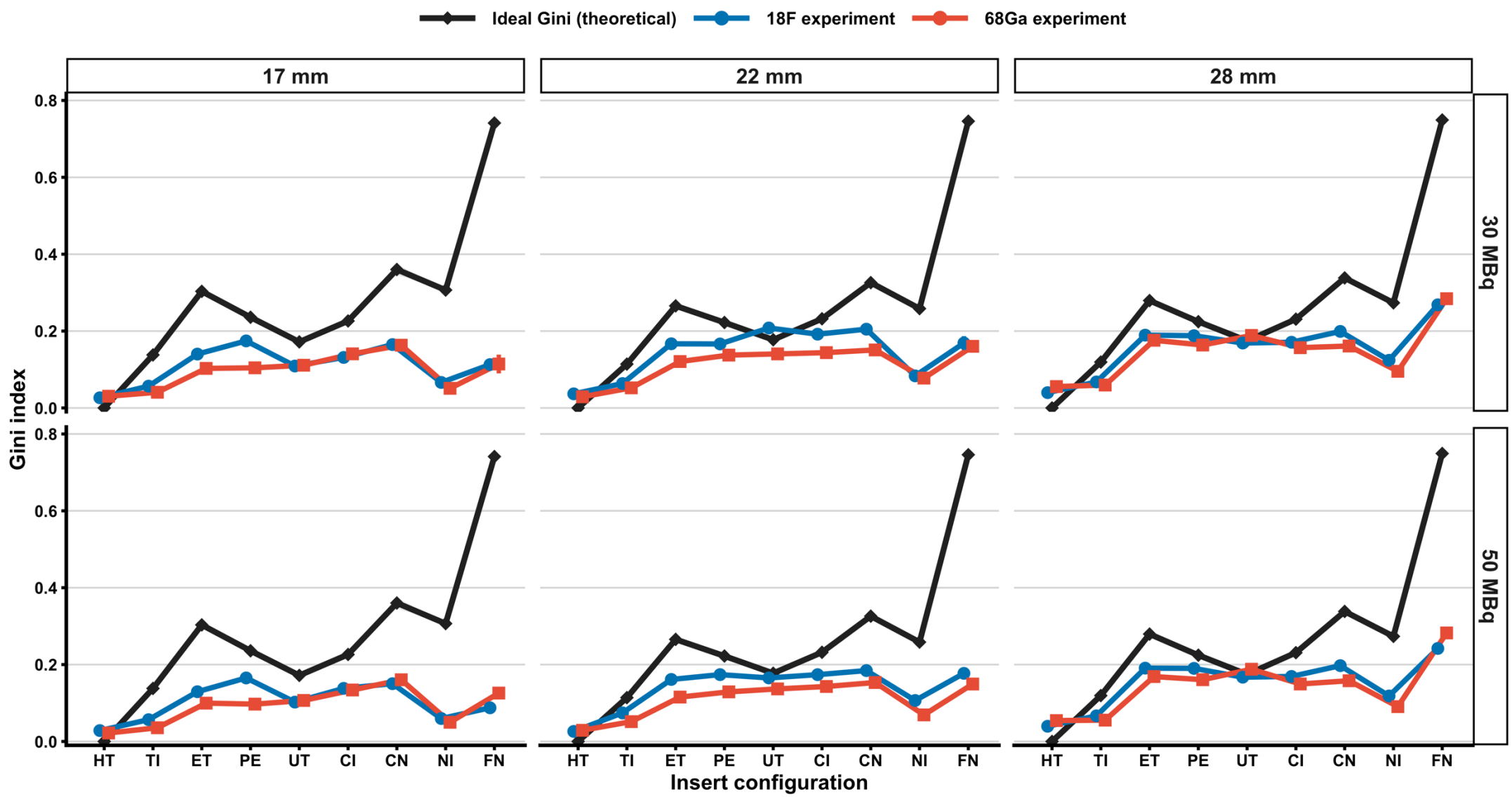




SUV Data

- ✓ Difference in SUV for all inserts
- ✓ PVE as function of diameter
- ✓ Close to Plateauing of SUV_{max} between ET till CI

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Gini Index Data

- ✓ Difference in GI from Ideal Value – expected due to system response
- ✓ Shape and trend agreement between tracers

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Cross-tracer ranks

Median ρ · strong % · Holm %

GLCM 0.84

74.1% strong · 63.0% Holm

GLRLM 0.92

91.7% strong · 88.5% Holm

GLSZM 0.77

67.7% strong · 45.8% Holm

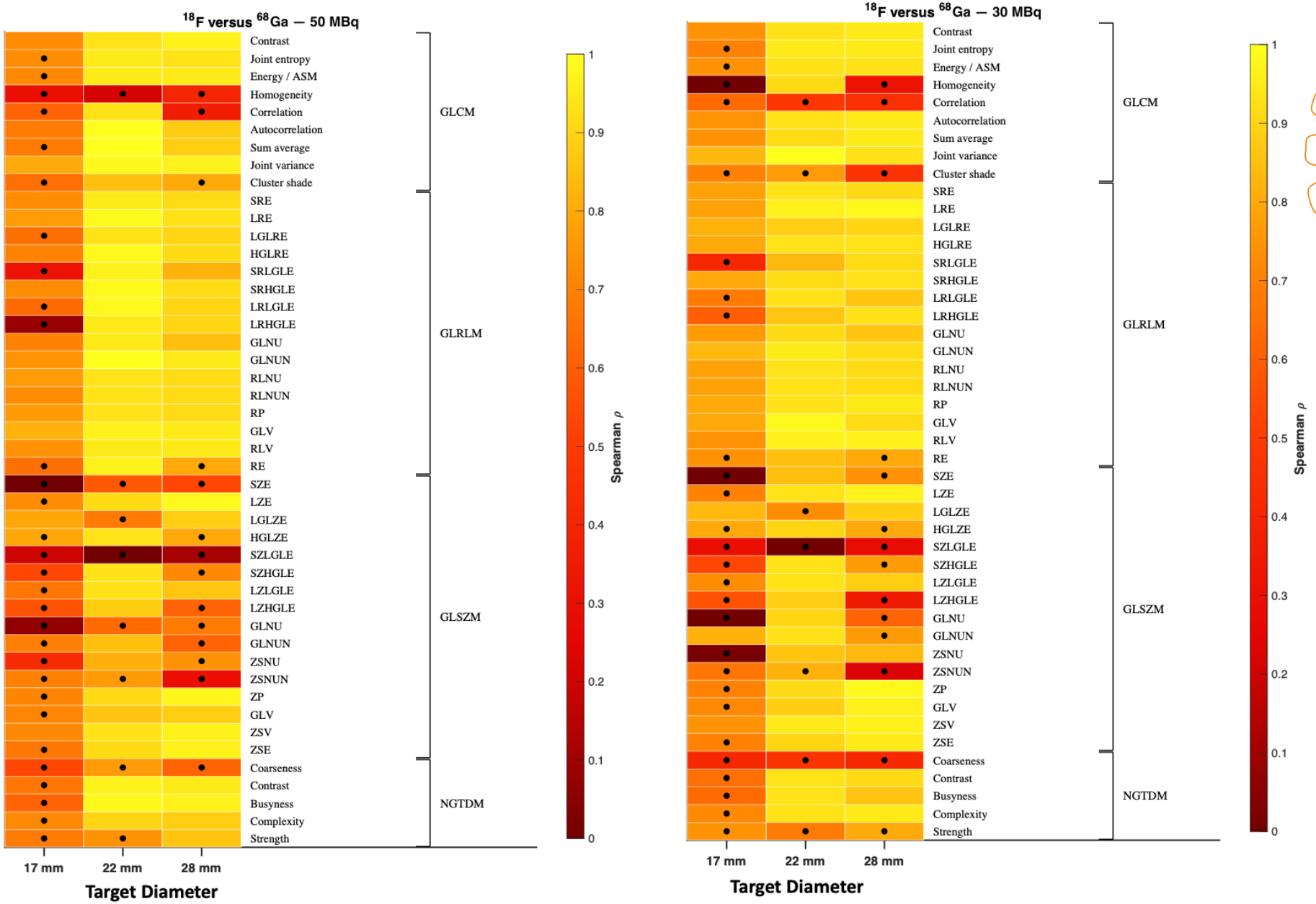
NGTDM 0.76

63.3% strong · 43.3% Holm

Strong: ρ ≥ 0.75
Holm: adjusted p < 0.05

Size trend: 22 > 28 > 17

- no rank correlation



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Second Order Radiomics Metrics

Discussion

Designed uptake patterns were distinguishable both by first order and second order metrics

LIMITATIONS

- Full IBSI benchmark verification was not performed
- Small targets, limited background and preprocessing affected texture values
- Nominal ROIs did not measure printed geometry errors

RECOMMENDATIONS

- Verify feature extraction with IBSI reference cases.
- Perform acquisitions at isotropic voxel spacing to avoid resampling

Presented at

EANM 2025

using the Bayesian penalized likelihood (Q.Clear) algorithm (20it15ss). Three reconstruction configurations were used: 1. CT-based attenuation correction with traditional SC, 2. CT-based attenuation correction with novel SC, and 3. CT-based attenuation correction with novel SC and TC. For the combined peak (113+208 keV), only the second and third configurations were applied. At one time point, reconstruction effects were evaluated step-by-step, adding one correction at a time. Total counts were measured using a cylindrical VOI. ***Results:*** The mean CFs for the 113 and 208 keV peaks using the first configuration were 69.58 ± 1.63 and 71.42 ± 2.38 cps/MBq (activity ranges: 3700-140 MBq and 3700-16 MBq), respectively. For the second configuration, CFs were 113.89 ± 2.97 and 64.07 ± 2.99 cps/MBq, and for the third, 148.84 ± 4.26 and 80.04 ± 3.57 cps/MBq (range: 3700-16 MBq). For the combined peak, mean CFs were 177.94 ± 5.18 and 228.94 ± 7.09 cps/MBq for the second and third configurations, respectively. ***Conclusion:*** The novel SC significantly improved CFs at low activity levels, especially for the 113 keV peak, where CFs remained stable over the same activity range as those acquired with the 208 keV peak. This improves the quantification capability of the 113 keV peak in late post-therapy dosimetry and supports potential clinical use of the combined peak.

EP-0947

Cross calibration of a dose calibrator with a gamma counter for ^{68}Ga and ^{18}F activity measurements

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Aim/Introduction: Gallium-68 (^{68}Ga) is used in theranostics, mainly for PET imaging in neuroendocrine and prostate cancers, as a surrogate to the same pharmaceutical agent marked with Lutetium-177 used for therapy. Accurate PET quantification requires precise activity measurements from dose calibrators. However, differences of activity measurements have been reported between similar devices for ^{68}Ga. Common verification methods to find the correct dose calibrator setting, such as the use of a traceable source or the PET/CT cross calibration, are not always available or reliable. In this study, we propose to use a gamma counter together with Monte Carlo (MC) simulations to overcome these limitations. ***Materials and Methods:*** Samples of about 23 and 10MBq of ^{18}F-FDG and $^{68}GaCl_3$ were measured using a dose calibrator, then diluted by a factor of 1000 to be measured on a gamma counter for 60 minutes. MC simulations were performed using GATE to model the gamma counter's response. The measured and simulated spectra for both isotopes were compared for validation. Sensitivity factors were calculated by dividing the total number of disintegrations by the number of counts in the 511 keV and 1022 keV energy windows ([450-570]keV and [900-1150]keV). These factors were then used to evaluate the samples activities and compared them to dose calibrator values. ***Results:*** The simulated energy spectra were in good agreement with experimental data. Sensitivity factors computed from the simulations allowed an accurate recovery of the samples activities from gamma counter data, with predicted activities being within less than 5% of the dose calibrator values. The sensitivity factors derived from the 511keV energy window displayed variations below 2% for changes of the samples volume up to 50%. ***Conclusion:*** This study presents a method to verify the dose calibrator settings for PET isotopes based on an activity measurement with a gamma counter. The method does not require traceable source, nor cross calibration with a PET scan, and is little dependent on the samples volume. This approach is not limited to PET isotopes: its extension to Lutetium-177 is a work in progress.

EP-0948

Exploring Activity and Scan Time Reduction of [^{18}F]PSMA-1007 within EARL2 Standards

L. Kloosterboer, *L. Heijmen, J. W. J. van Brecht, P. Dibbets-Schneider, T. Ruytenberg, L. de Geus-Oei, D. D. D. Rietbergen, F. H. P. van Velden;*
Leiden University Medical Center, Leiden, NETHERLANDS.

Aim/Introduction: Ongoing advancements in PET/CT detector technology significantly improved image quality and sensitivity, enabling reduction of scan duration and/or the administered radiopharmaceutical activity. Van de Burgt et al. (Physica Medica, 2015) have shown in a phantom study that reduction in scan duration or administered activity of at least a factor four is possible for [^{18}F]FDG on an Omni Legend 32. Our aim is to evaluate on patient data whether either scan duration or the administered activity of [^{18}F]PSMA-1007 can also be reduced, while complying with the EARL Fluorine-18 standards 2 accreditation (EARL2). ***Materials and Methods:*** Fifteen subjects received a [^{18}F]PSMA-1007 PET/CT scan (3 MBq/kg, 120 sec/bed) on an Omni Legend 32 PET/CT between February, 2025 and April, 2025. Four administered activity regimes are simulated by varying the time/bed (120, 60, 45 and 30 sec) in each EARL2 (4 mm) reconstruction. To obtain the coefficient of variation (COV) within healthy tissue as an indication of the noise level, a 2-cm diameter sphere is manually drawn inside the spleen. Lesions (n=31; median volume: 6.5 mL, range: 1.1-35.1 mL) are delineated semi-automatically using an absolute threshold of SUV of 4 and a lower volume limit of 1 mL to assess SUV_{max}, SUV_{mean} and SUV_{peak}. Statistically significant differences ($p \leq 0.05$) between these quantitative measures are assessed using the Wilcoxon signed-rank test. ***Results:*** No statistical differences were found for lesion SUV_{mean} and SUV_{peak}, comparing the simulated reconstructions to the 120 sec/bed reconstruction. Lesion SUV_{max} for the simulated reconstructions were statistically different from the 120 sec/bed reconstruction. Median COV of the spleen were 4.7% (2.6-7.8%), 5.0% (3.7-11.8%), 5.5% (3.7%-12.7%) and 6.0% (4.0%-12.8%) for 120, 60, 45 and 30 sec/bed, respectively. ***Conclusion:*** These preliminary results suggest that the scan duration or administered activity of [^{18}F]PSMA-1007 can be reduced by at least a factor four. The noise level increases while decreasing scan duration (resulting in increased SUV_{max}), but remains below 15% as required by EARL for [^{18}F]FDG images. Further research should demonstrate whether this assumption can be translated to [^{18}F]PSMA-1007. Analysis of additional data is necessary to obtain definite conclusions and should at least include the assessment of: a larger patient group, EARL1 and EARL2 (2 mm) reconstructions, lower time/bedposition(s) and a qualitative image analysis. ***References:*** Van de Burgt et al. "[^{18}F]FDG administered activity reduction capabilities of a 32-cm axial field-of-view solid-state digital bismuth germanium oxide PET/CT system while maintaining EARL compliance." Physica Medica 131 (2025): 104935.

EP-0949

3D Printed PET/CT Phantom with Mixed Target Uptake Characteristics to Investigate Lesion Development

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[1]*Oncopole Claudius REGAUD - IUCT Oncopole, Toulouse, FRANCE,* [2]*Tachyons SAS, Toulouse, FRANCE,* [3]*Universite de Toulouse, Toulouse, FRANCE,* [4]*Rambam Health Care Campus, Haifa, ISRAEL,* [5]*Technion Israel Institute of Technology, Haifa, ISRAEL.*

Aim/Introduction: Precise lesion characterisation is a common challenge in practice due to heterogeneity within tumours. Conventional quality control practices, such as fillable acrylic spheres have limitations in assessing the heterogeneity. In contrast, 3D printing enables the creation of versatile phantoms that are customisable to include complex features such as fillable printed grids which can simulate multitude of contrasts within

a phantom using single tracer injection. This work builds upon a previous 3D printed phantom inserts design [1] to introduce a more complex design including heterogeneous uptake patterns with minimal phantom preparation time. ***Materials and Methods:*** 4 cm gyroid cubic grids were 3D printed (75% Infill) in Acrylonitrile butadiene styrene (ABS). Spherical voids of 17mm and 28mm were incorporated centred on the cube. A second region with different infill (50% or 100%) was added leading to primary targets (Target Background Ratio ~ 4) of either a sphere or crescent shape. Heterogeneity was either multifocal (TBR ~ 2) or necrosis (TBR ~ 0) based on second regions density. These inserts were placed in a conventional QC phantom and filled with water, soap and [^{18}F] FDG (96 MBq). Acquisitions were performed on two different PET/CT scanners (β:650). Recovery coefficient (RC) was evaluated for quantitation of primary lesion along with TBRMax of overall insert. Additionally, standardised uptake value (SUV) profiles were extracted on Transaxial slice for heterogeneity impact analysis. ***Results:*** Heterogeneity was apparent visually on PET images for all 28mm targets whereas only primary lesion was predominant for 17mm targets due to partial volume effects on secondary regions. TBRMax for 17mm set were in the range of 1.22 - 4.21 and for 28mm targets were 1.34 - 4.52. Mean RC for primary lesions ranged from 0.71 - 0.82 (PET 1) and 0.61 - 0.73 (PET 2) for 17mm, and 0.81 - 0.88 (PET 1) and 0.74 - 0.82 (PET 2) for 28mm. Maximum RC ranged from 0.65-1.02. SUV profiles showed reduced / shifted peak values and evident transitions between multiple regions. Reference cubes without targets yielded SUV values of 0.19-0.27 against expected 0.25 (SUVWater ~1). SUVNecrosis were slightly higher than 0. ***Conclusion:*** Our results highlight that 3D printed inserts effectively model mixed uptakes while demonstrating the impact of lesion size and heterogeneity on PET/CT visual interpretation and quantitation. ***References:*** [1] Nalla, S.K.K., Maronnier, Q., Kennedy, J.A., Palchan-Hazan, T. and Caselles, O., 2025. Enhancing PET/CT Assessment with Porous 3D-Printed Grids-A Pilot Study. medRxiv, pp.2025-04.

EP-0950

Quantitative accuracy of clinical PET systems in the UK (2019-2023)

A. Fenwick, *A. Robinson, A. Denis Bacelar, K. Ferreira, D. Deidda, D. Roddy, W. Heetun;*
National Physical Laboratory, Teddington, UNITED KINGDOM.

Aim/Introduction: Quantitative accuracy in PET is a critical factor when considering multi-centre comparability. To assess the comparability of PET systems in the UK a specialised traceably calibrated solid phantom was designed and distributed to 16 centres. The results were analysed centrally and are reported. ***Materials and Methods:*** The phantom was adapted from the NEMA NU-2 Image quality phantom which consists of 6 spheres ranging in size from 10 mm to 37 mm diameter. A seventh sphere with a 5 mm diameter was added to the phantom in a position optimised using monte carlo techniques. The phantom was prepared with ^{68}Ge spiked epoxy resin with a total activity of 99.0(30) MBq (k=2) with a sphere:background ratio of 4:1. The Phantom was shipped to 16 sites for measurement between 2019 and 2023 with images acquired using a decay-corrected acquisition time simulating a 3 min per bed position scan. An additional scan with a single frame centred over the sphere axis was also obtained at most sites. The phantom was periodically returned to NPL and scanned to confirm stability of the phantom. Images were reconstructed using typical clinical protocols at the participating site, and image analysis was performed centrally by two individuals (independently). Results were calculated for quantitative accuracy of background and sphere activities and recovery coefficient curves were generated for each scan made. ***Results:*** The results showed good agreement between sites with all participants complying with EARL recommended recovery coefficient curves. uncertainty analysis was included to provide comparative data. ***Conclusion:*** This study has demonstrated the comparability between clinical PET/CT sites in the UK using a standardised phantom and has created a dataset which can be used in future studies when investigating performance capability and for validation.

EP-0951

Development and Evaluation of a 3D-Printed Thyroid Phantom for Simulating Hot and Cold Nodules in SPECT Imaging

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[1]*Division of Radiology and Imaging Science, Department of Medical Imaging, Faculty of Medicine, University of Debrecen, Debrecen, HUNGARY,* [2]*Mediso Ltd., Budapest, HUNGARY.*

Aim/Introduction: Thyroid scintigraphy is among the most frequently performed procedures in nuclear medicine; therefore, robust quality assurance protocols tailored to SPECT imaging are essential. These protocols often rely on phantom-based evaluations. However, commercially available thyroid phantoms typically represent only healthy tissue, limiting their relevance for simulating pathological conditions such as hot or cold nodules. [1,2] Recent advances in 3D printing technology enable the design and fabrication of customized phantoms, including those that mimic pathological features. In this study, we present a modified anthropomorphic thyroid phantom capable of simulating both hot and cold thyroid nodules. ***Materials and Methods:*** A commercially available 3D thyroid phantom was redesigned using 3D modeling software to alter anatomical volumes and incorporate pathological structures. Spherical nodules (1 cm diameter) were added: a solid (cold) nodule was embedded in the left lobe, and a fillable hollow sphere (0.3 ml) representing a hot nodule was placed in the right lobe. The models were printed in two sizes (20 ml and 40 ml fillable thyroid volumes) using PLA material via fused deposition modeling (FDM) technology. Reproducibility of the manufacturing process was assessed through repeated printing and CT imaging. For realistic attenuation and scatter conditions, the thyroid phantoms were placed within a standard neck phantom. SPECT/CT acquisitions using Iodine-131 were performed on a triple-head SPECT/CT system equipped with a multi-pinhole collimator. The activity concentration ratio between the thyroid tissue and the hot nodule was set at 1:10. ***Results:*** The fabrication process demonstrated high reproducibility, with no leakage observed during phantom preparation. The printed phantoms were fully compatible with conventional neck phantoms. Reconstructed SPECT images clearly depicted both nodules, with the hot and cold lesions showing 88% higher and 28% lower uptake, respectively, compared to the uniform thyroid background. ***Conclusion:*** Results confirm that fillable 3D-printed thyroid phantoms can reproducibly simulate pathological conditions, providing a valuable tool for image quality assessment in SPECT imaging. ***References:*** 1. https://doi.org/10.1097/HP.0000000000000995 ; 2. https://doi.org/10.1016/j.radphyschem.2019.108462.

EP-0952

Manufacturer improvement of ^{99m}Tc scatter correction for its long bore collimator CZT ring SPECT-CT

A. Seret[1], *K. Di Bartolomeo*[2], *C. Bernard*[3];
[1]*University of Liege, Liege, BELGIUM,* [2]*GE Healthcare, Brussels, BELGIUM,* [3]*CHU of Liege, Liege, BELGIUM.*

Aim/Introduction: The manufacturer of the long bore collimator CZT-based ring SPECT-CT is currently releasing an upgraded version of its reconstruction software where dual energy window scatter correction is performed in the iterative loop of the reconstruction whereas it was previously a subtraction at the projection level. Phantom data of our past study of this system

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Compositional Anthropomorphic Spatial Synthesizer for ground truth Lesion Export

Why?

Simulation of heterogenous targets (especially with necrotic regions) or Patient derived shapes was not practically possible in the version of DUETTO available during the thesis

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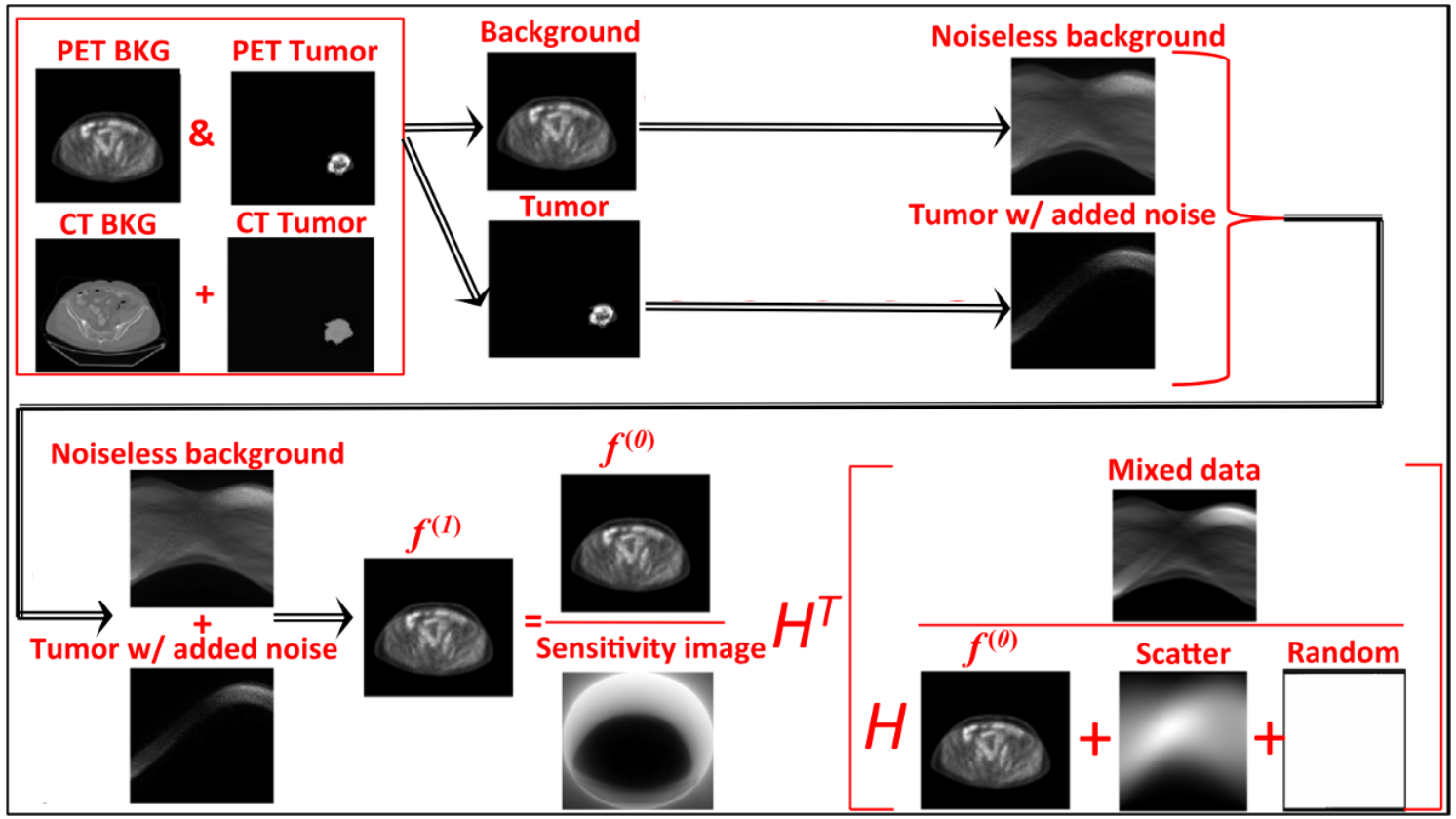
Cassoulet Sim follows core architecture as PET STEP

- Scatter Correction implemented as a fixed 10cm gaussian
- Randoms implemented as percentage of trues

Designed for newer scanners

The scanner models are designed after GE DMI 5R, GE OMNI 32cm and Siemens Quadra

(Based on published NEMA and opensource data)

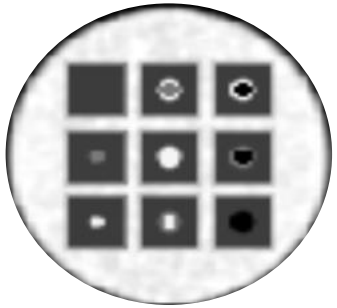


- Insertion uses same fractional mask logic as DUETTO SLI
- CT simulation performed in image domain
- Features from SMART PET such as noise factor (for count calibration), crystal normalization

Can CASSOULET simulator reproduce the PET response of the previously measured heterogeneous inserts?

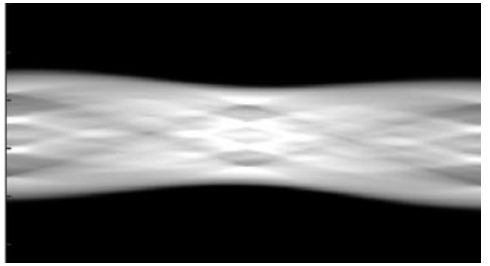
METHODS : Based on 27 inserts with 9 pattern design in each layer

- Simulations performed for an approximated model of OMNI legend PET/CT system
- **OSEM** 2.5D reconstruction with 2 iterations x 10 subsets, PSF correction, 5mm post smoothing
- Simulations performed only for 18F-FDG phantom data



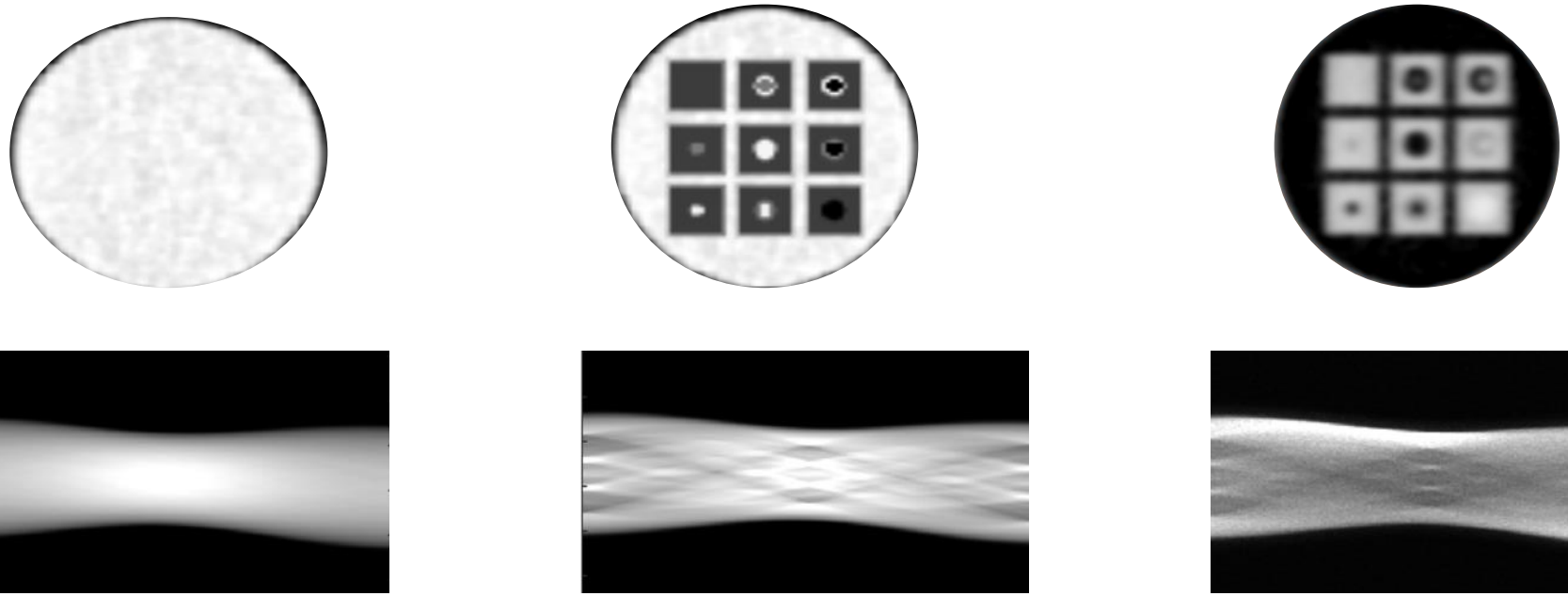
ANALYSIS

- Visual comparison of PET images
- VOI extraction and preliminary metrics (RC, GI)
- 60 panel radiomic feature analysis – same procedure as experimental study



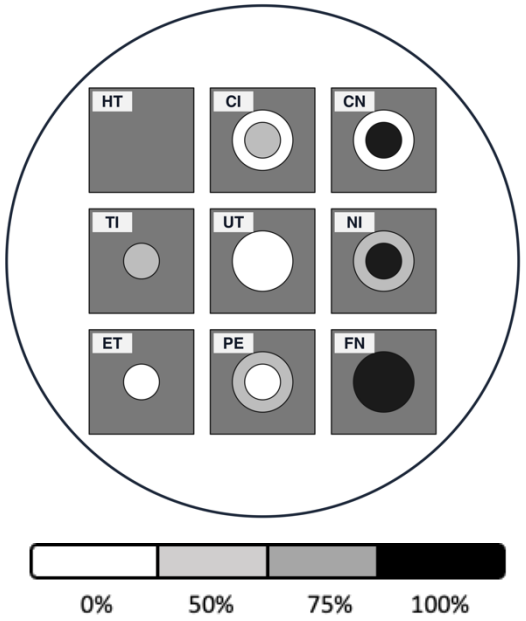
Simulations performed retrospectively using existing FDG phantom data.

Insertion of Synthetic Data for Cassoulet in phantom

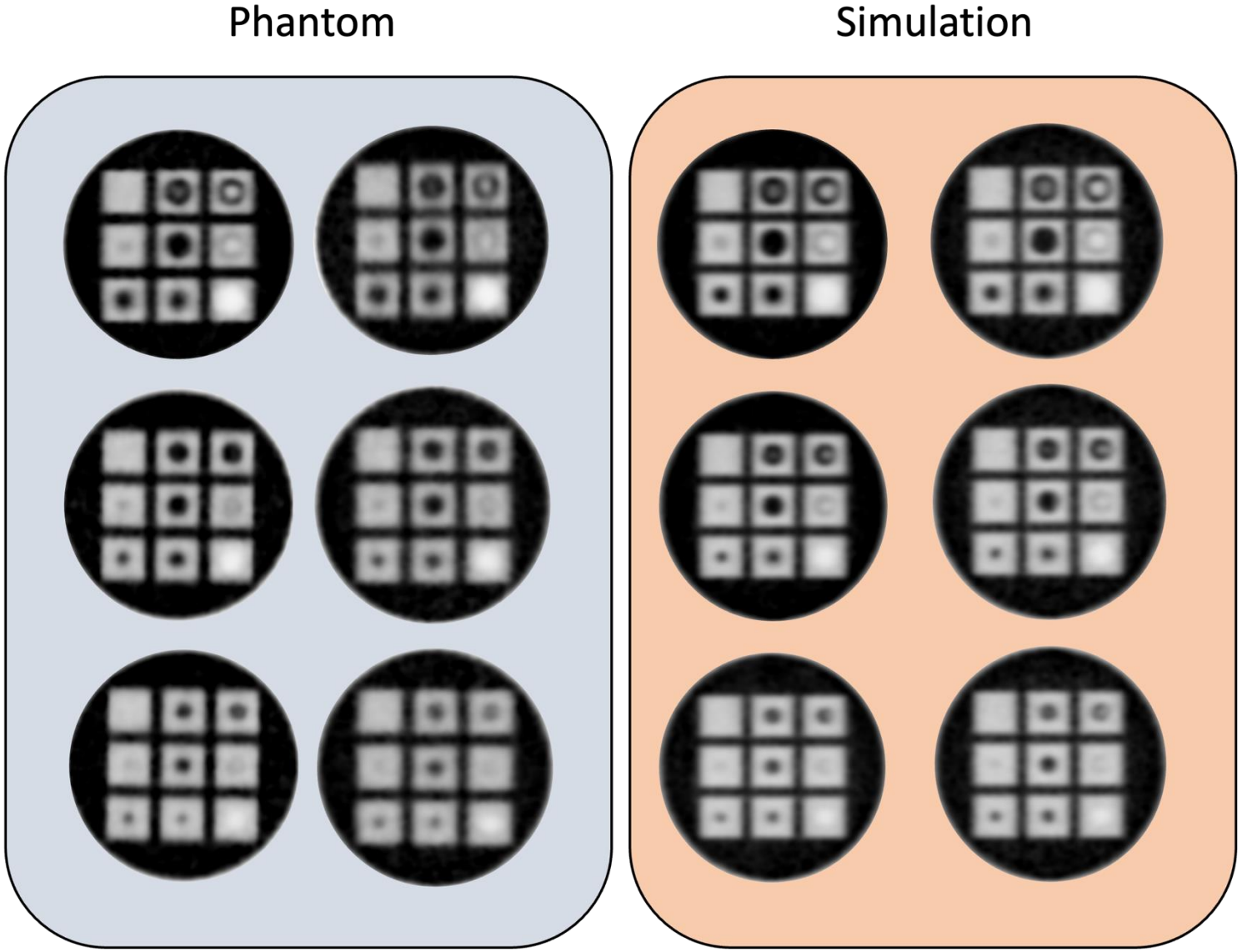


- Baseline
- Post Data Insertion
- Post Noise and Reconstruction

- DUETTO inserts synthetic lesions / targets on homogenous insert background in the phantom
- CASSOULET inserts the entire cube with target void on the reference solution in the phantom
- Key difference is modelling dilution feature (contrast < 1) is practically not possible on Duetto SLI



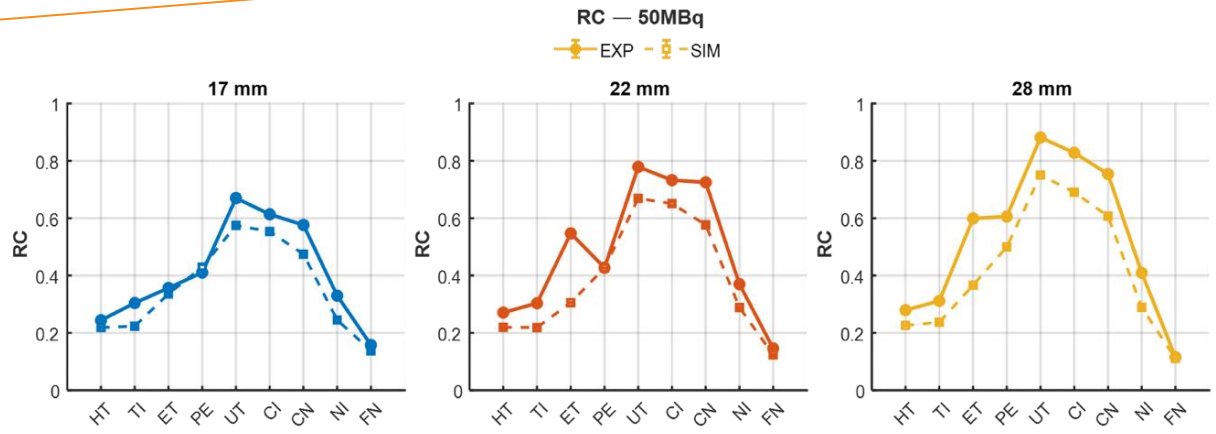
Simulation Images are slightly sharper i.e. less information lost

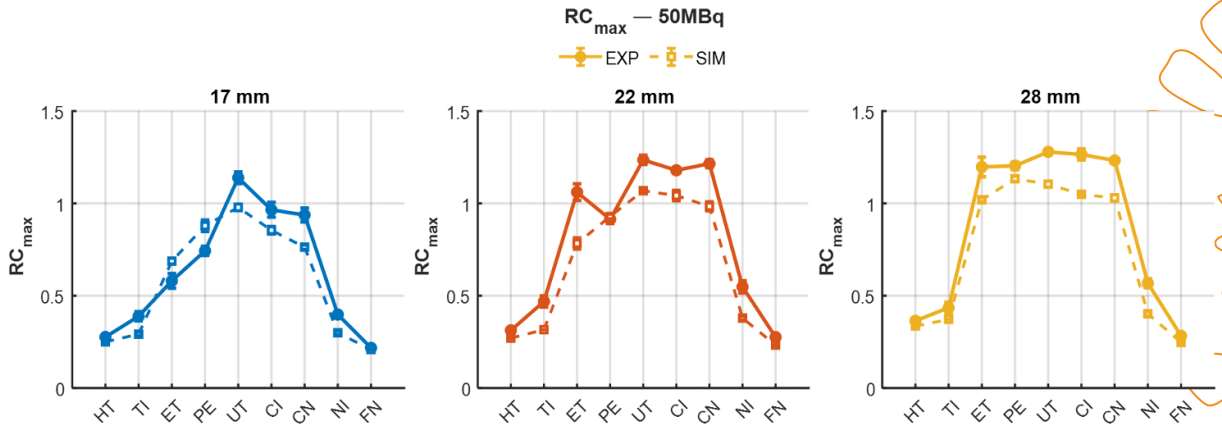


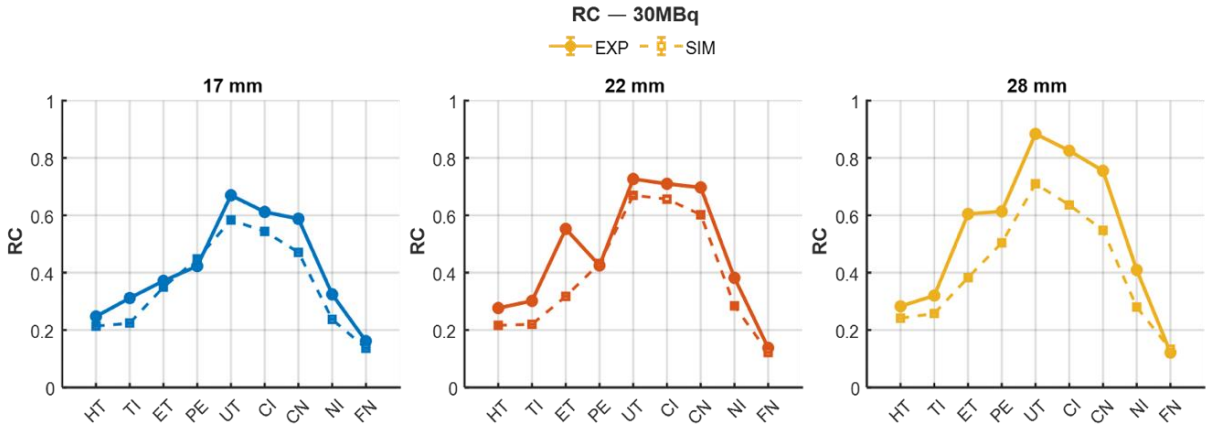
PET Central Slice

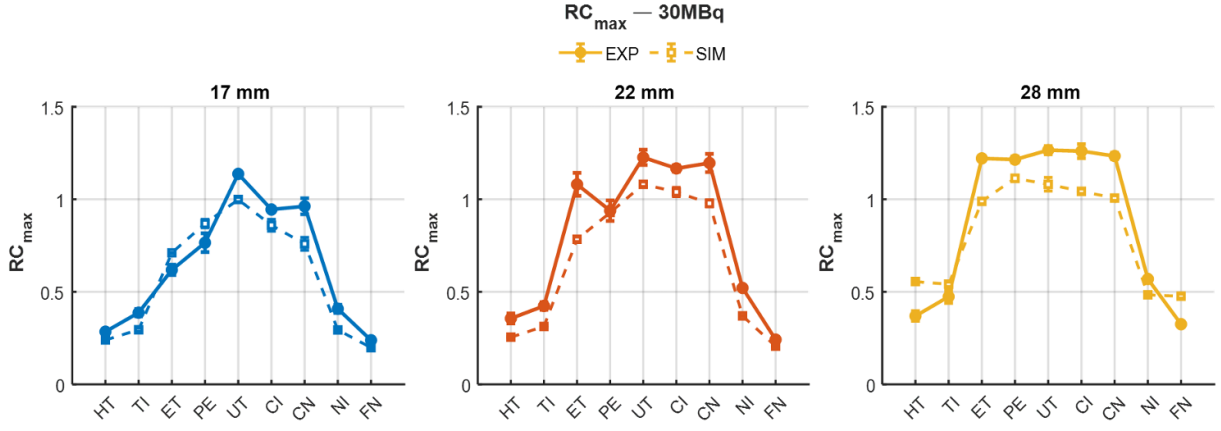


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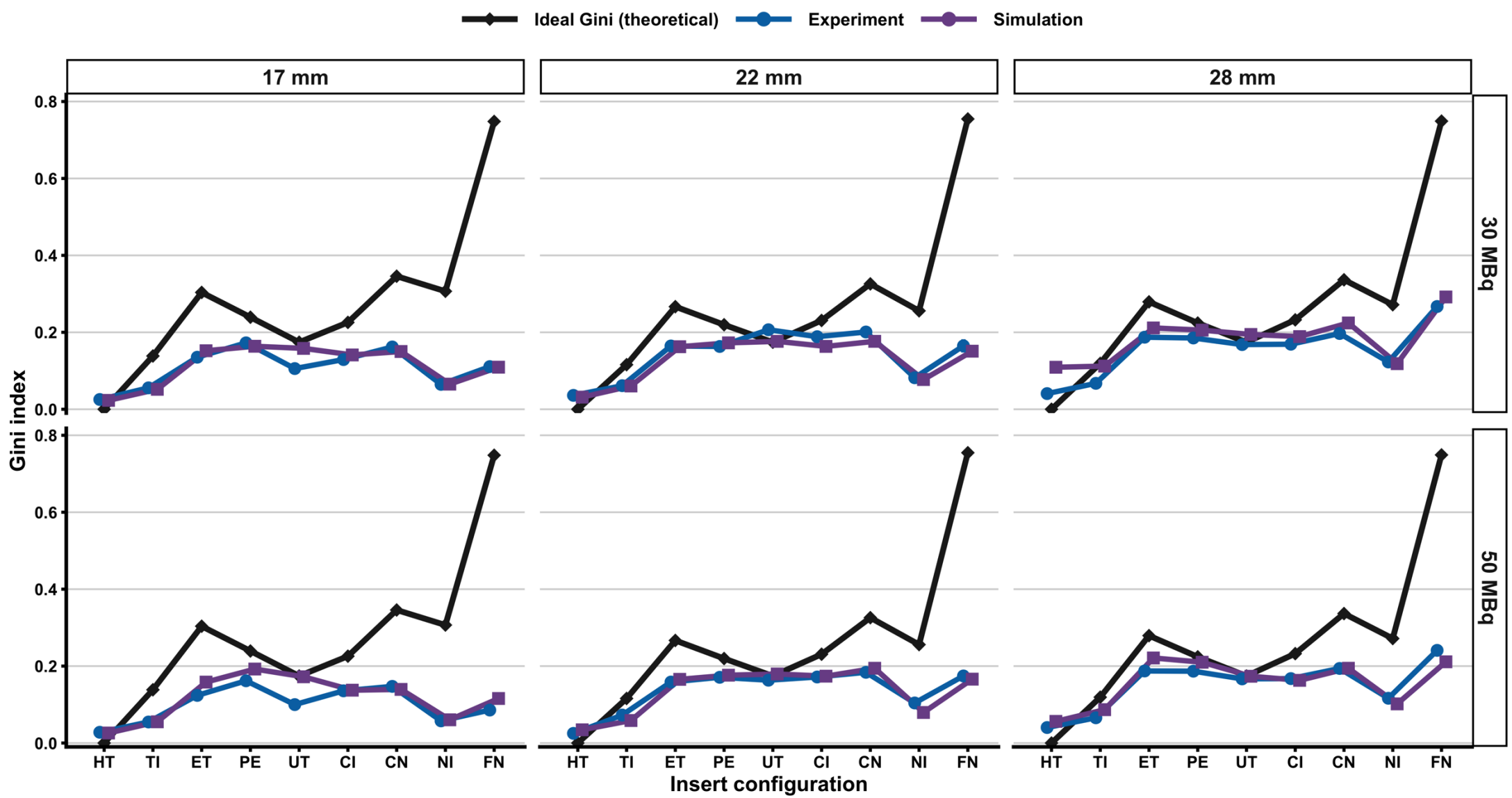




Recovery Coefficient

- Simulation generally underestimates recovery (OSEM vs BSREM)
- Overall trends are broadly reproduced, most for 28mm

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Gini Index Data

- ✓ **Difference in GI from Ideal Value**
- ✓ **Agreement between simulation and experiment**

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Cross-tracer ranks

Median ρ · strong % · Holm %

GLCM 0.84

88.9% strong · 75.9% Holm

GLRLM 0.92

88.5% strong · 82.3% Holm

GLSZM 0.82

79.2% strong · 58.3% Holm

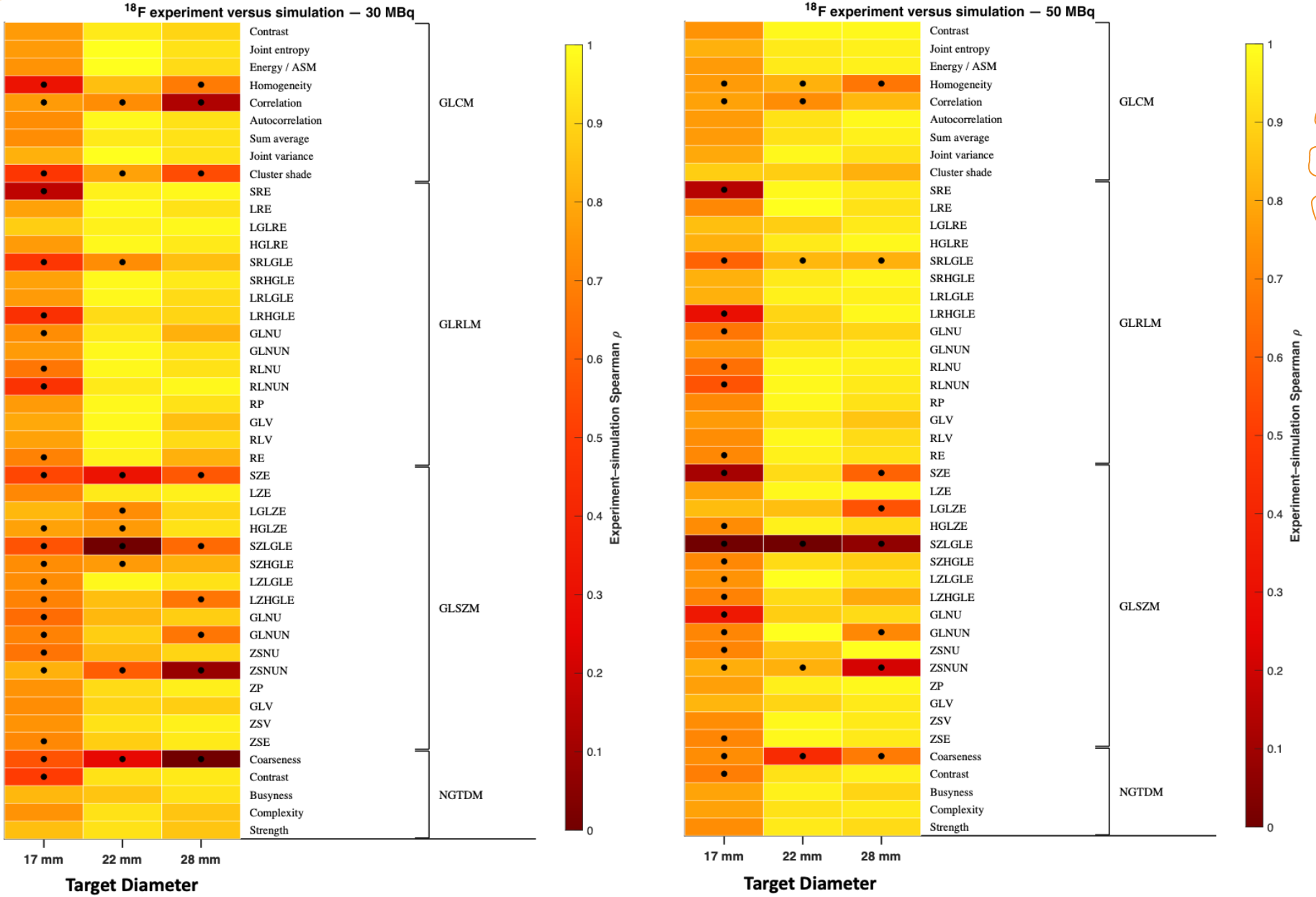
NGTDM 0.87

76.7% strong · 73.3% Holm

Strong: ρ ≥ 0.70
Holm: adjusted p < 0.05

Size trend: 22 > 28 > 17

- no preserved ordering



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Experiment vs simulation: uptake and features

Uptake metrics: distribution tests

Metric	30 MBq	50 MBq
RCmean	**0.005**	**0.009**
RCmax	**0.035**	**0.013**
CoV	0.606	0.233
Gini	0.494	0.214

Mann Whitney p values

Median absolute feature error (%)



Experiment used as reference

Discussion

Simulation reproduced heterogeneity and retained insert trends more reliably than absolute texture values

LIMITATIONS

- Experimental BSREM and simulated OSEM produced different noise texture
- The 30 MBq count/noise match needs closer tuning

RECOMMENDATIONS

- Counts based calibration of reconstruction + noise factor

This Work

Design Evaluation of Porous 3D printed Grids: Experiment + Simulation

Pilot study · multi-contrast encoding · single centre

This Work

Dual-Centre Transferability of Multi contrast Phantom : Experiment

Same phantom · two independent institutions · consistent results

This Work

Heterogeneous Uptake in Single Encoded grid: Experiment + Simulation

Gini index as physical ground truth for intra-tumoural heterogeneity

This Work

Hybrid Clinical implementation workflow for Anthropomorphic Lesions

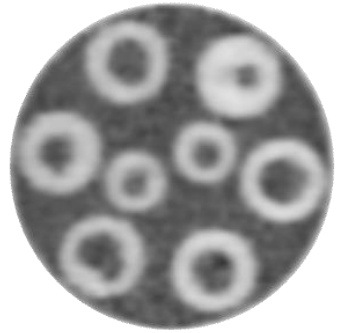
8 clinical shapes · hybrid framework · proof of concept validated

Can patient-derived lesion shapes be compared in physical phantom PET and retrospective synthetic insertion?



METHODS : Eight TCIA-derived lesion geometries (40–66 mm)

- Phantom: 8 Cylindrical PLA inserts in Jaszczak phantom · 100 MBq ^{18}F-FDG as tracer
- **Phantom Acquisitions on Omni PET/CT system** – BSREM $\boldsymbol{\beta = 650}$
- Synthetic: Lesions inserted in healthy volunteer dataset (derived from open source)
- **Sim Reconstructions on approximated Omni model** – OSEM 2.5D (3 iterations, 16 subsets, 5mm post smoothing)

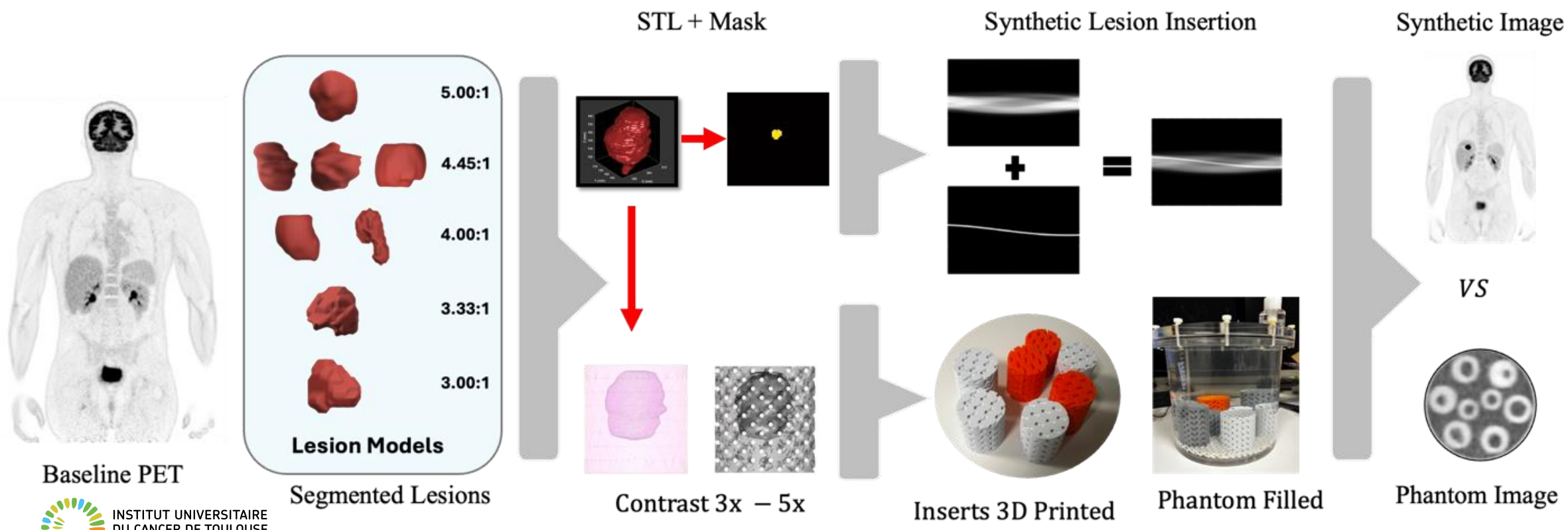


ANALYSIS

- Visual analysis of PET images separately
- VOI data extraction (same size for sim and phantom) followed with fixed segmentation threshold (42% SUV_{max})
- Shape metrics, Point Cloud Registration and dice scores, 3 PET metrics
- Due to limited sample size only MAPE

Workflow

- PLA was used for 3D printing instead of ABS
- Healthy volunteer data comes for total body quadra scanner (OSEM reconstruction)
- Image matrix and FoV adjusted for OMNI prior to forward projection



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Data Analysis & Phantom Acquisition

1 VOI (Same fo

Mean and Max SUV

CoV

Coefficient of Variation in %

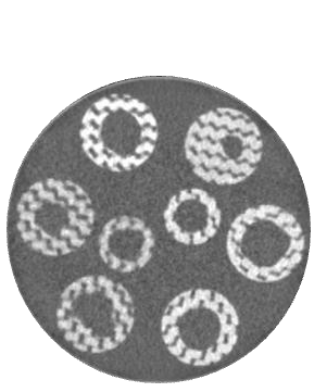
$$\mathrm{CoV} = \frac{SUVmean * 100}{SUV\ SD} \quad (9)$$

tion, 3 PET

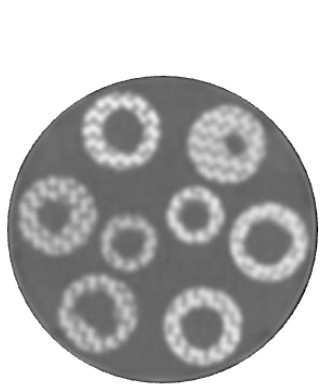
CRC

Contrast Recovery Coefficient

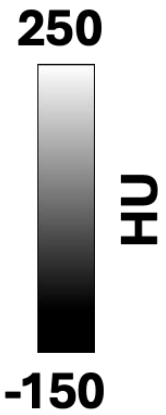
$$CRC_{\text{mean/max}} = \frac{SUV_{\text{mean/max, target}} - SUV_{\text{mean/max, background}}}{(TBR_{insert} - 1) * SUV_{\text{mean, background}}} \quad (10)$$

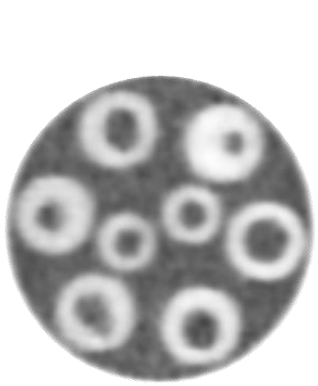


CT Standard

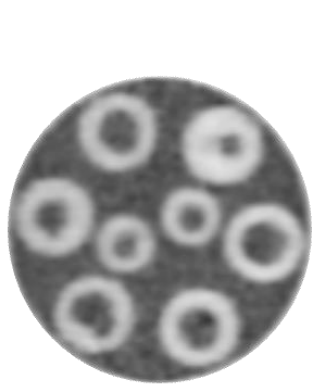


CT Attenuation



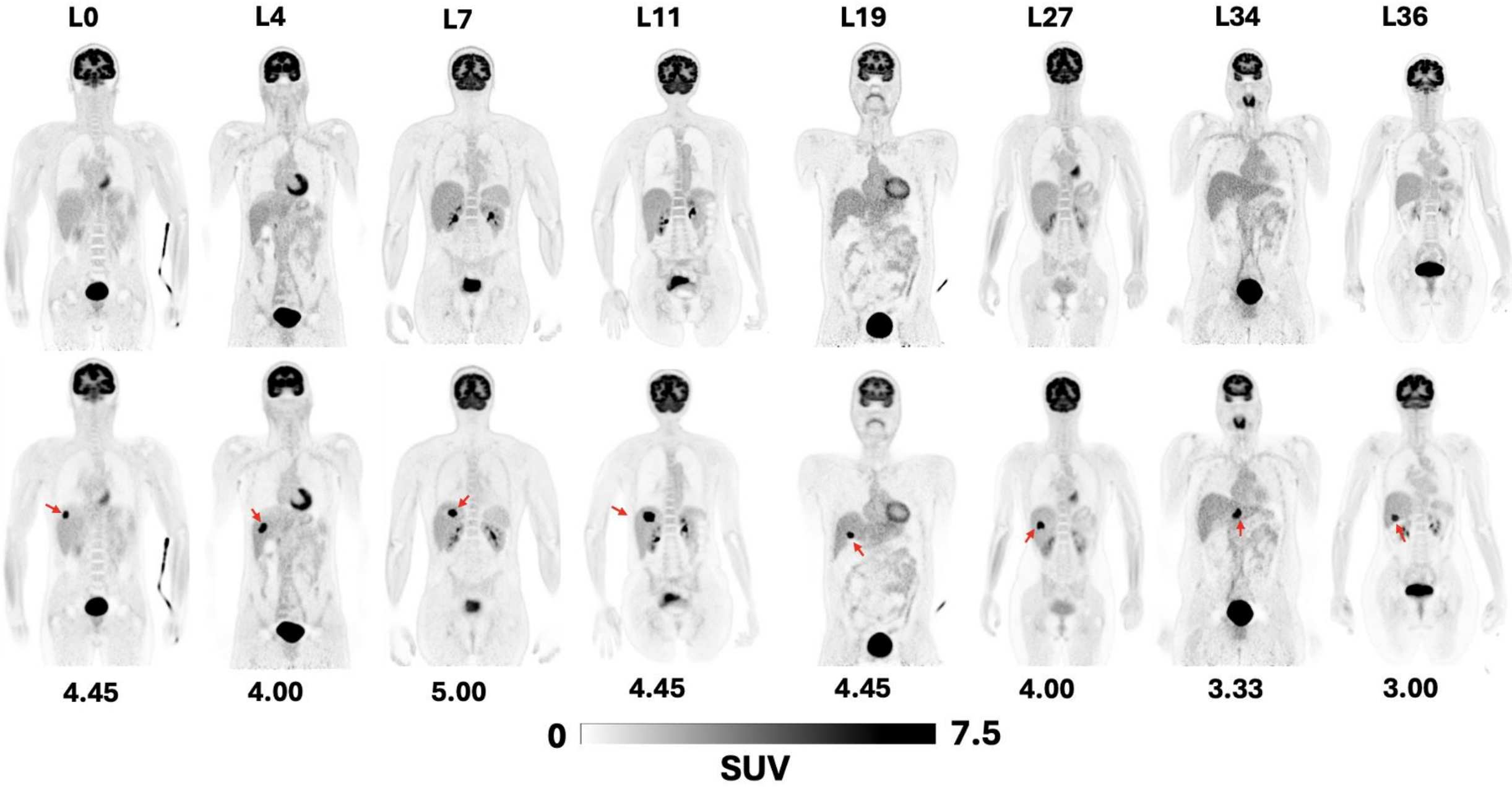


PET ($\beta \sim 650$)



PET ($\beta \sim 850$)

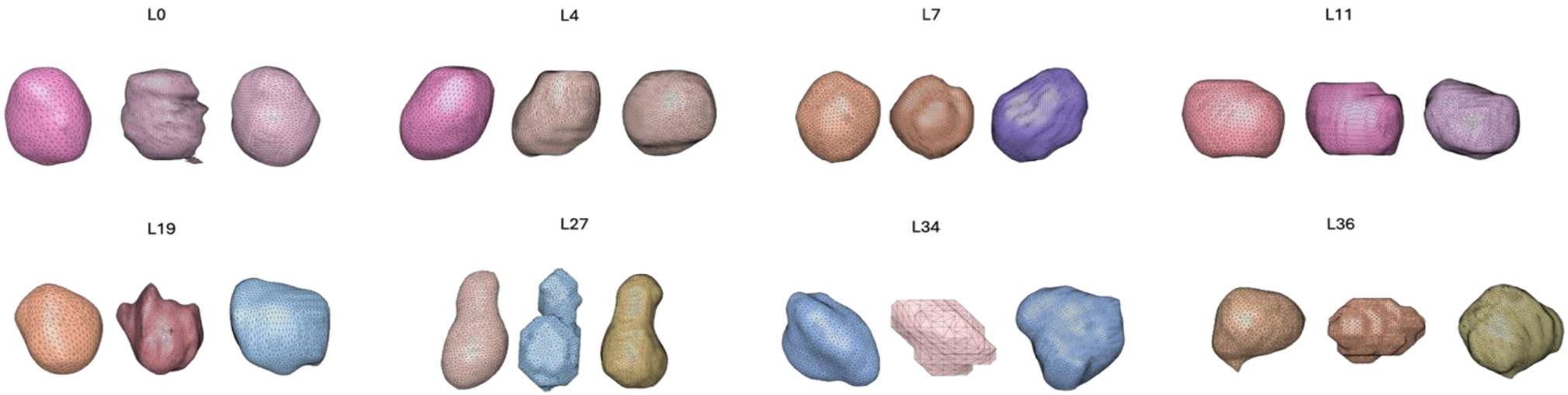
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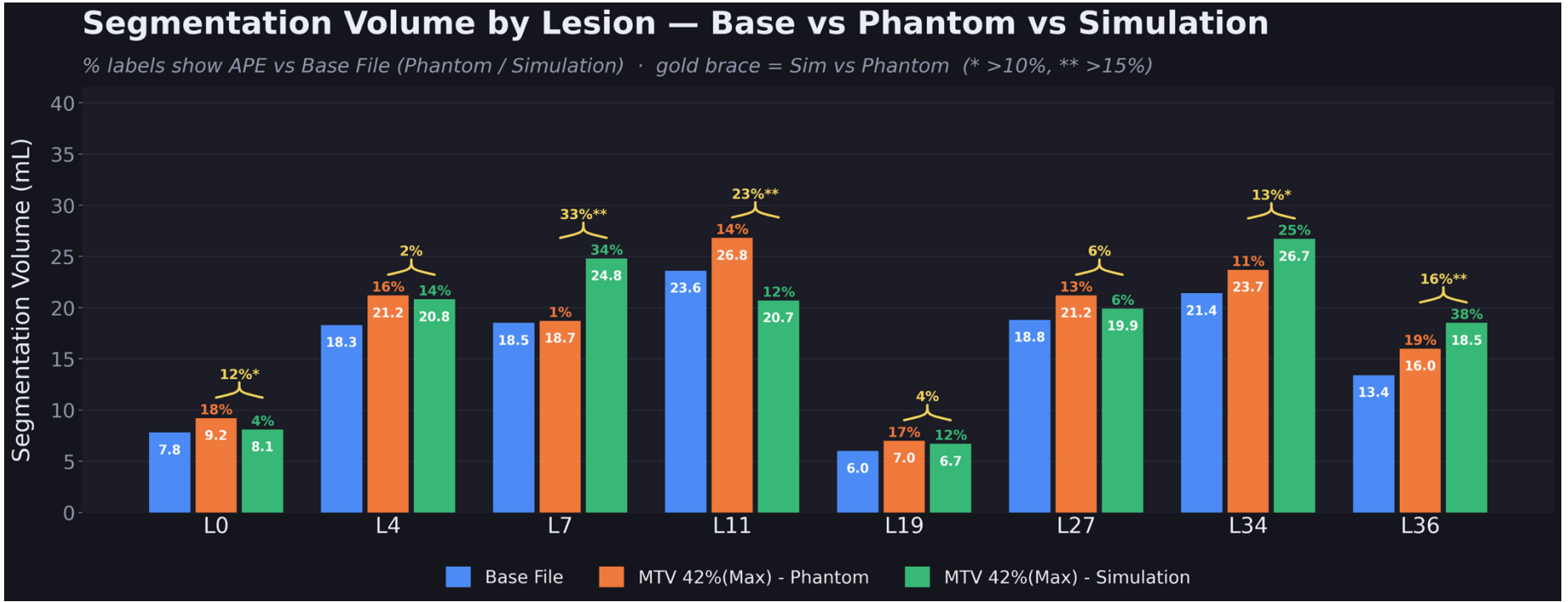


Synthetic Lesion Insertion

Metabolic Tumour Volume

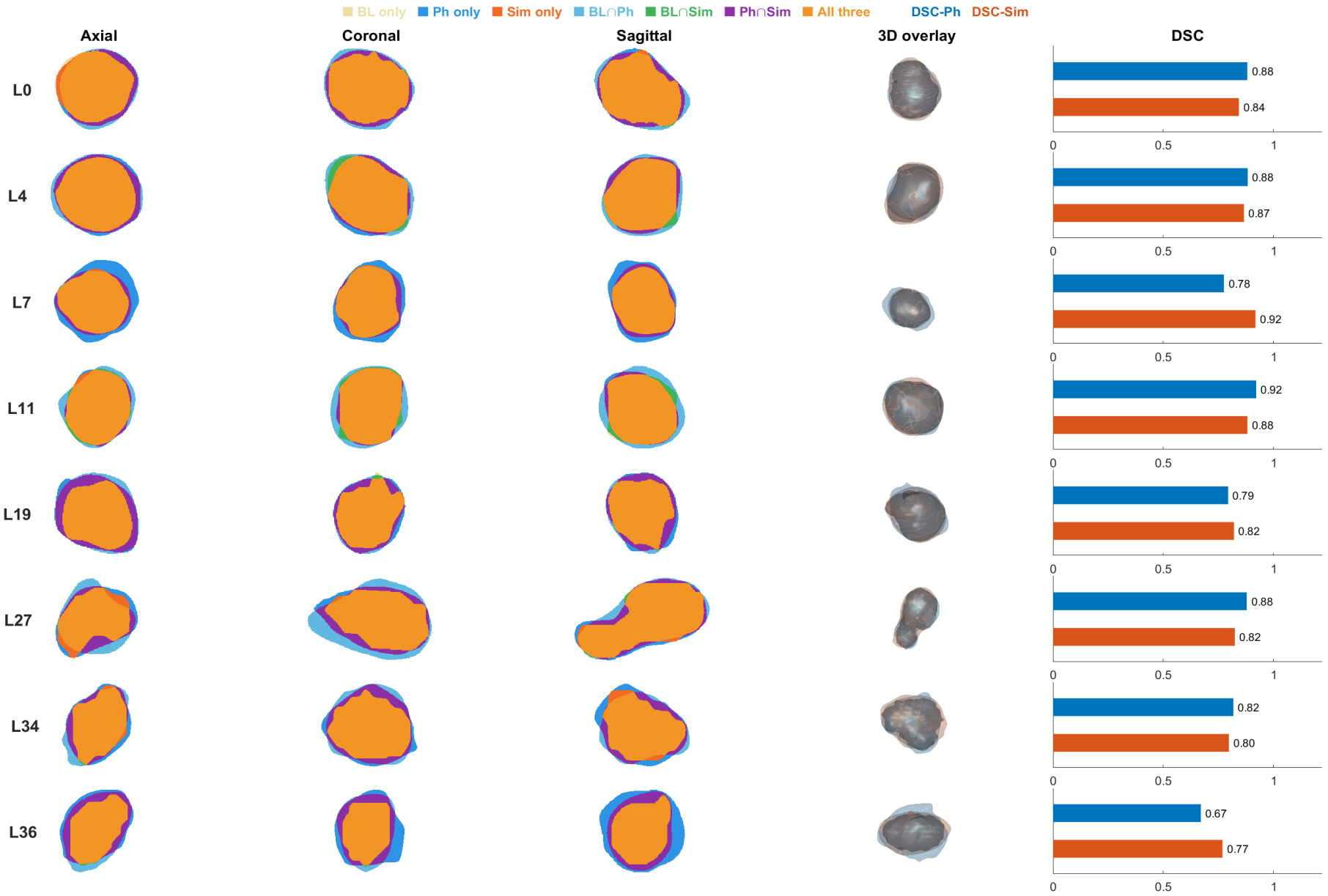
Simulation **Baseline** **Phantom**





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Segmentation Registration and Dice

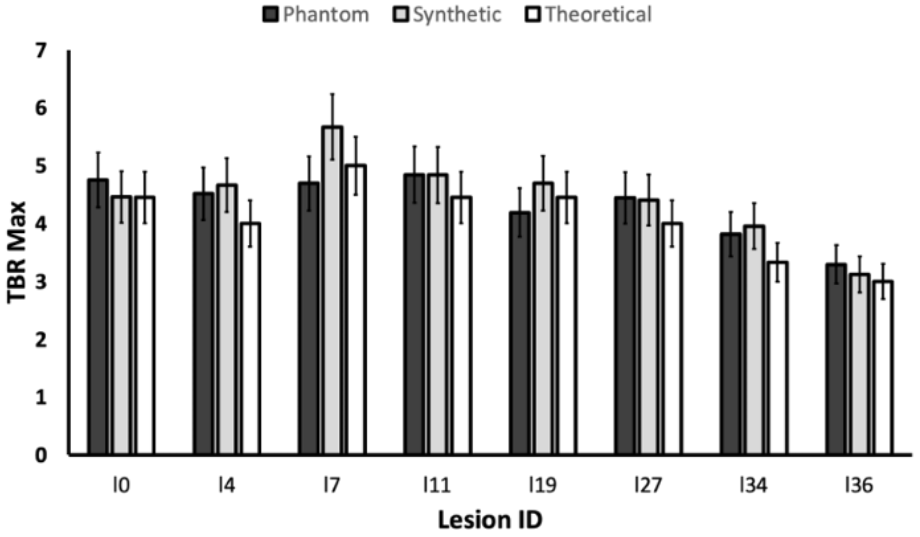


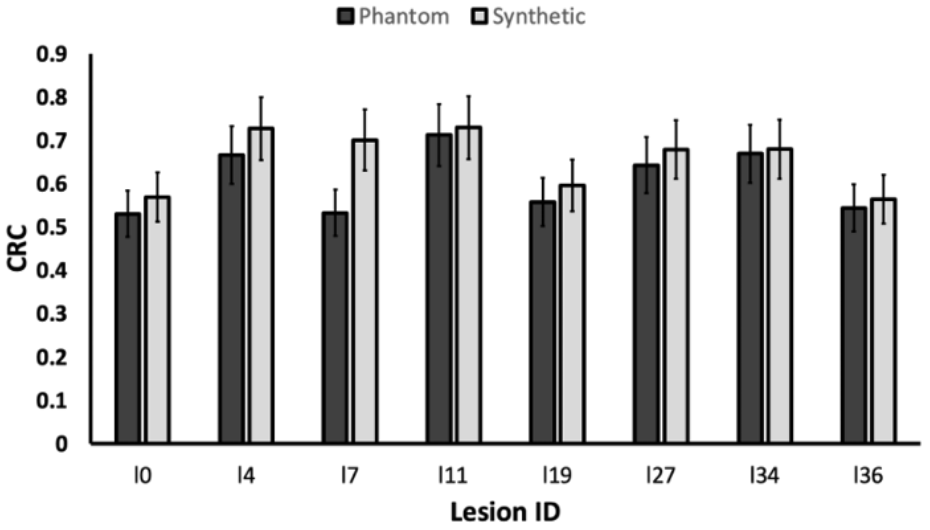
3 Step Registration

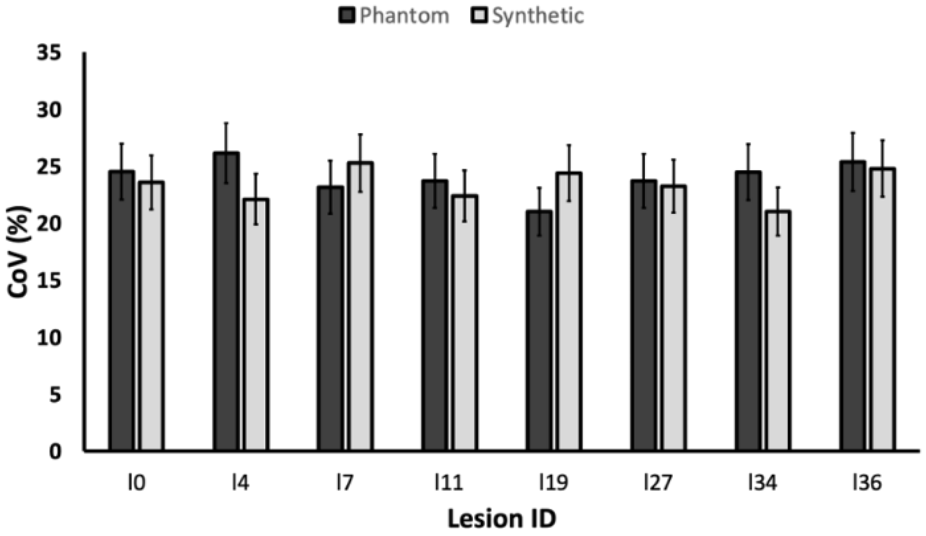
1) Centroid Matching
2) Axes Matching
3) Iterative Closest Point (ICP)

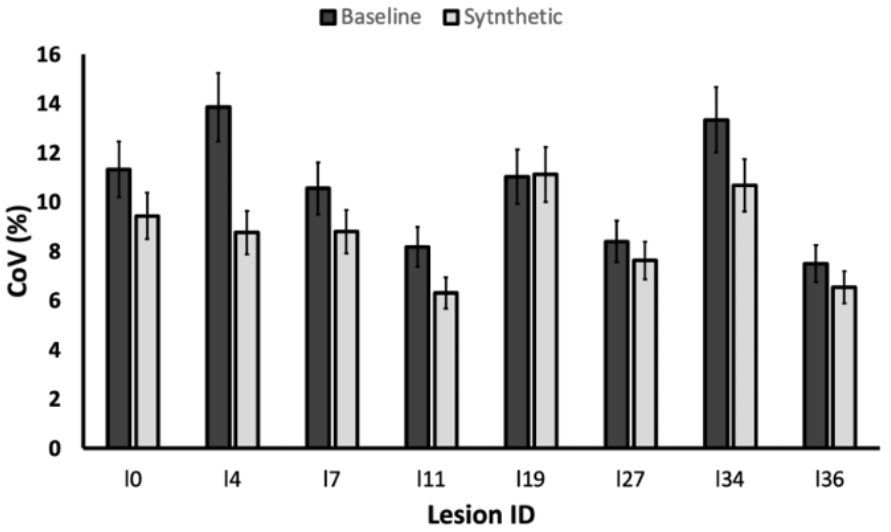
- High overlap, orange-dominant
- DSC varies per lesion (0.67 – 0.92)
- Mean DSC-Ph: 0.83
- Mean DSC-Sim: 0.84
- Least for L36

Quantitative PET Metrics









BACKGROUND CHECK for Liver

PHANTOM vs SYNTHETIC

Within ±10%

TBR_{max} 6/8 lesions

CRC 7/8 lesions

CoV 5/8 lesions

LARGEST DIFFERENCES

L7: TBR_{max} 17.2%; CRC 24.0%

CoV: L4 18.3%, L19 13.9%, L34 16.4%

BACKGROUND CHECK

Simulated liver CoV is lower than baseline; scanner differs

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Shape metrics: absolute percentage differences (%)

Phantom vs baseline

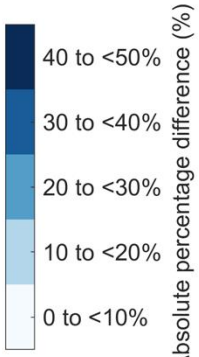
	Sphericity	LA	MA	SA	Surface area	Eq. diameter	Flatness	Elongation	Compactness	Surface / volume
L0	7.6	1.0	12.1	5.6	7.1	7.4	4.6	11.0	24.7	13.5
L4	3.5	8.2	4.4	14.4	3.7	3.6	24.6	13.7	10.8	6.7
L7	0.5	28.9	9.7	10.1	34.6	16.3	14.6	14.9	1.5	14.5
L11	1.4	1.4	2.7	8.2	3.9	1.3	9.5	1.2	4.2	0.1
L19	7.4	19.1	9.6	7.7	20.5	13.7	9.6	8.0	23.8	18.1
L27	9.8	1.3	8.7	5.7	5.5	7.7	4.4	7.3	32.4	15.4
L34	11.2	5.8	11.1	16.0	14.4	12.8	9.7	5.1	37.5	20.3
L36	7.5	10.6	20.8	45.6	46.5	25.5	31.6	9.2	24.2	25.9
Mean	**6.1**	**9.5**	**9.9**	**14.2**	**17.0**	**11.0**	**13.6**	**8.8**	**19.9**	**14.3**

Simulation vs baseline

	Sphericity	LA	MA	SA	Surface area	Eq. diameter	Flatness	Elongation	Compactness	Surface / volume
L0	7.5	6.7	15.2	4.2	12.5	10.0	2.3	8.0	24.3	15.4
L4	2.2	6.1	14.8	6.0	17.1	9.4	0.1	8.2	6.6	10.5
L7	2.4	10.2	0.2	4.0	7.9	5.1	5.6	9.5	7.2	7.0
L11	3.0	4.4	6.1	8.6	10.7	6.8	4.0	1.6	9.3	9.1
L19	9.2	17.1	8.4	7.2	16.2	12.6	8.4	7.4	30.0	18.6
L27	10.0	2.5	6.6	11.8	8.7	9.3	9.1	4.0	33.1	16.9
L34	10.7	9.1	13.2	8.4	14.3	12.5	0.6	3.8	35.8	19.7
L36	7.9	15.1	14.7	18.0	26.9	17.0	2.6	0.3	25.5	20.7
Mean	**6.6**	**8.9**	**9.9**	**8.5**	**14.3**	**10.3**	**4.1**	**5.4**	**21.5**	**14.8**

Simulation vs phantom

	Sphericity	LA	MA	SA	Surface area	Eq. diameter	Flatness	Elongation	Compactness	Surface / volume
L0	0.1	5.6	2.8	1.3	5.0	2.4	6.6	2.7	0.3	2.3
L4	1.3	15.5	10.0	7.4	12.9	5.6	19.8	4.8	3.8	4.1
L7	1.8	14.5	9.0	5.5	19.8	9.6	10.5	6.4	5.6	8.7
L11	1.6	3.0	3.3	18.3	15.1	8.1	14.9	0.3	4.9	9.0
L19	1.6	1.7	1.0	0.5	3.6	1.0	1.3	0.7	5.0	0.6
L27	0.2	1.2	1.9	5.7	3.0	1.6	4.5	3.0	0.6	1.7
L34	0.4	3.1	1.9	6.5	0.1	0.3	9.4	1.2	1.2	0.7
L36	0.3	4.0	5.1	19.0	13.4	6.8	22.1	8.7	1.0	6.9
Mean	**0.9**	**6.1**	**4.4**	**8.0**	**9.1**	**4.4**	**11.1**	**3.5**	**2.8**	**4.2**



< 10% - 68/80

<15% - 75/80

SHAPE INTERPRETATION

Both PET segmentations are larger and smoother than baseline.
Median volume error:
Ph 34.2%; Sim 31.9%.

SIMULATION vs PHANTOM

Mean difference: 0.9% sphericity, 4.2% volume, 11.1% flatness.

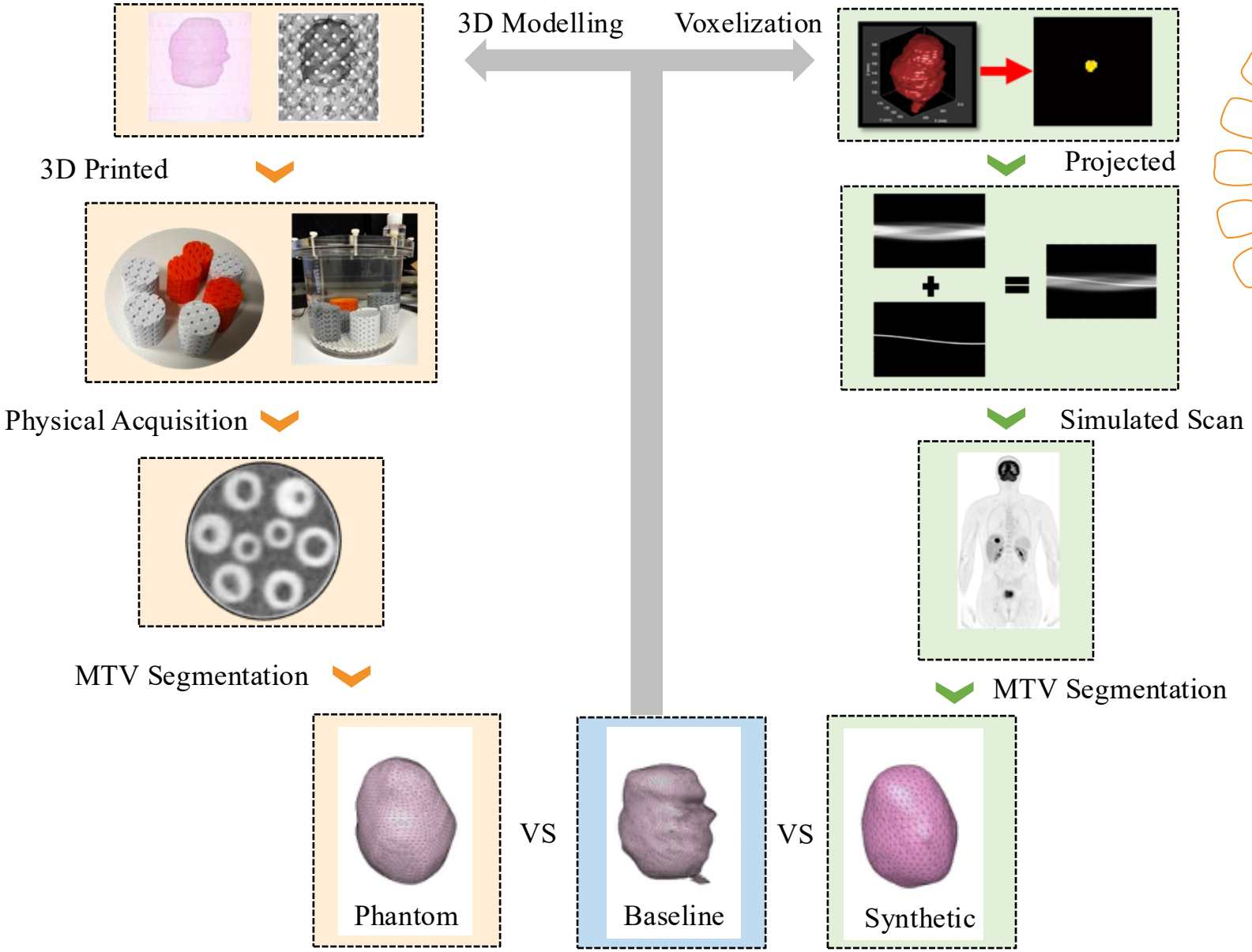
LESION-SPECIFIC LIMIT

L7/L36 differ more at boundaries; similar global shape does not ensure contour agreement.

Discussion

Shared lesion geometry enabled paired physical and synthetic tests.

- Multiple organs to be tested
- Volume matched sphere analysis
- AFoV for Phantom vs Clinical Data



- **Proof of concept for segmentation and shape-recovery benchmarking**

This Work

Design Evaluation of Porous 3D printed Grids: Experiment + Simulation

Pilot study · multi-contrast encoding · single centre

This Work

Dual-Centre Transferability of Multi contrast Phantom : Experiment

Same phantom · two independent institutions · consistent results

This Work

Heterogeneous Uptake in Single Encoded grid: Experiment + Simulation

Gini index as physical ground truth for intra-tumoural heterogeneity

This Work

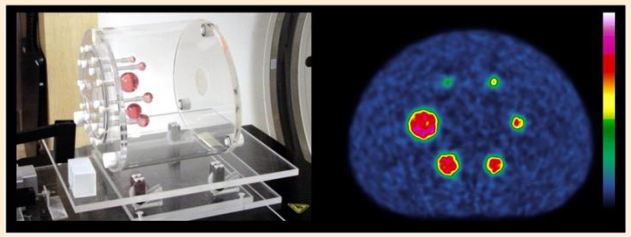
Hybrid Clinical implementation workflow for Anthropomorphic Lesions

8 clinical shapes · hybrid framework · proof of concept validated

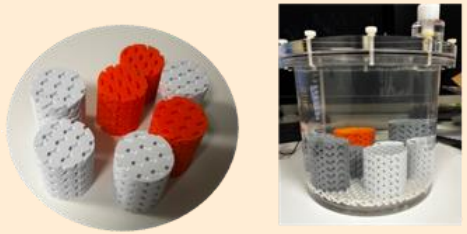
Conclusion & Perspectives

Gap in validation methods

Standard Phantoms



Custom Phantoms



Real Clinical Data

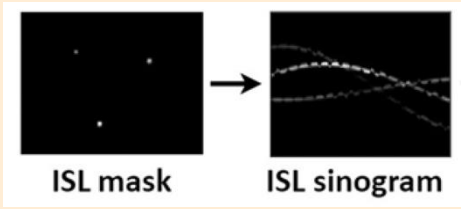


✓ Model Link Feasibility Displayed

✓ Model Validated

✓ Model Applied

Synthetic Lesion Insertion



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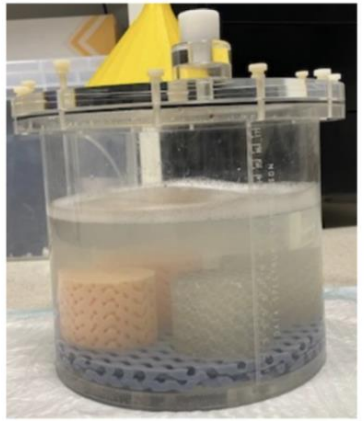
Limits of the evidence

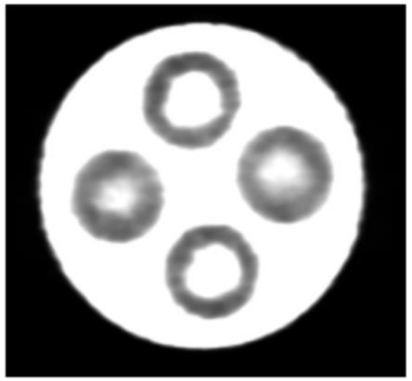
The studies support controlled cross-validation within a defined experimental scope

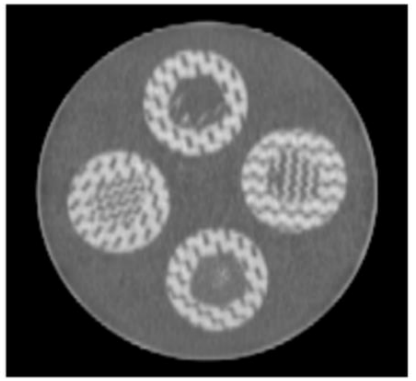
	VALIDATED HERE	NEXT TEST
SCANNERS	2 GE platforms	More centres + vendors
MATERIALS	Printed ABS	CT-matched resin / composites
LESIONS	17–28 mm; 8 patient shapes	Real Lesions + heterogeneous
PATIENT CONTEXT	One background / region	Multiple regions + bed positions
CONTRASTS	2:1 – 5:1	Higher contrasts to find limits of 3D printing

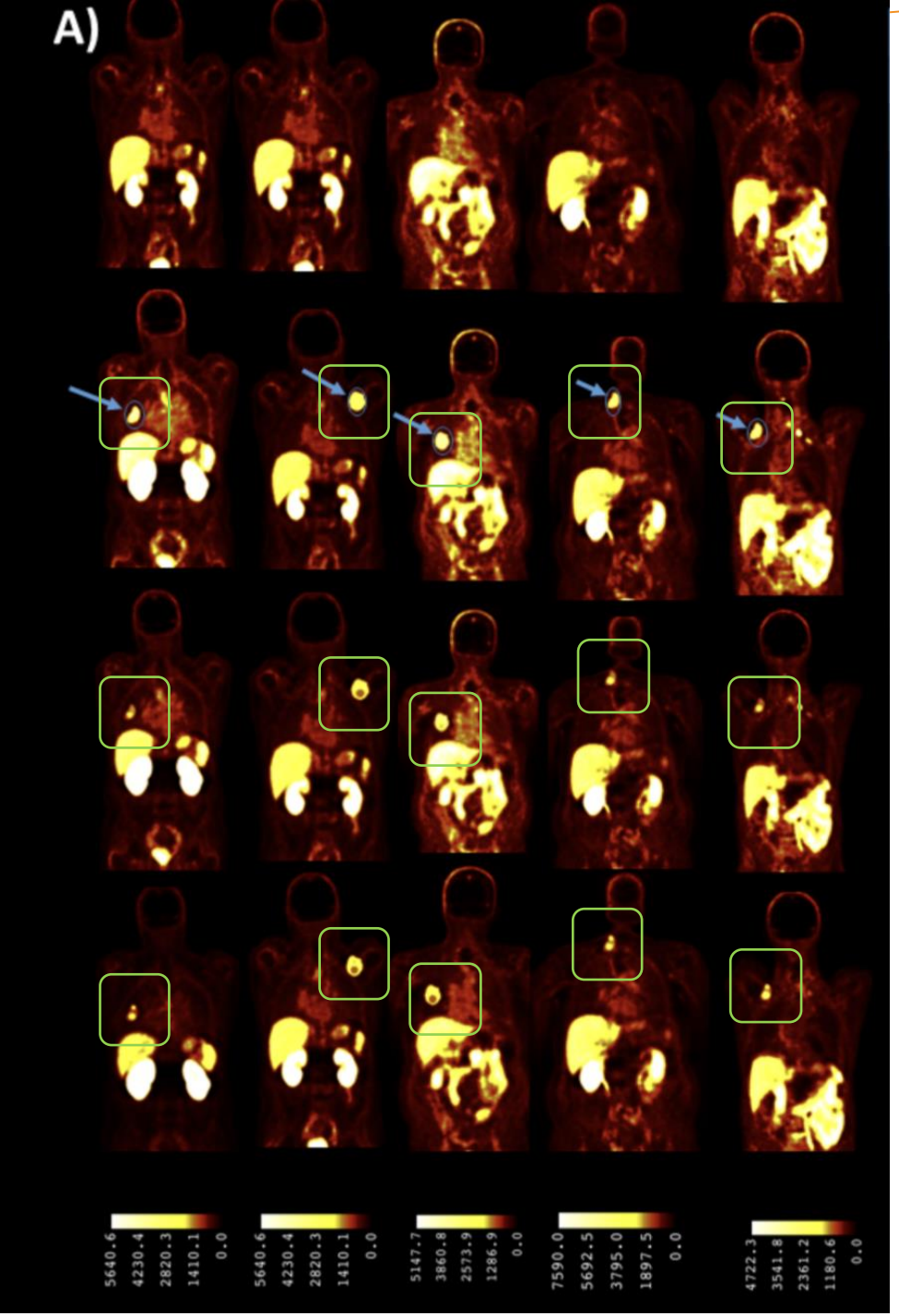
Modified Patient Derived Heterogenous Lesion Inserts

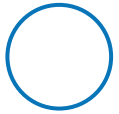
Limited Study Preformed by Tahina DELIGNE

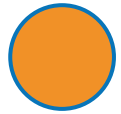






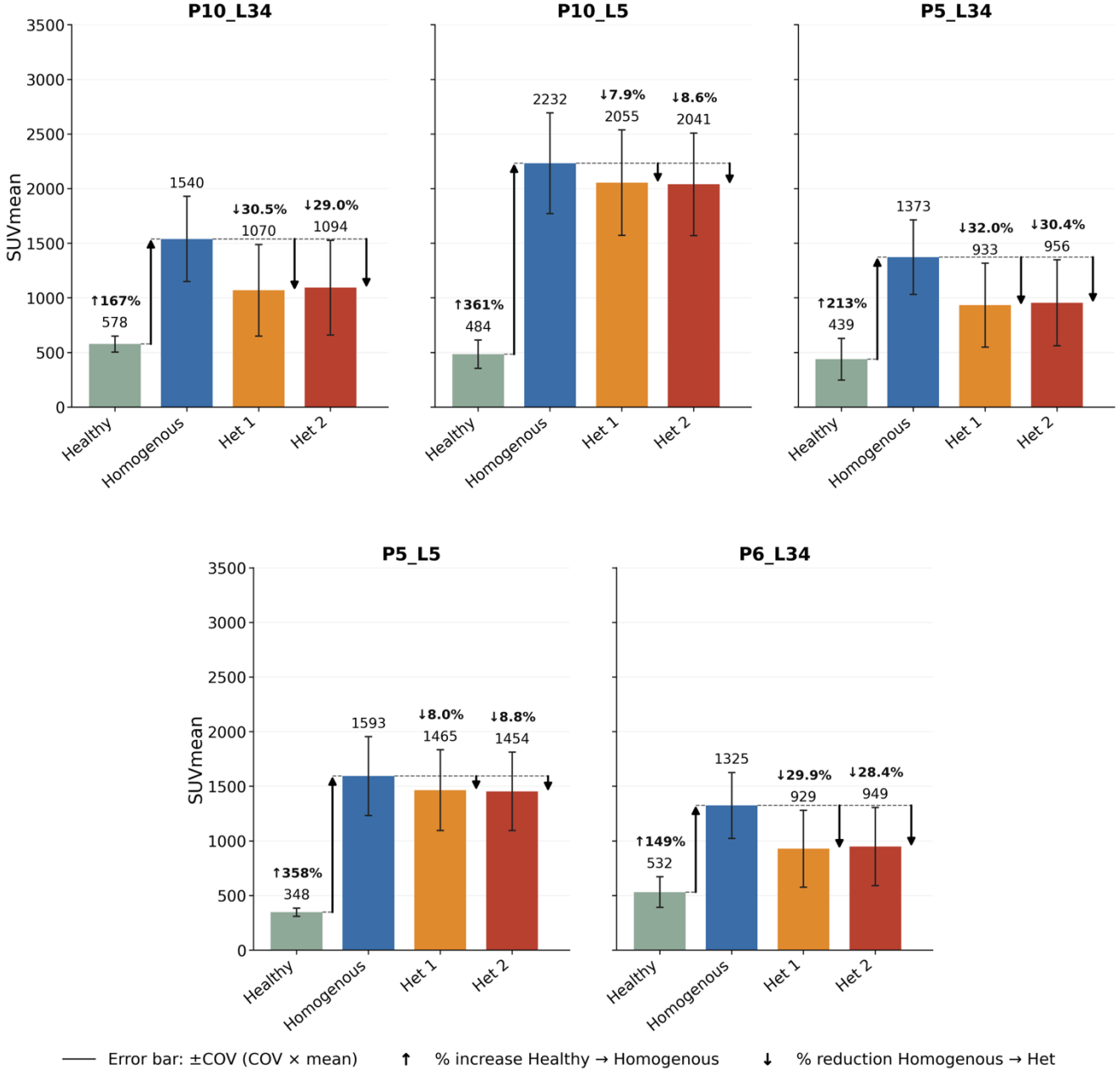




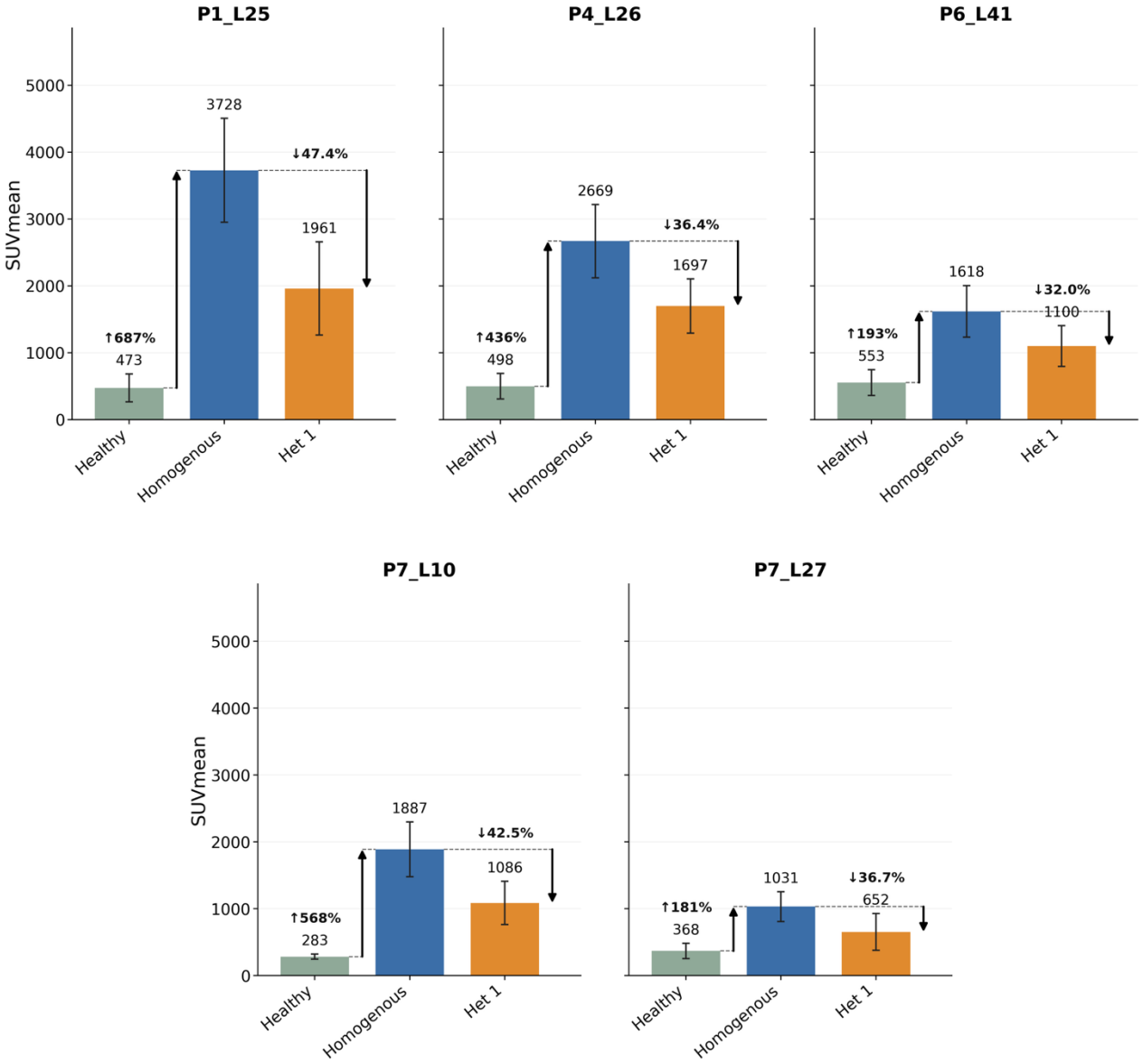


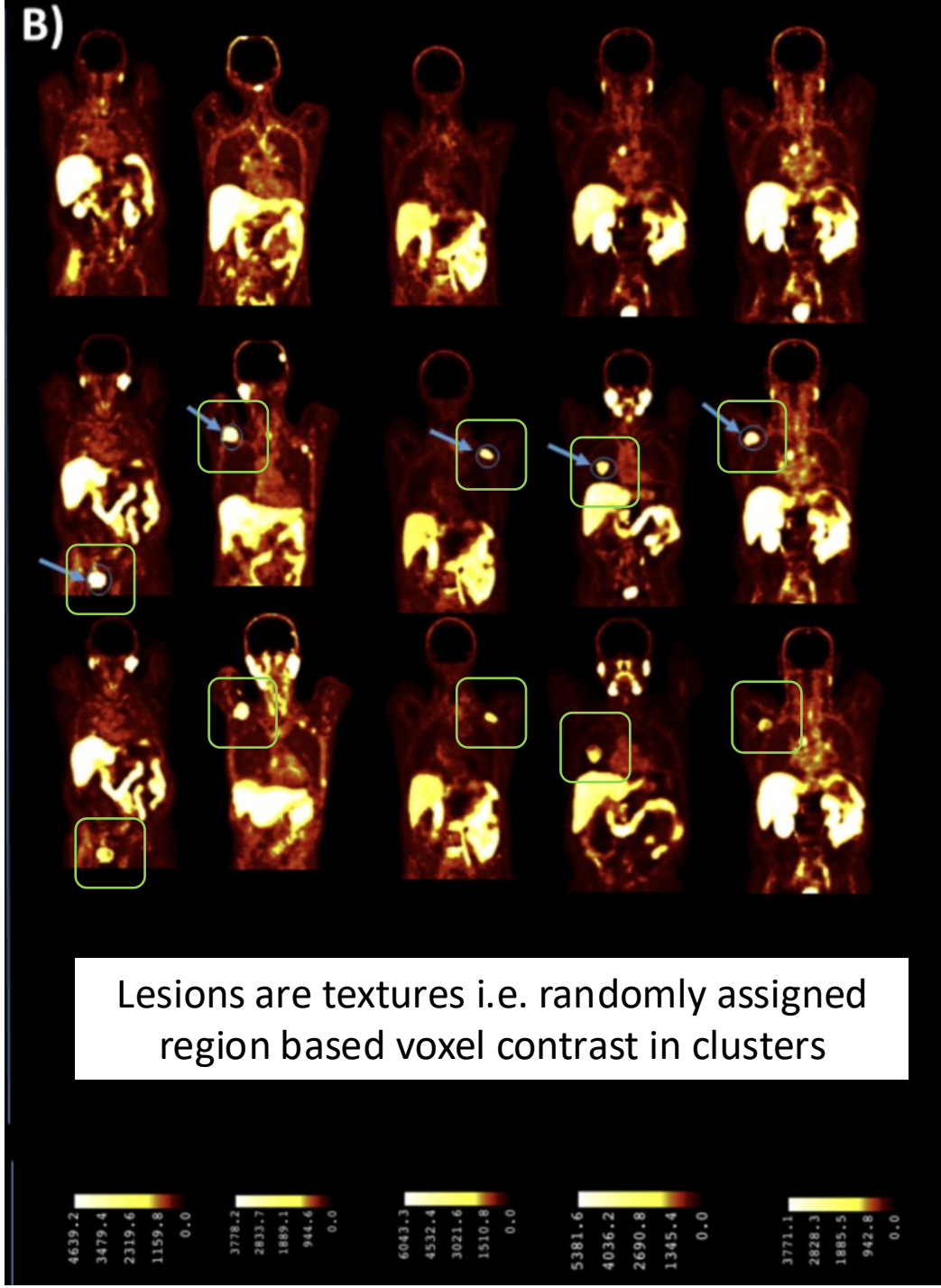






FULL RADIOMIC FEATURE ANALYSIS NOT PERFORMED





Lesions are textures i.e. randomly assigned region based voxel contrast in clusters

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FULL RADIOMIC FEATURE ANALYSIS NOT PERFORMED

CHANGED IN V2

PROJECTION

2D → **3D parallel-beam**

RECONSTRUCTION

OSEM → **BSREM** via PyTomography

VOXEL SAMPLING

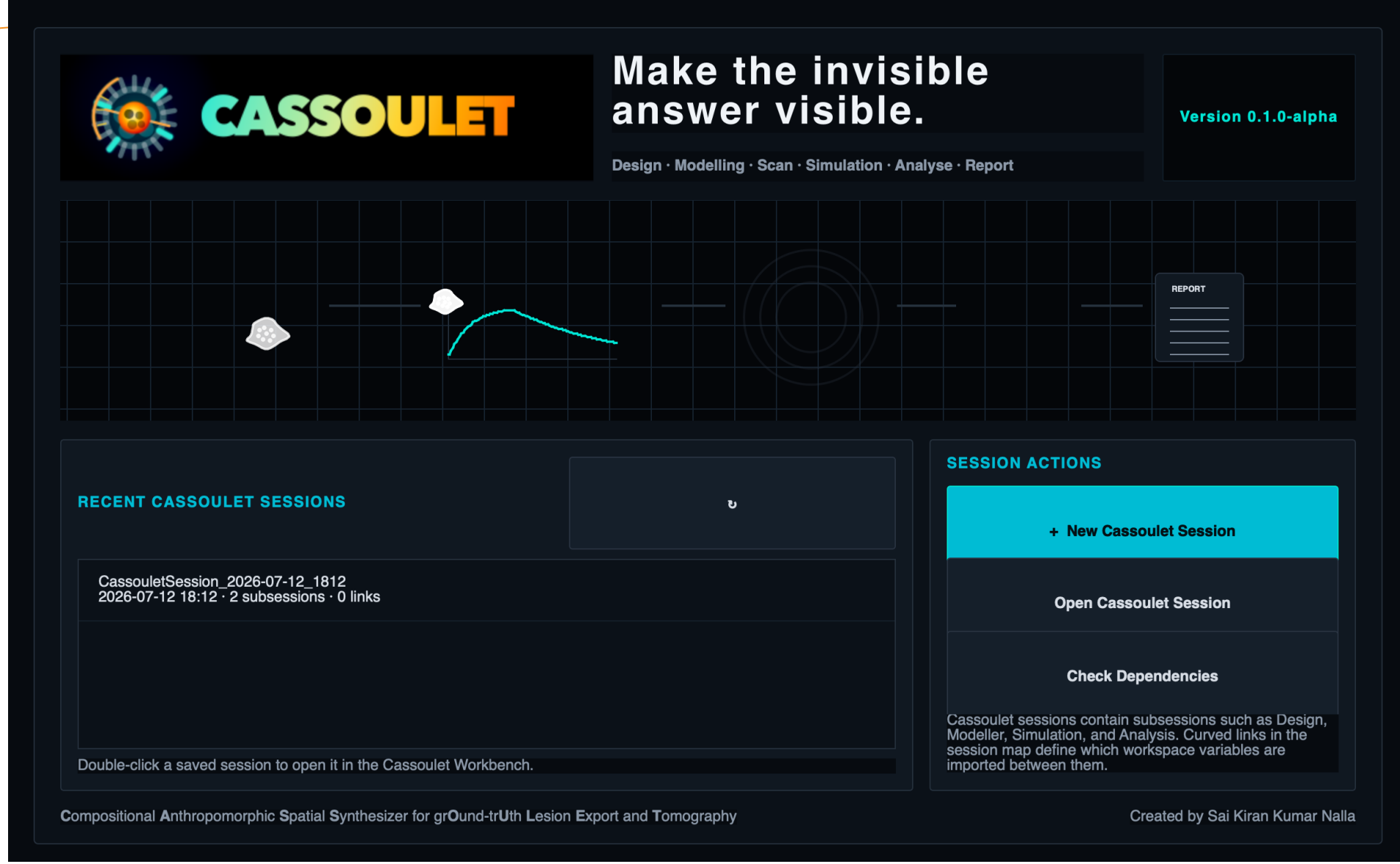
2D slice-wise → **3D ray tracing**

GEOMETRY

No CAD model → **3D CAD model**

RETAINED WORKFLOW

- ✓ **Meshing**
- ✓ **Dynamic PET**
- ✓ **Bed positioning**
- ✓ **Corrections**



CASSOULET Version V2 – Python + Java Script

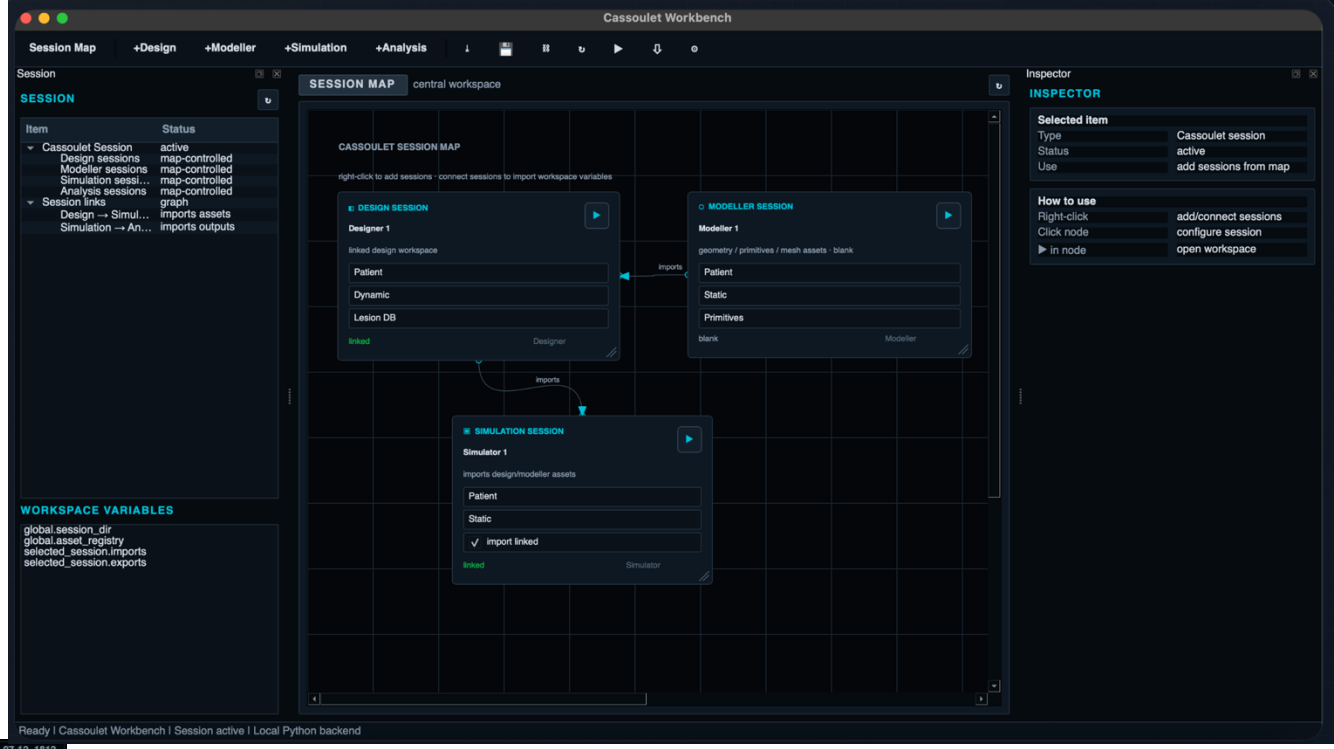
Currently under development

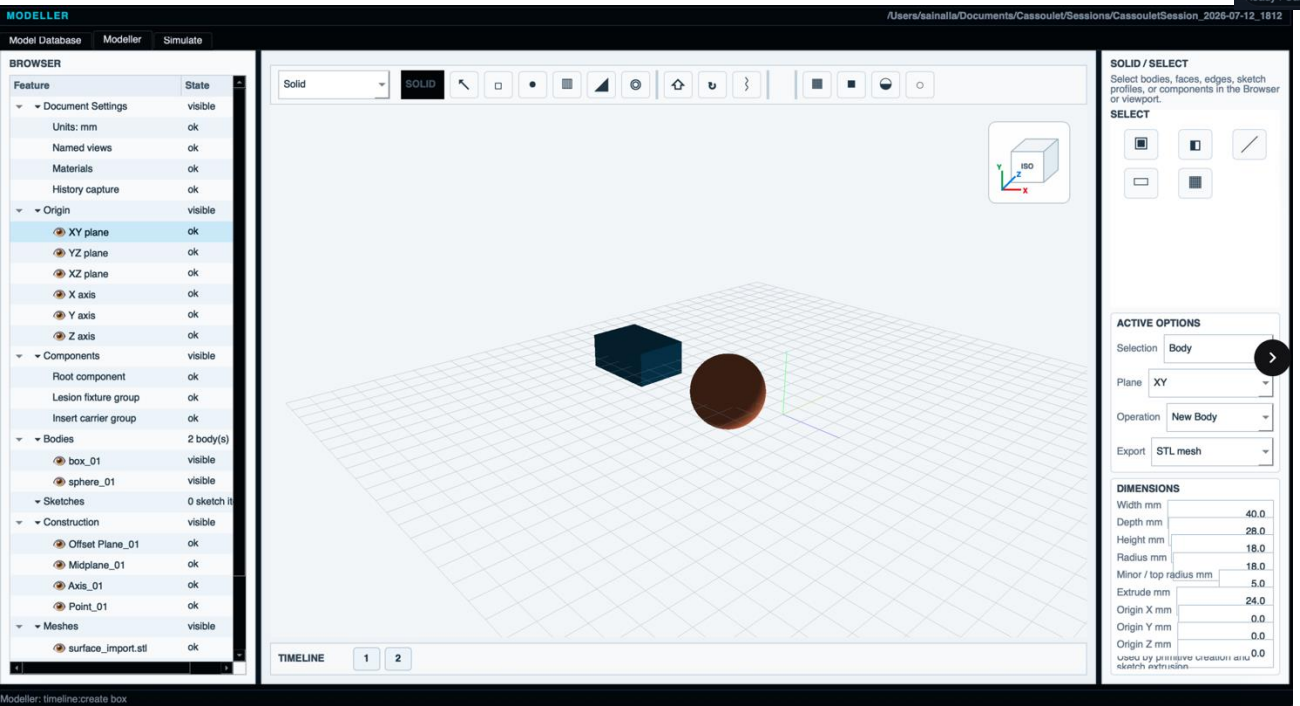
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CASSOULET WORKBENCH

Main session controller

Links design, modelling, simulation and analysis sessions through one shared workspace.





3D MODELLER

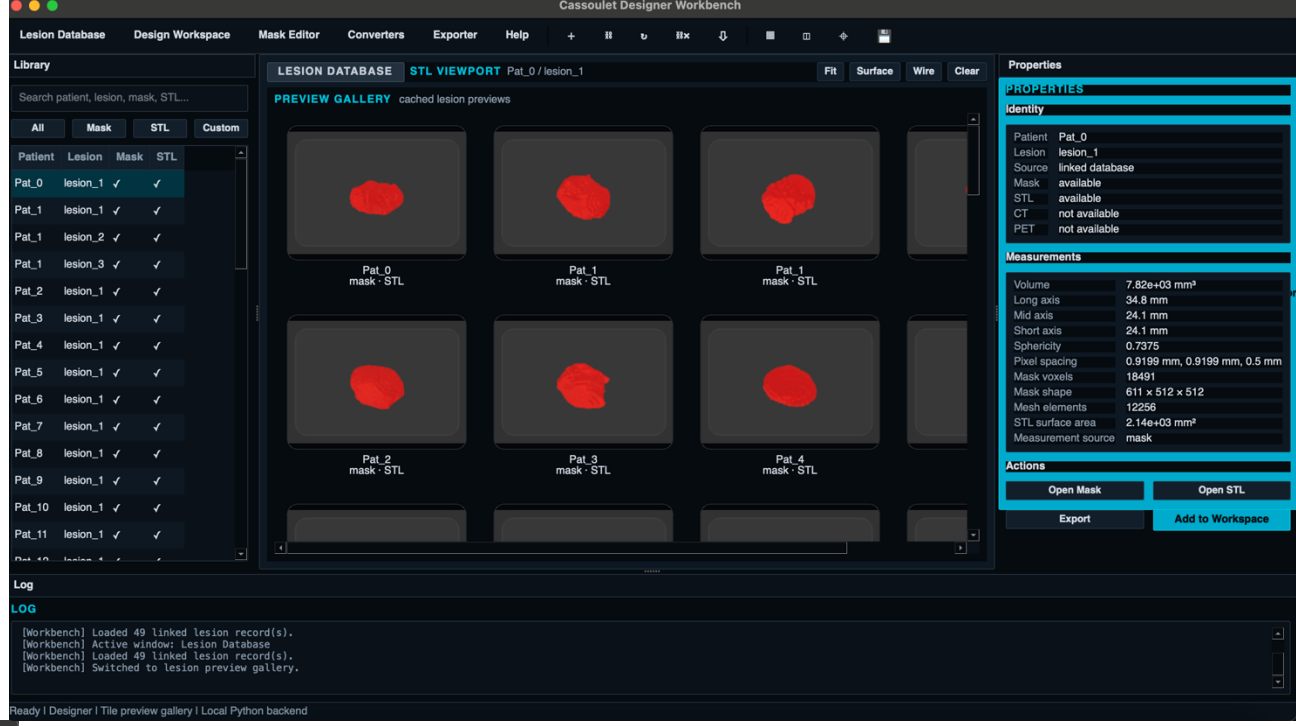
Geometry and mesh workspace

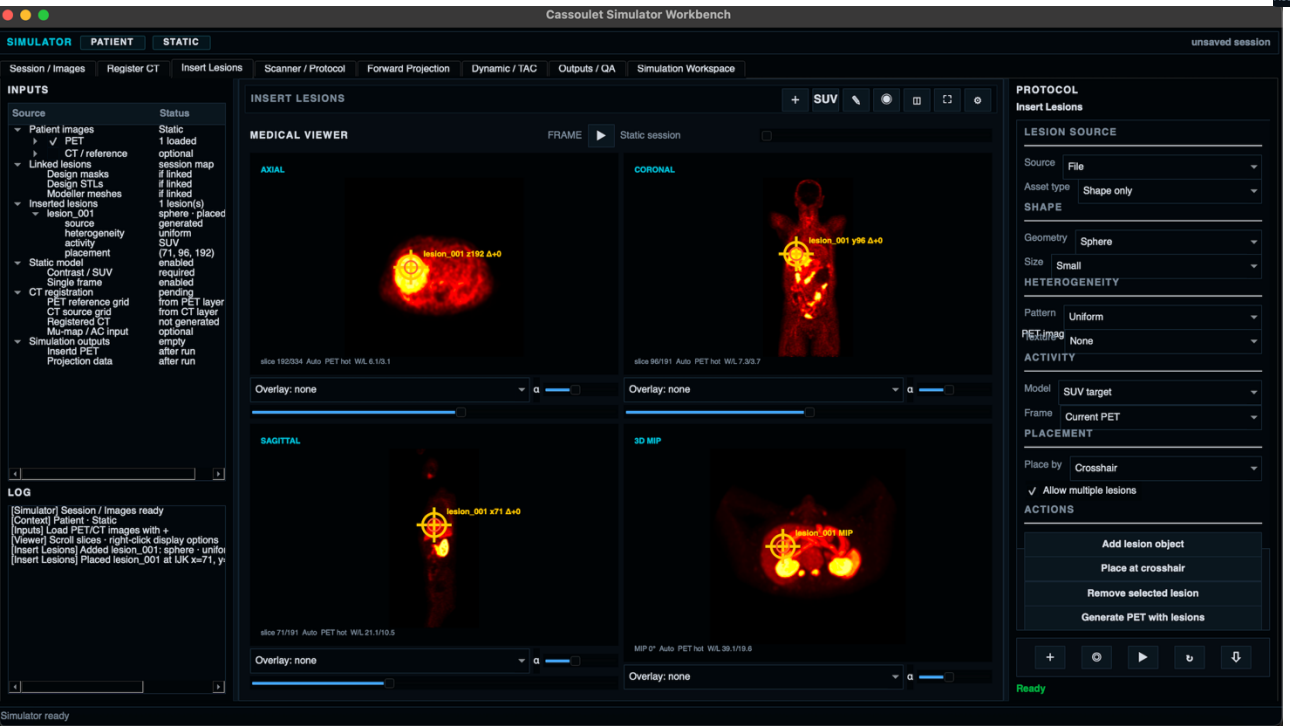
Build and position solids, inspect construction geometry, then prepare meshes for export.

LESION DESIGNER

Database + design workspace

Browse patient-derived masks and STL models, STL to Mask (Ray Tracing), Mask to STL (marching cubes)





SIMULATOR GUI

Place and configure lesions

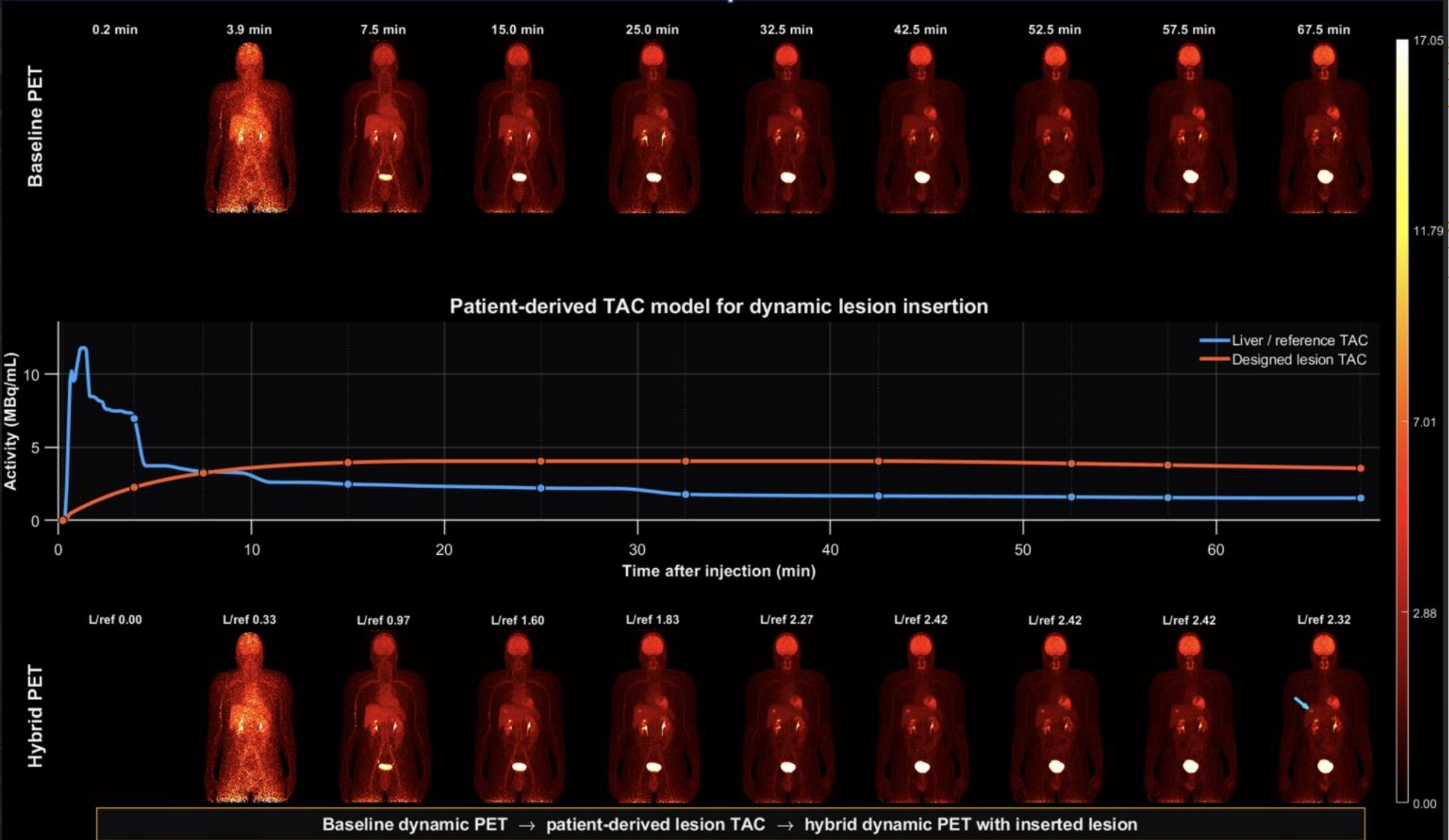
Set location, activity and protocol; review the axial, coronal and sagittal PET views, project, reconstruct

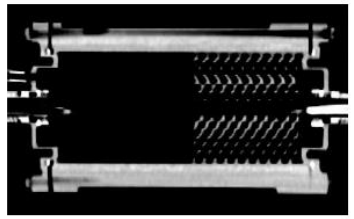
➢ In the thesis

PET VCAR
GE HealthCare
3D Slicer
AUTODESK Fusion
UltiMaker Cura
UltiMaker
MATLAB
MathWorks
R Studio
JASP

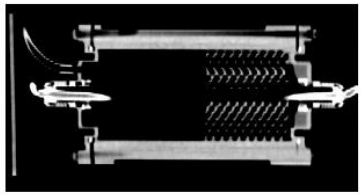
➢ Potential

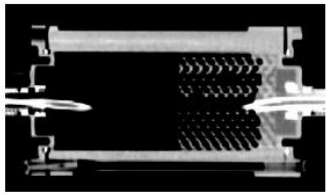
PET VCAR
GE HealthCare
CASSOULET
UltiMaker Cura
UltiMaker

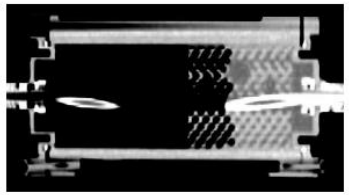

Baseline PET
0.2 min
3.9 min
7.5 min
15.0 min
25.0 min
32.5 min
42.5 min
52.5 min
57.5 min
67.5 min
17.05
11.79
7.01
2.88
0.00
Patient-derived TAC model for dynamic lesion insertion
Liver / reference TAC
Designed lesion TAC
Activity (MBq/mL)
0
5
10
0
10
20
30
40
50
60
Time after injection (min)
Hybrid PET
L/ref 0.00
L/ref 0.33
L/ref 0.97
L/ref 1.60
L/ref 1.83
L/ref 2.27
L/ref 2.42
L/ref 2.42
L/ref 2.42
L/ref 2.32
Baseline dynamic PET → patient-derived lesion TAC → hybrid dynamic PET with inserted lesion

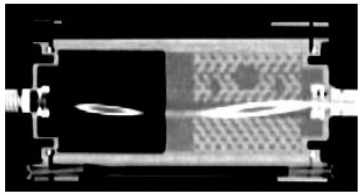






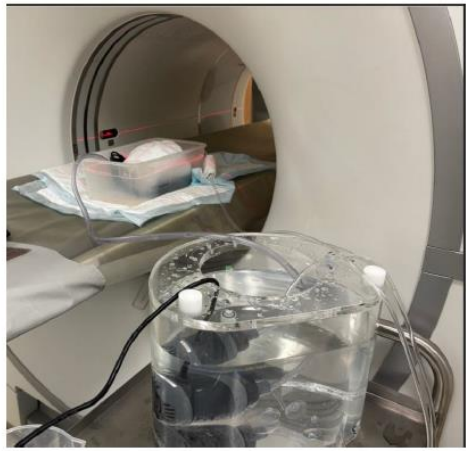






- Porous grids can be adapted by dynamic phantoms (Ex: Calibration Phantom)











- 3D Printing can be extended for large AFOV dynamic phantoms

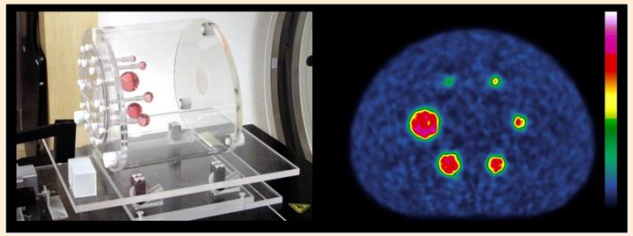


Thank You For Your Attention

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How is TRI / RC different for our design ?

Standard Phantom



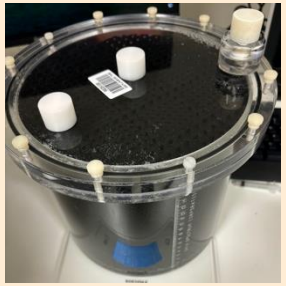
✓ Volume of the fillable sphere

✓ Activity injected into each compartment from dose calibrator

$$RC_{mean/max} = \frac{AC_{target;\,mean\,/max}}{AC_{theoretical}}$$

Porous Phantom





✓ Volume of the sphere

x Activity injected into each insert

✓ Total fillable volume of reference solution

$$RC_{mean/max} = \frac{AC_{target;\,mean\,/max}}{AC_{reference\ solution;\,mean}}$$

$$In\ the\ ideal\ case\ for\ the\ porous\ phantom$$

$$AC_{theoretical} = \frac{A_{inj}}{V_{Phantom} - V_{Inserts}}$$

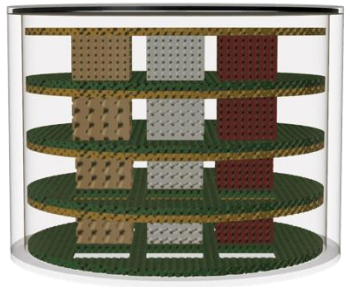
HYPOTHESIS

This assumes uniform mixing of tracer in which case both are equivalent

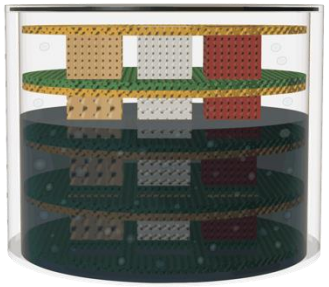
ASSUMPTION

Image derived value of reference layer is used as substitute for theoretical value

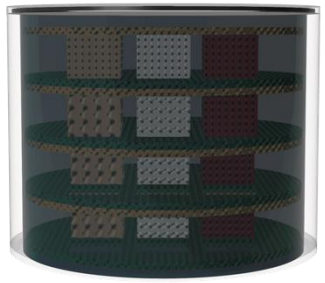
Phantom Preparation Protocol



Soap + Water

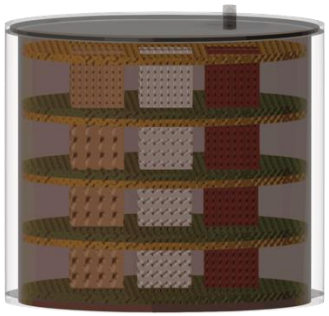


Relaxation

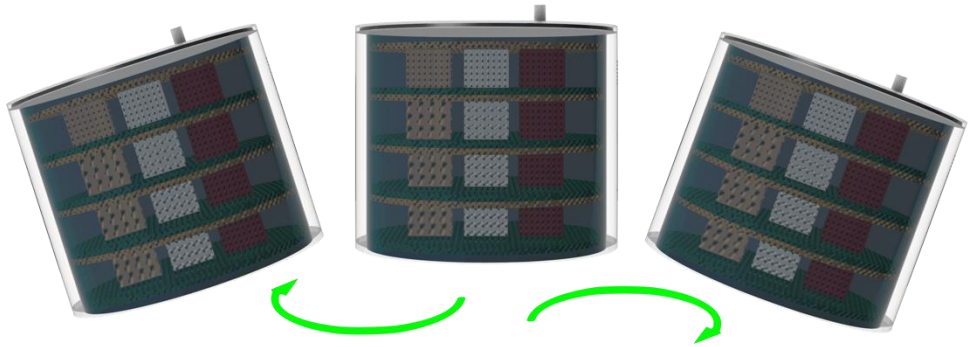


Injection

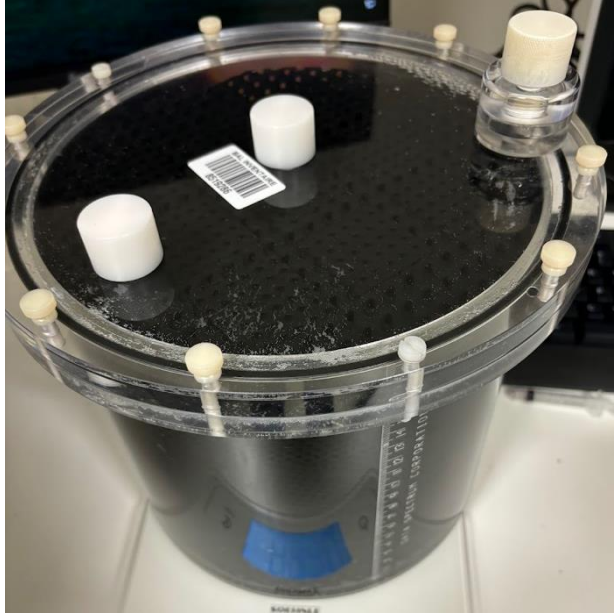




Ready for Imaging



Mixing / Shaking



Total Measured Weight
MPhanotm = 8kg

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Methods : Injection and Acquisition Parameters

- 5 Acquisitions – **Thursday** (D0), **Friday** (D1), **Monday** (D4), **Tuesday** (D5), **Wednesday** (D6)
- Phantom left undisturbed across days – Except during **injection**

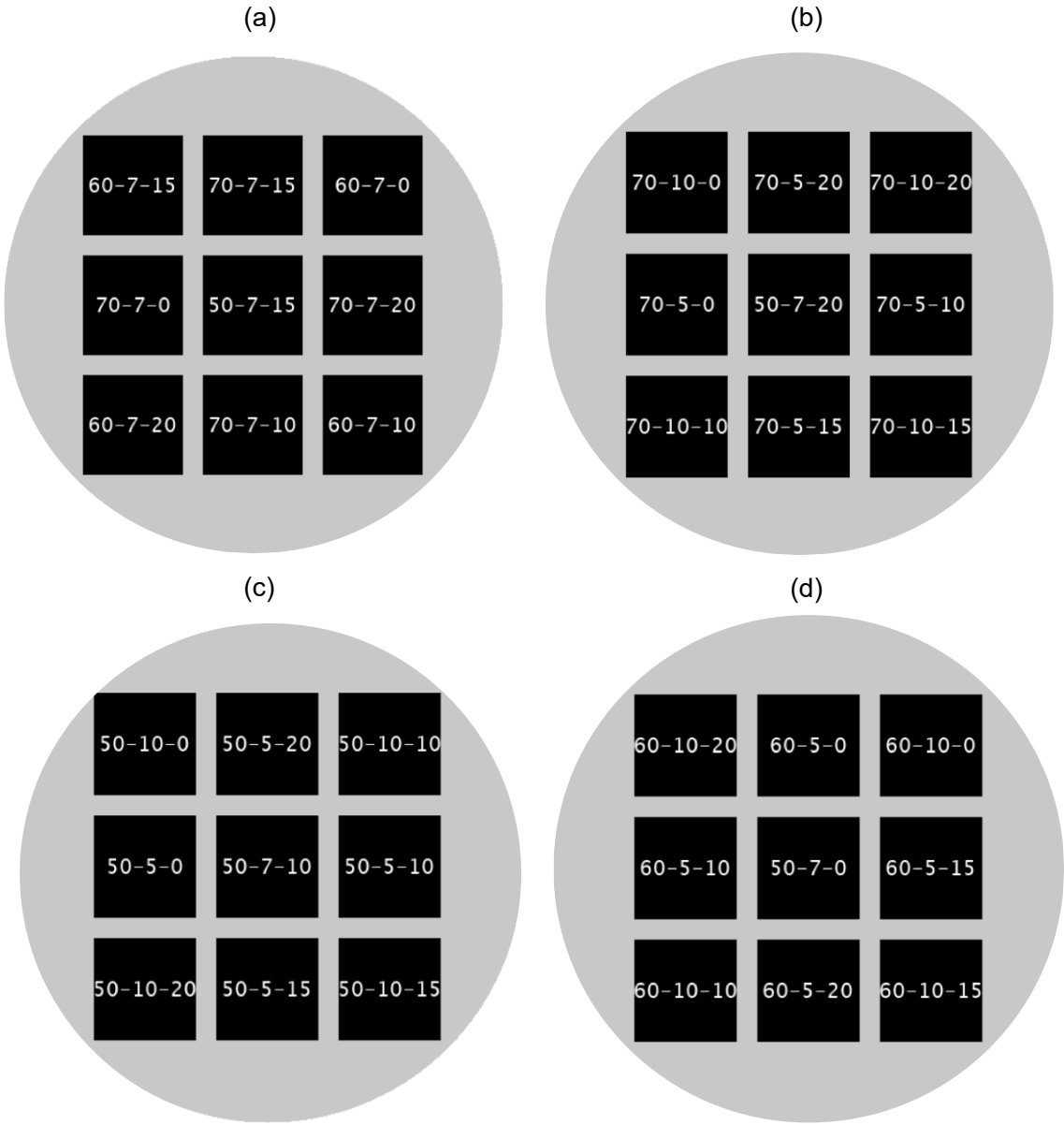
Injected Activity (MBq) (measured on dose calibrator)	Decay Time (min)	Decay corrected at first scan (MBq)	Day
82.8	60	56.1	0
66	73	41.2	1
64.3	45	47.5	4
60.4	30	47.5	5
54.4	25	45.4	6

Different preparation protocols tested across days to identify pitfalls

- 3 CT acquisitions @ **1.25, 2.5, 3.75** mm (Attenuation CT)
- PET acquisition on each day : **5 x 300 s** with 1 bed position
- PET reconstruction on each day : **256** × **256 x 89** with reconstruction $\boldsymbol{\beta = 550}$

Arrangement of the Layers

- **Dp : 5 mm and 10 mm together in bottom layers as**
  - **Layer 2 – S: 70%**
  - **Layer 3 – S: 50%**
  - **Layer 4 – S: 60%**
- **Central cube in each layer is S: 50% - Dp: 7mm**
- **Layer 1 – corresponds to 7 mm of S: 60 & 70%**



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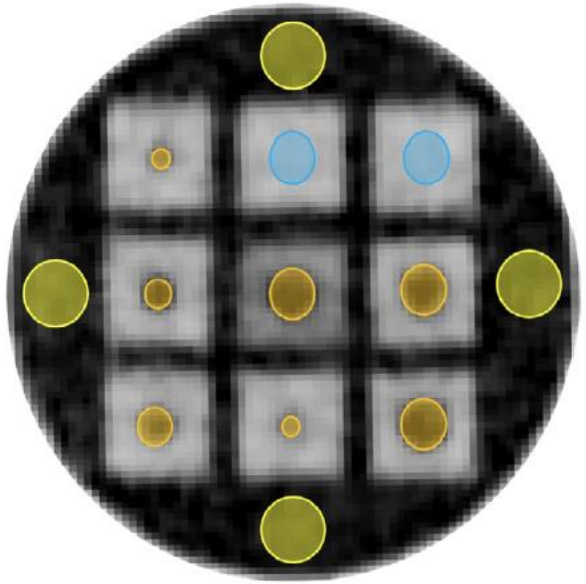
Methods : Injection and Acquisition Parameters

- TOF - 5R Discovery MI PET/CT system (DMI) with AFOV of 25 cm
- 3 CT acquisitions @ 1.25 mm, 2.5 mm, 3.75 mm (Attenuation CT)

Key PET Acquisition Parameters	
No. of Bed Positions	1
Matrix Size	256 × 256 x 89
Scan Duration	5 × 300 s
Reconstruction Diameter	600 mm
Voxel Dimensions	2.34 × 2.34 × 2.79 mm
Voxel Volume	15.43 mm3
Reconstruction Algorithm	Q Clear (TOF BPL) - BSREM
Beta*	550

Data Extraction

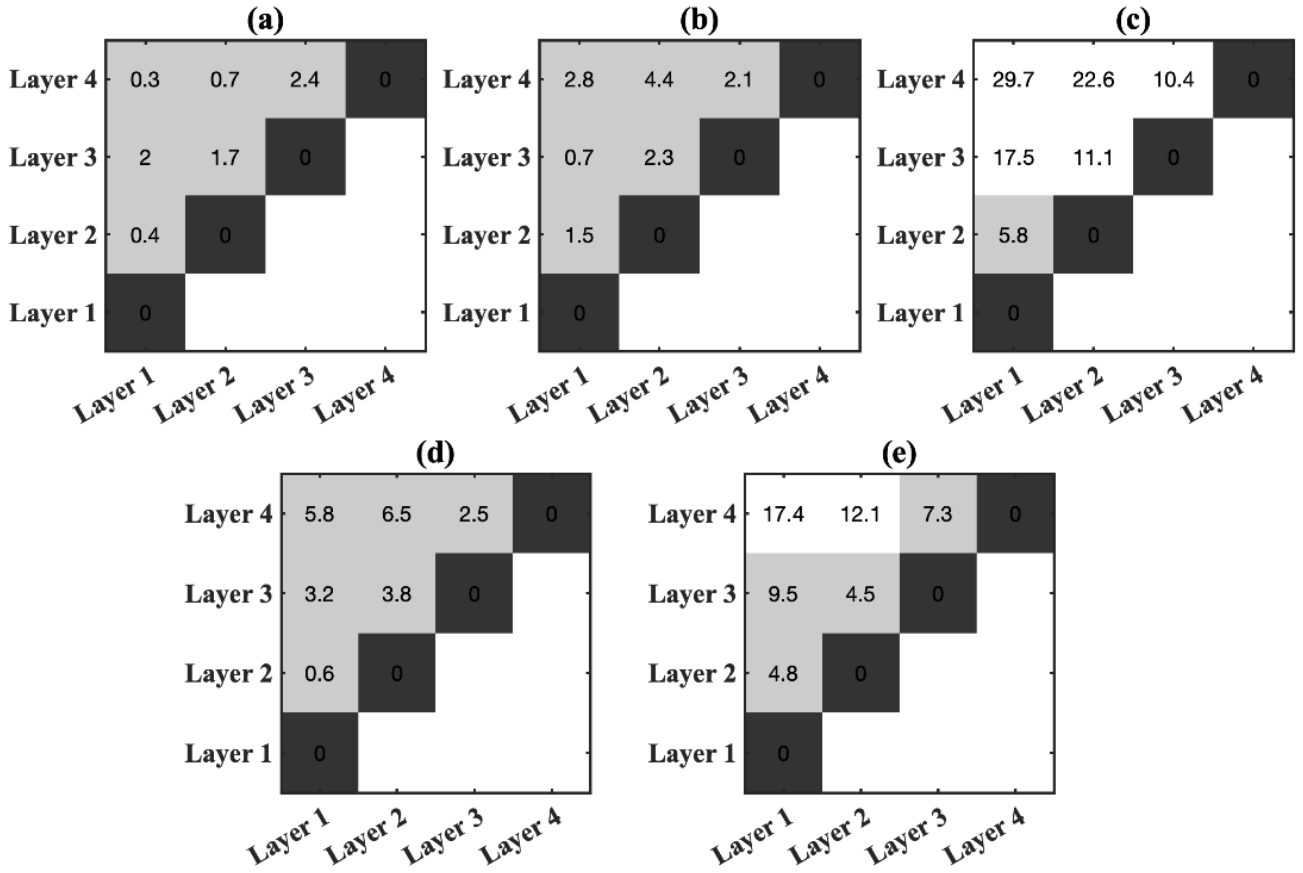
- Manual spherical VOI matching the nominal diameter of the target
- The corresponding voxels (approximated) in each VOI are
  - Ø 10 mm VOI ~ 29 voxels
  - Ø 15 mm VOI ~ 97 voxels
  - Ø 20 mm VOI ~ 229 voxels
  - Ø 30 mm VOI ~ 773 voxels
- Homogenous cubes use the 20mm VOI
- Reference solutions use the 30mm VOI



In each layer we place 13 VOIs

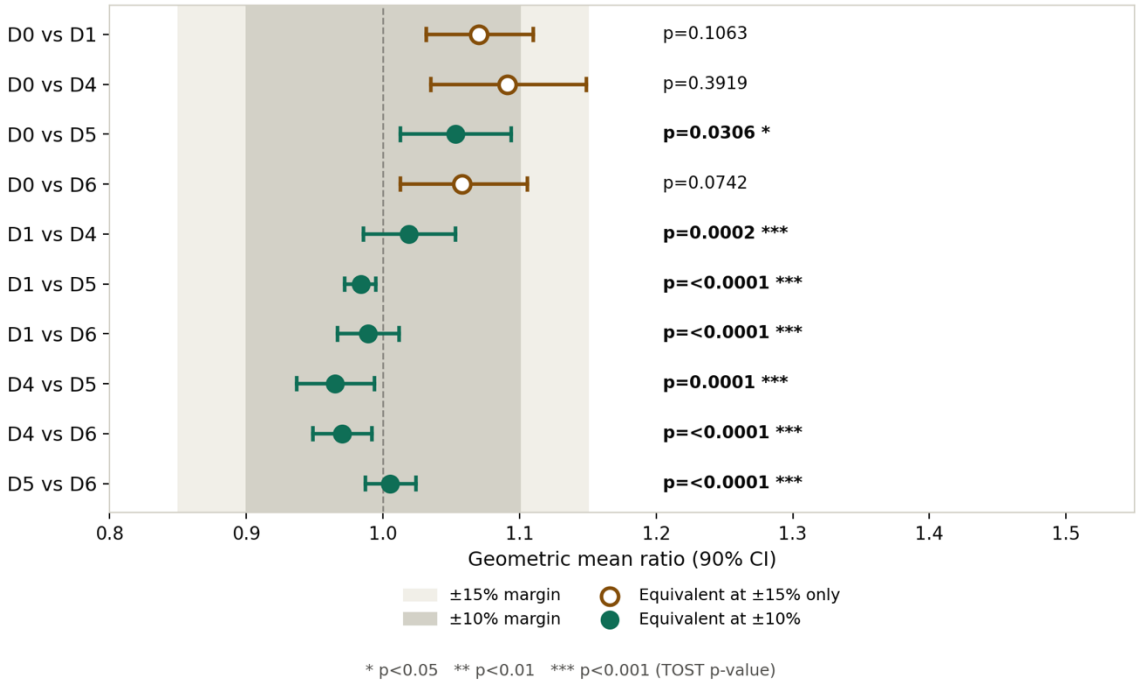
4 x Reference Solution
9 x Inserts

Layer Variability for Study - 1

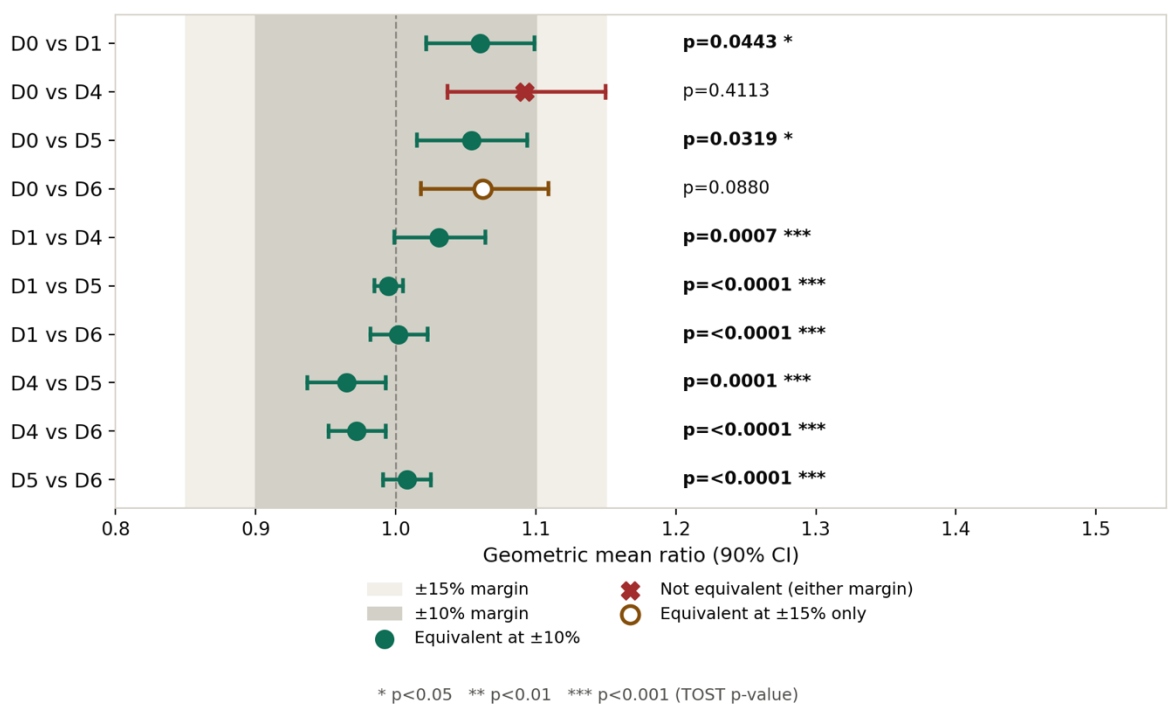


Results : Numerical Equivalence Tests on pairwise SUV ratio data

SUV_{max}



SUV_{mean}



- **D0 vs D4 is the worst pair with maximum deviation of 16% for SUV_{mean}**
- **D0 typically has 20-25% more counts or AC than other days due to higher injected activity**

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Numerical Equivalence Tests on pairwise TRI data

RCmax

	Day 0	Day 1	Day 4	Day 5	Day 6
Day 0	0.0% ± 0.0%	12.3% ± 11.1%	16.4% ± 16.7%	13.1% ± 12.0%	14.6% ± 13.6%
Day 1	12.3% ± 11.1%	0.0% ± 0.0%	6.7% ± 7.3%	3.0% ± 2.3%	4.9% ± 3.5%
Day 4	16.4% ± 16.7%	6.7% ± 7.3%	0.0% ± 0.0%	6.2% ± 5.3%	5.0% ± 4.8%
Day 5	13.1% ± 12.0%	3.0% ± 2.3%	6.2% ± 5.3%	0.0% ± 0.0%	4.1% ± 3.5%
Day 6	14.6% ± 13.6%	4.9% ± 3.5%	5.0% ± 4.8%	4.1% ± 3.5%	0.0% ± 0.0%

MAPE band
- 0–<5%
- 5–<10%
- 10–<15%
- 15–20%

RCmean

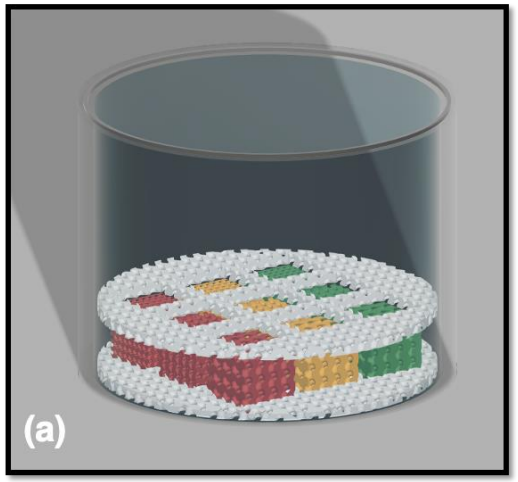
	Day 0	Day 1	Day 4	Day 5	Day 6
Day 0	0.0% ± 0.0%	12.9% ± 12.3%	17.0% ± 15.8%	13.5% ± 12.1%	12.8% ± 9.8%
Day 1	12.9% ± 12.3%	0.0% ± 0.0%	7.5% ± 5.9%	2.4% ± 2.5%	5.7% ± 4.4%
Day 4	17.0% ± 15.8%	7.5% ± 5.9%	0.0% ± 0.0%	8.0% ± 6.0%	9.8% ± 6.0%
Day 5	13.5% ± 12.1%	2.4% ± 2.5%	8.0% ± 6.0%	0.0% ± 0.0%	5.8% ± 4.6%
Day 6	12.8% ± 9.8%	5.7% ± 4.4%	9.8% ± 6.0%	5.8% ± 4.6%	0.0% ± 0.0%

MAPE band
- 0–<5%
- 5–<10%
- 10–<15%
- 15–20%

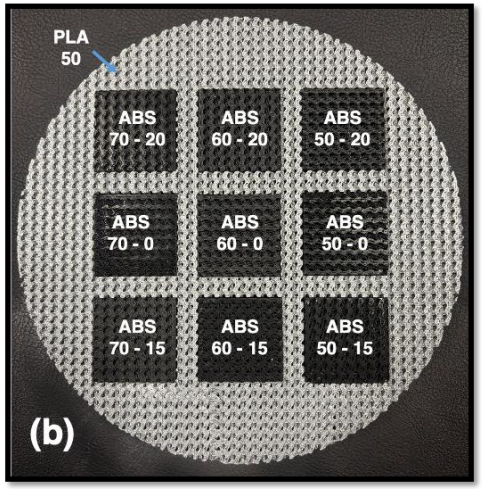
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➢ **D0 vs D4 is the worst pair with maximum deviation**

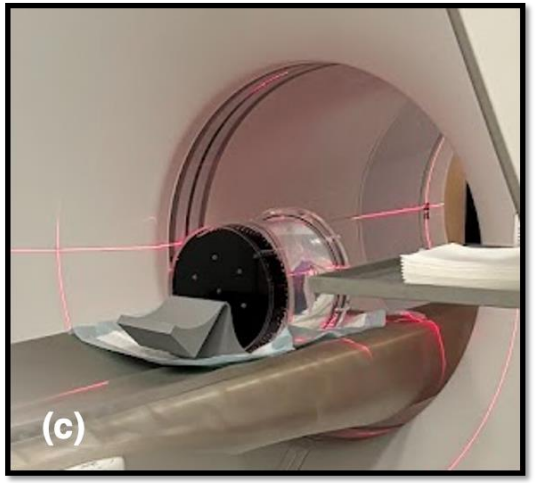
Lesion Insertion Evaluation Study



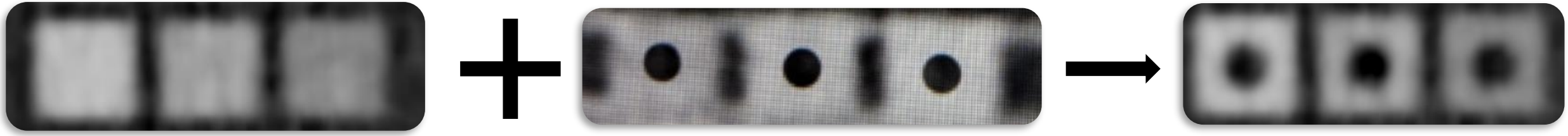
Envisioned Setup



Cube Arrangement



Setup on Scanner

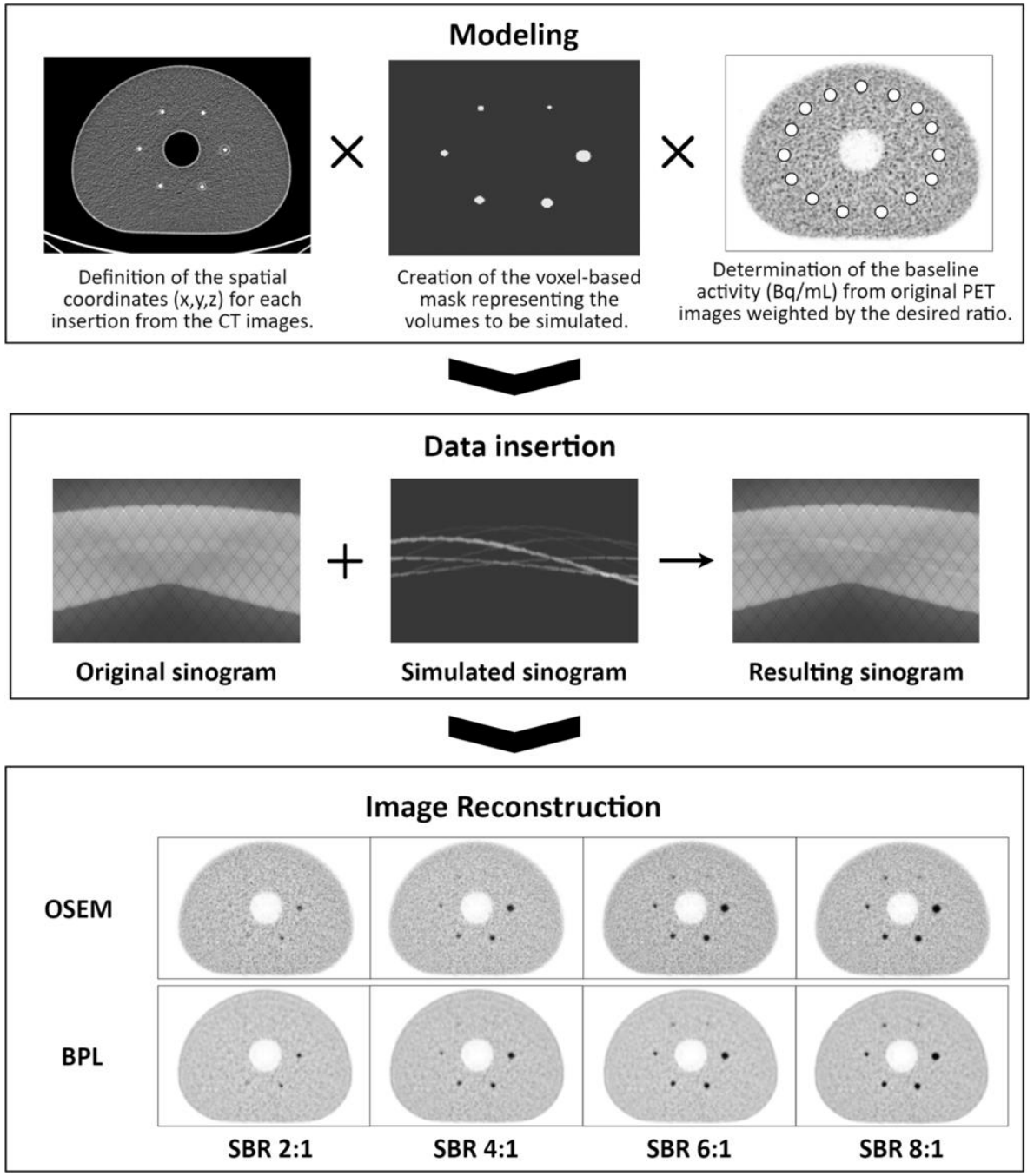


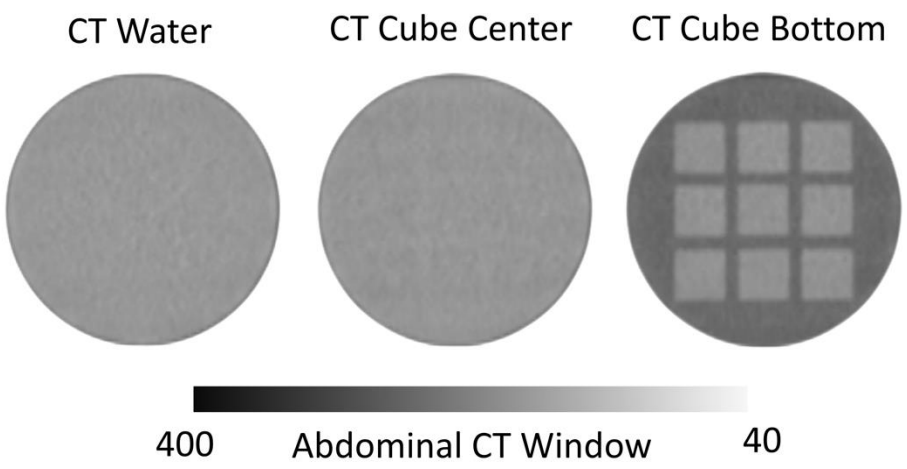
Baseline PET CT without Target

Baseline PET CT with Target Mask

Synthetic Lesion Inserted PET CT

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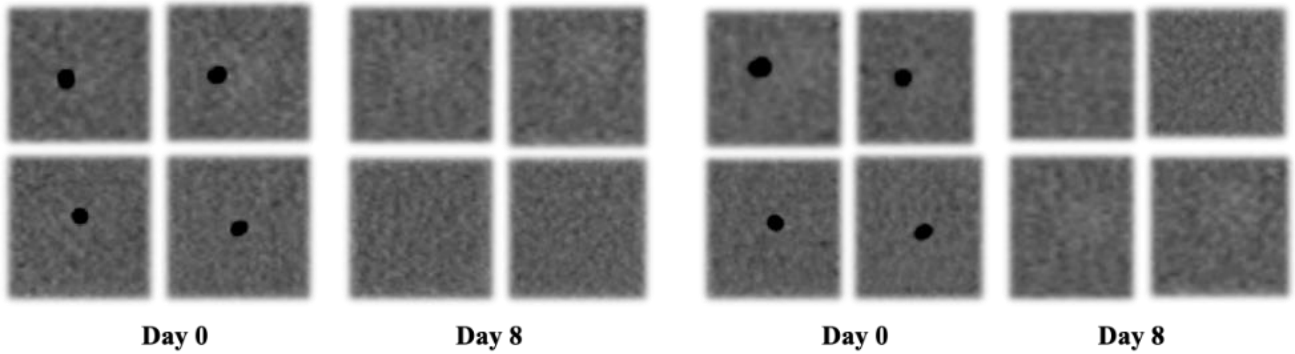

Modeling
Definition of the spatial coordinates (x,y,z) for each insertion from the CT images.
Creation of the voxel-based mask representing the volumes to be simulated.
Determination of the baseline activity (Bq/mL) from original PET images weighted by the desired ratio.
Data insertion
Original sinogram
Simulated sinogram
Resulting sinogram
Image Reconstruction
OSEM
BPL
SBR 2:1
SBR 4:1
SBR 6:1
SBR 8:1



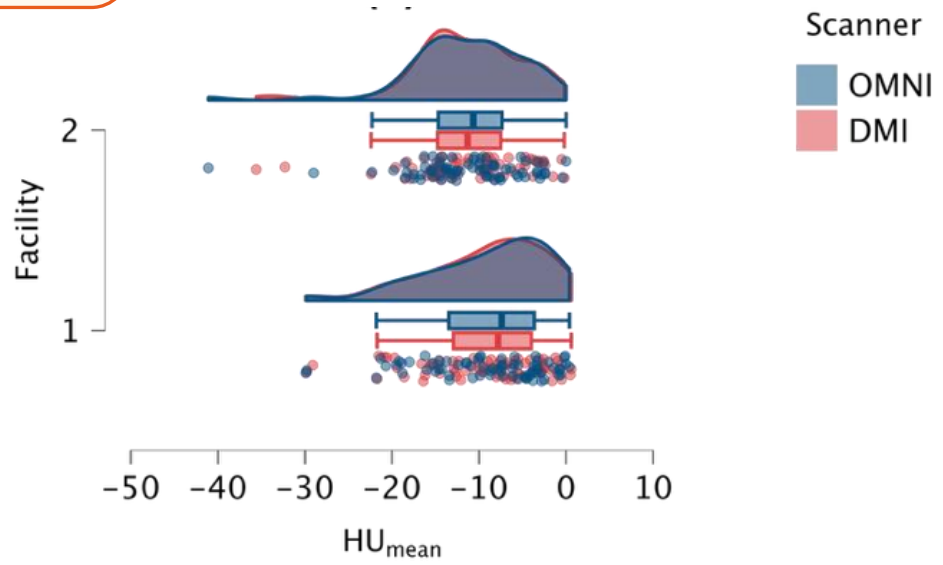
Error Percentage For Target to Background Ratio - Mean			
Cube	Theoretical VS Experimental	Theoretical VS Simulation	Experiment VS Simulation
50-0	9.68 %	-	-
60-0	11.5 %	-	-
70-0	8.02 %	-	-
50-15	25 %	21 %	5.4 %
60-15	25 %	22.5 %	3.58 %
70-15	30 %	28 %	1.99 %
50-20	21 %	16 %	6.10 %
60-20	21 %	15 %	6.87 %
70-20	25 %	21 %	4.77 %

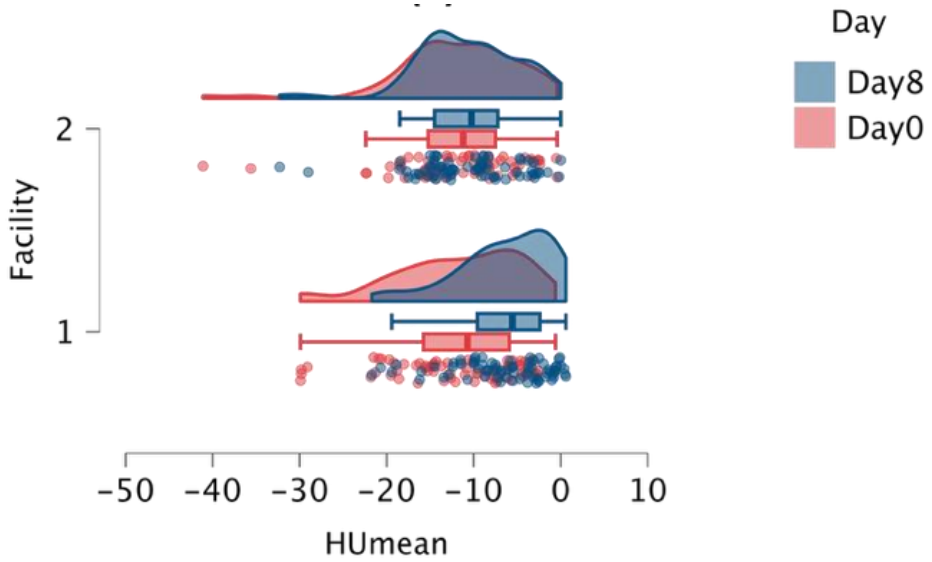
Error Percentage For Target to Background Ratio - Max			
Cube	Theoretical VS Experimental	Theoretical VS Simulation	Experiment VS Simulation
50-0	-	-	-
60-0	-	-	-
70-0	-	-	-
50-15	4.5 %	6.5 %	11.19 %
60-15	4 %	14.5 %	9.83 %
70-15	3.9 %	11%	6.69 %
50-20	2 %	13 %	10.93 %
60-20	9 %	22 %	13.82 %
70-20	5.7 %	15 %	8.91 %

CT at 2.5mm for cross facility

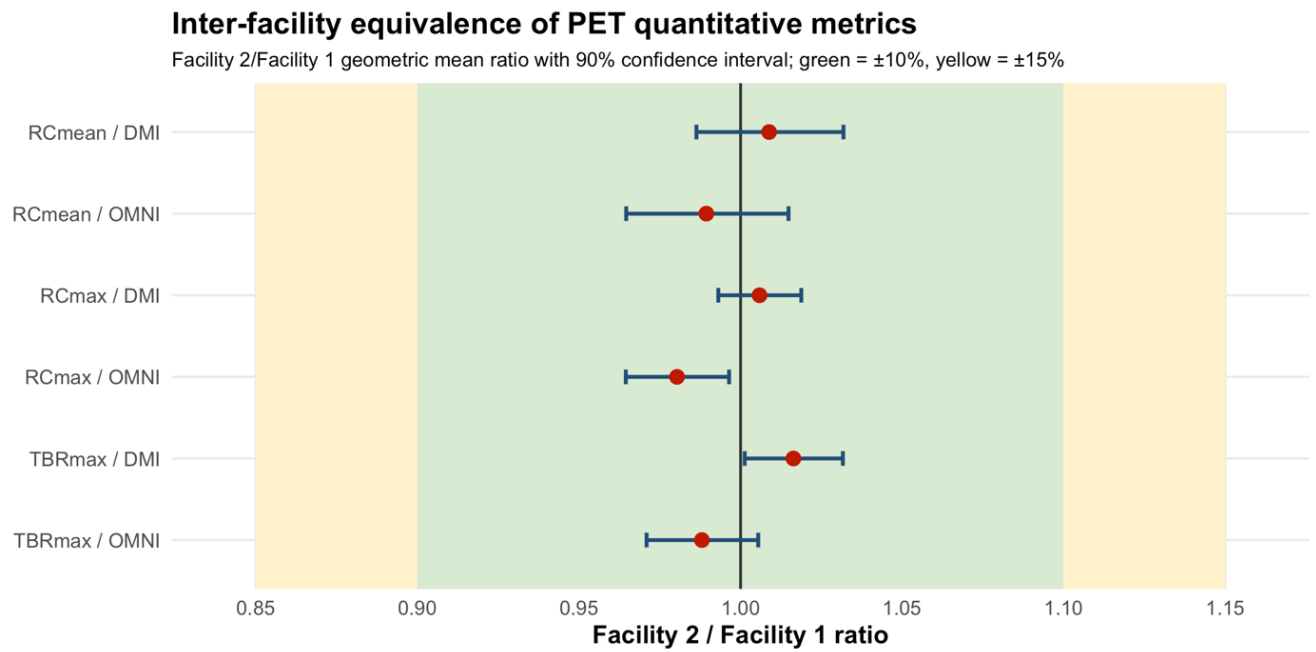


- Improvement of HU (D0 vs D8)
- $\Delta\,\mathbf{HU}_{\mathbf{Facility,max}} = 12$





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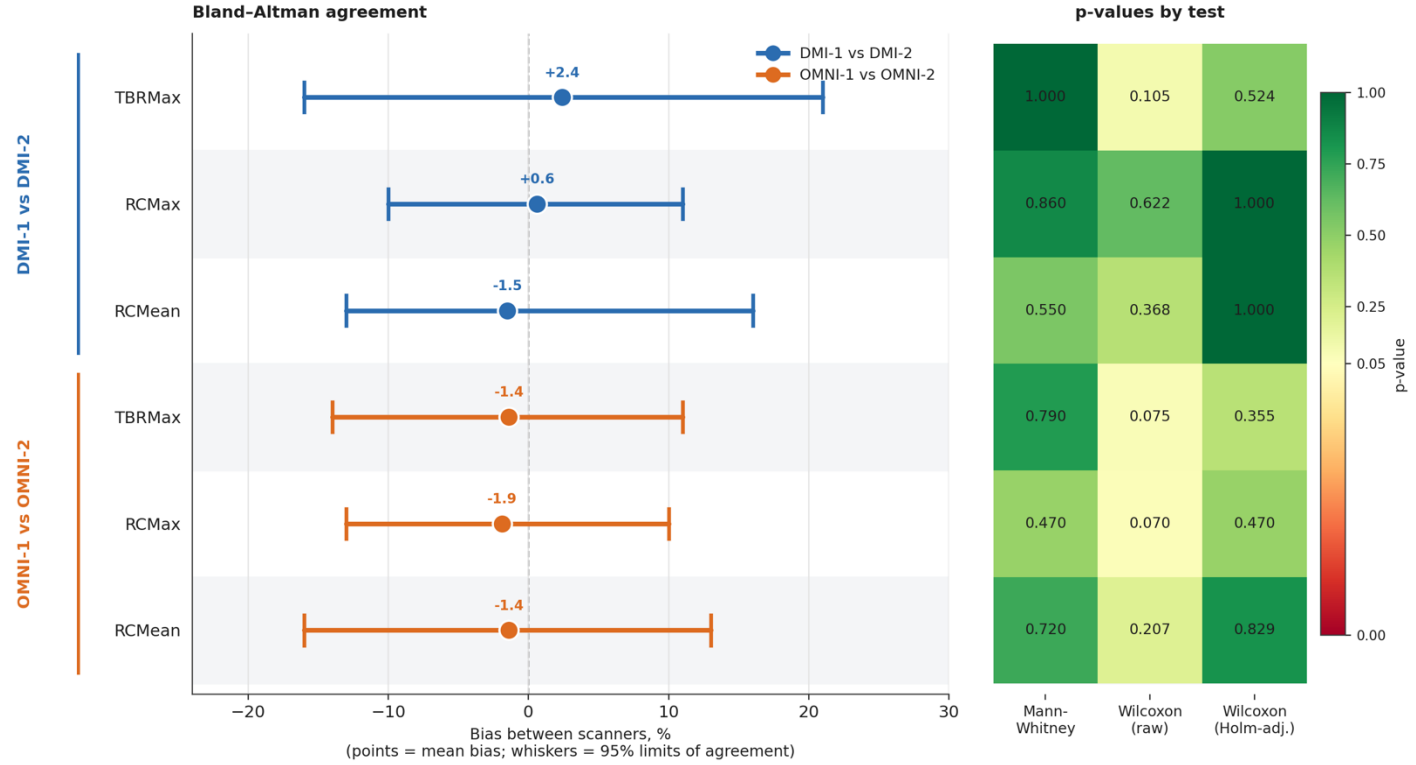


✓ **TOST show that all GMR comparisons fall within the 10% bounds including the 90% CI (p < 0.05)**

✓ **All Bland Altman mean Bias < 5%**

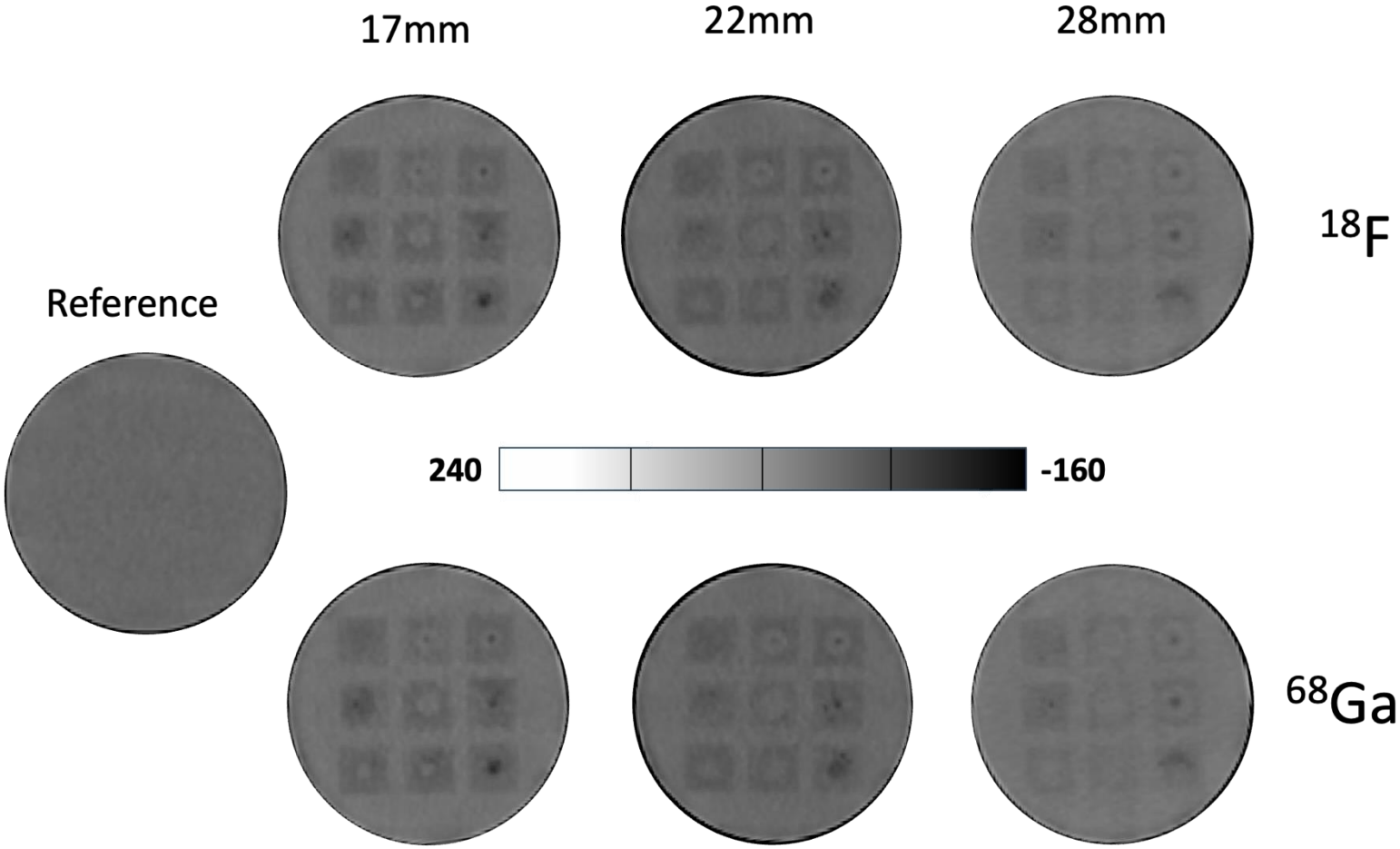
! **Limits of Agreement (LoA) are wider**

✓ **All pairwise p values are > 0.05 (no significance)**

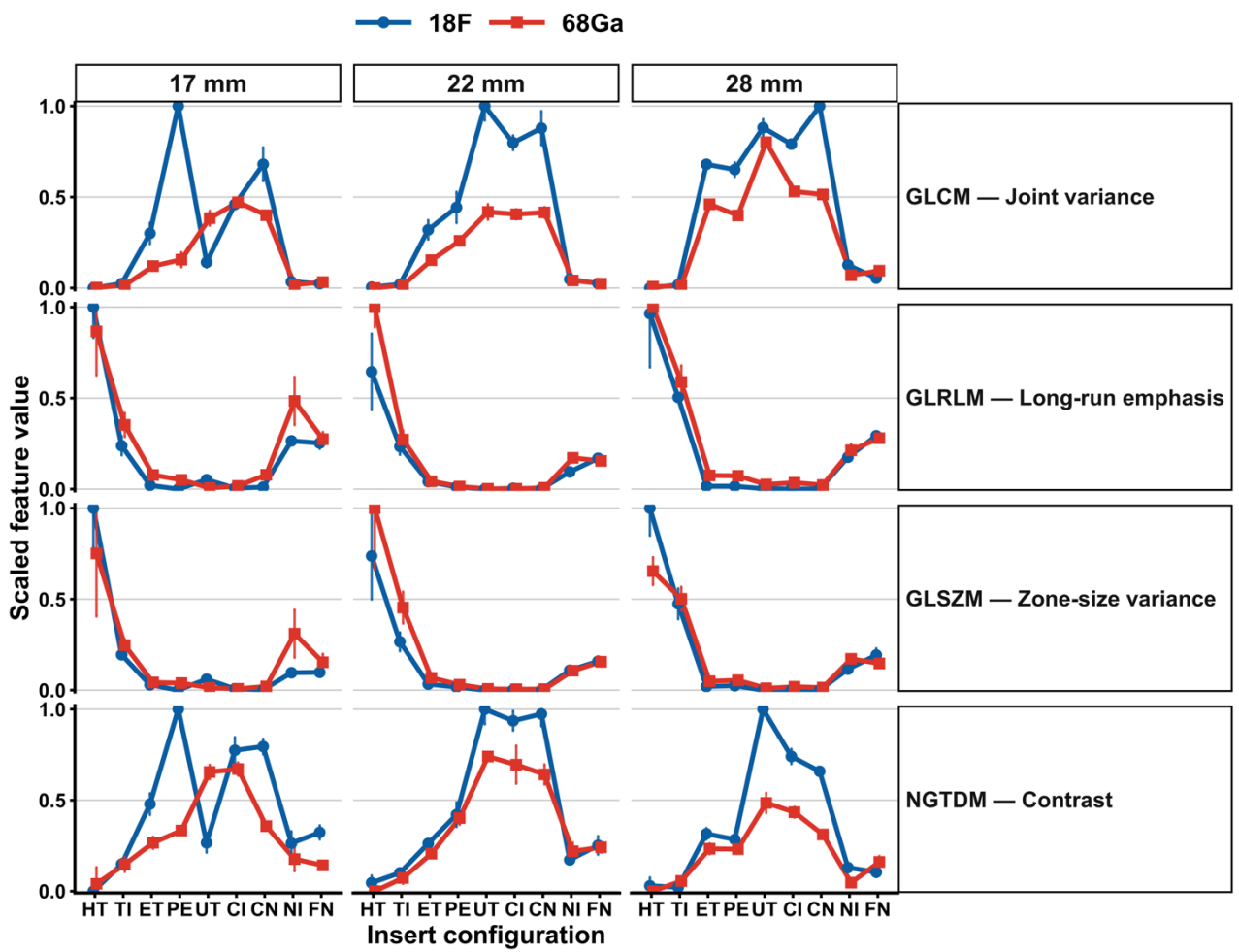


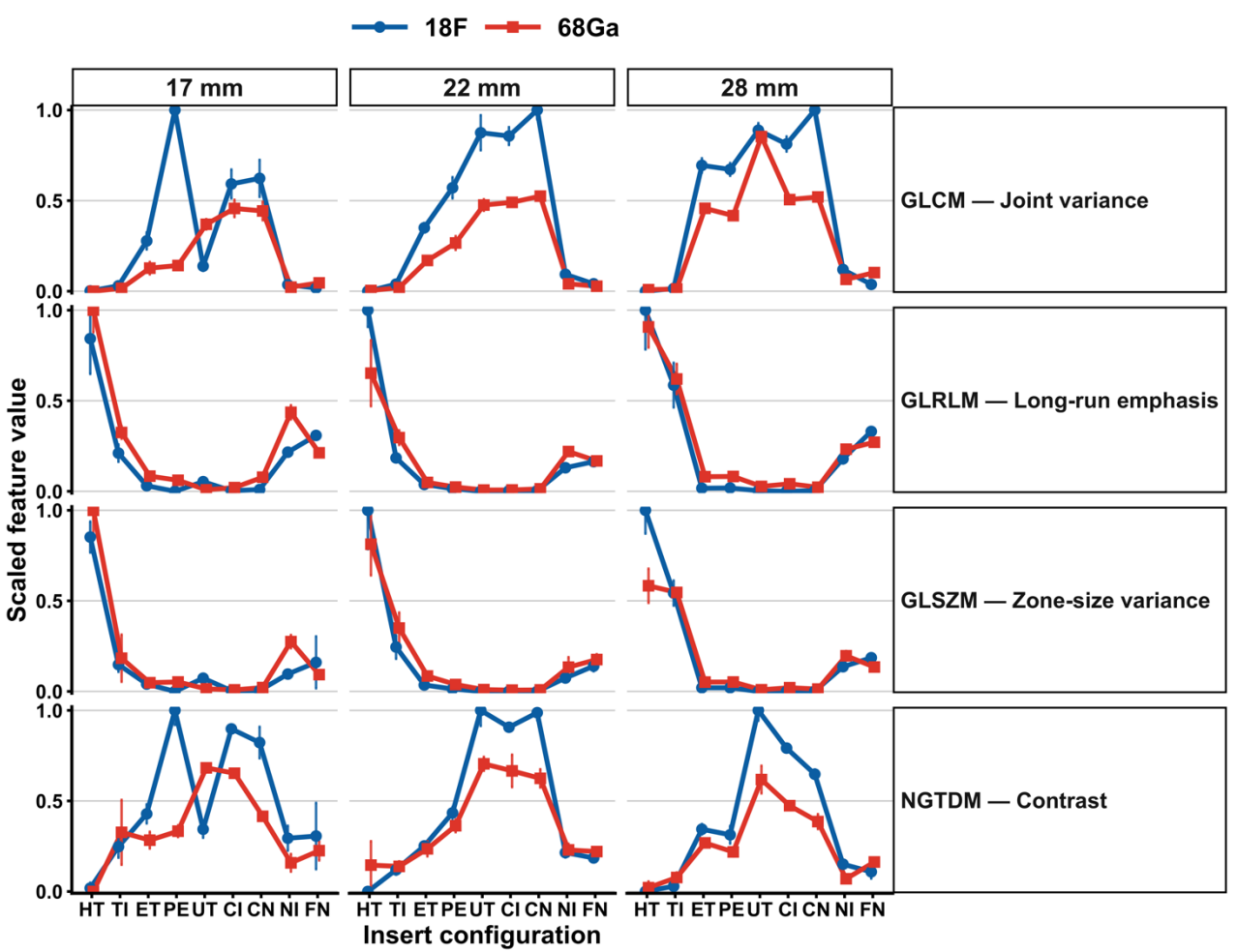
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CT at 2.5mm for Heterogeneity Phantom



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SCALED AXES

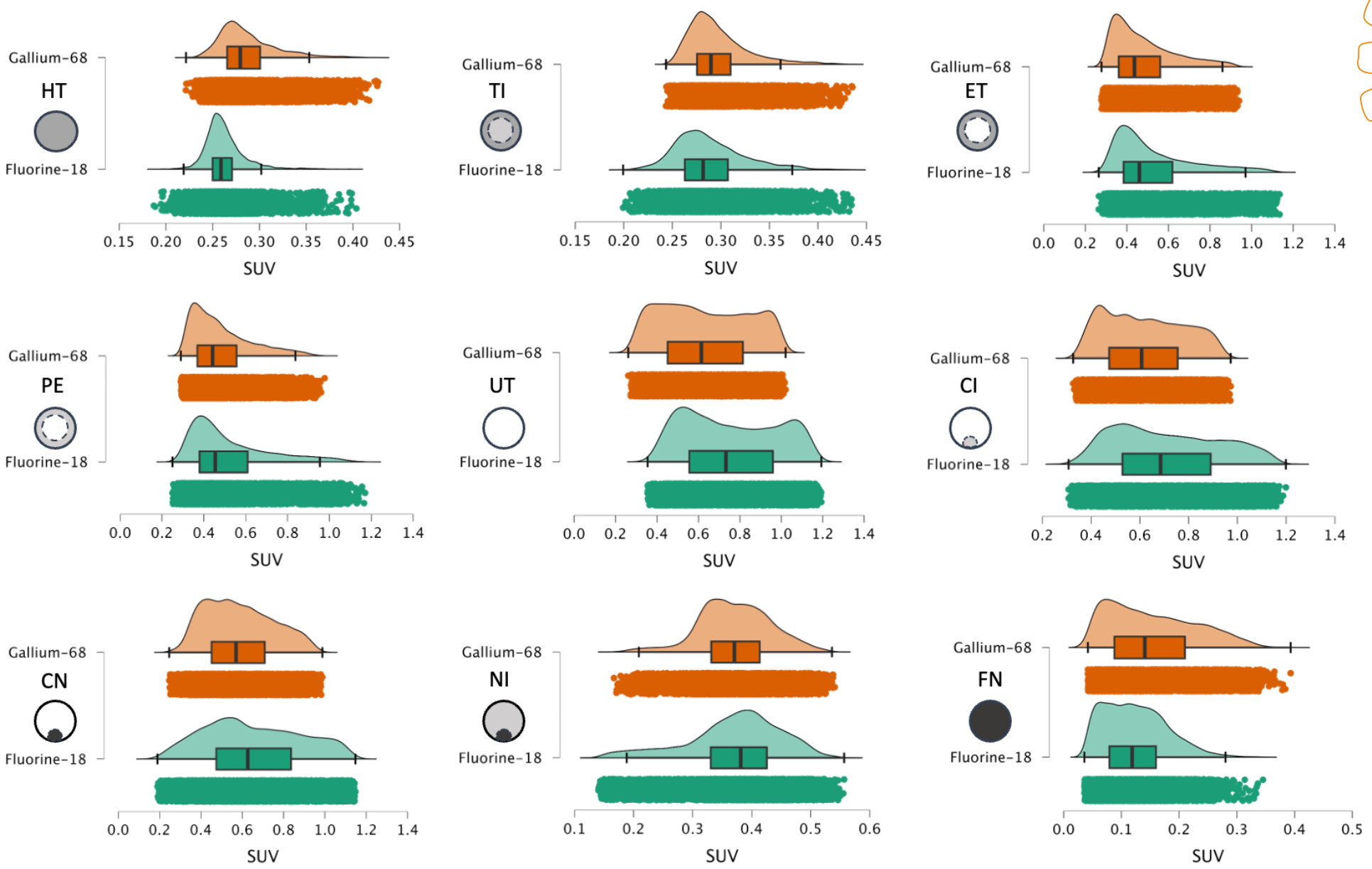
Similar curve shapes do not imply equal feature magnitudes.

CROSS-TRACER RESPONSE

Similar F/Ga pattern shapes with outliers

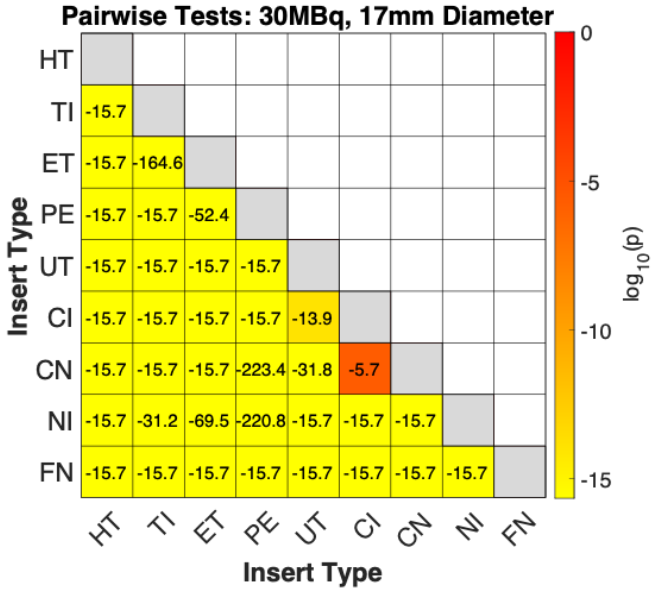
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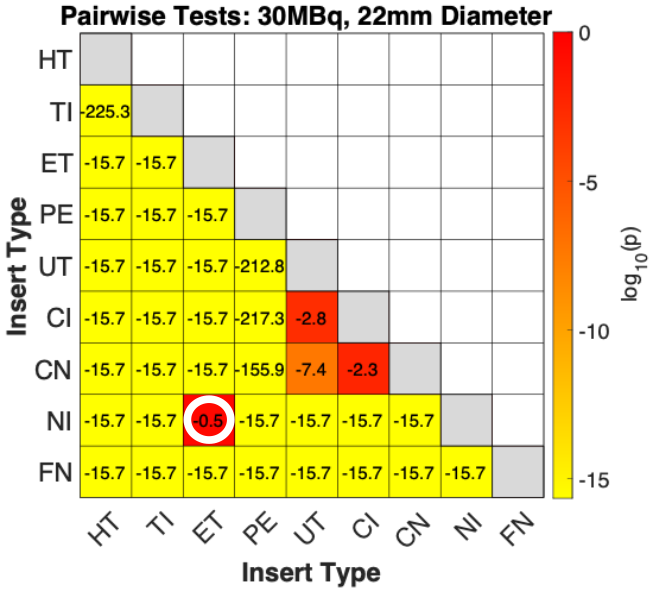
Highest correlation feature in each family

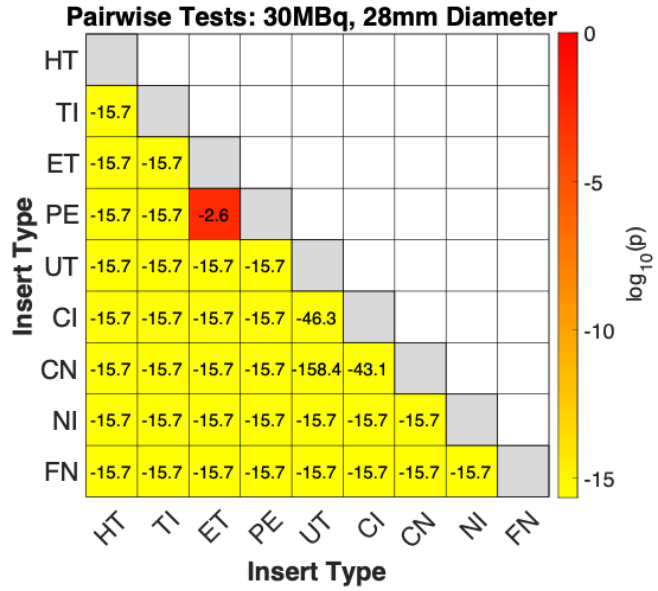


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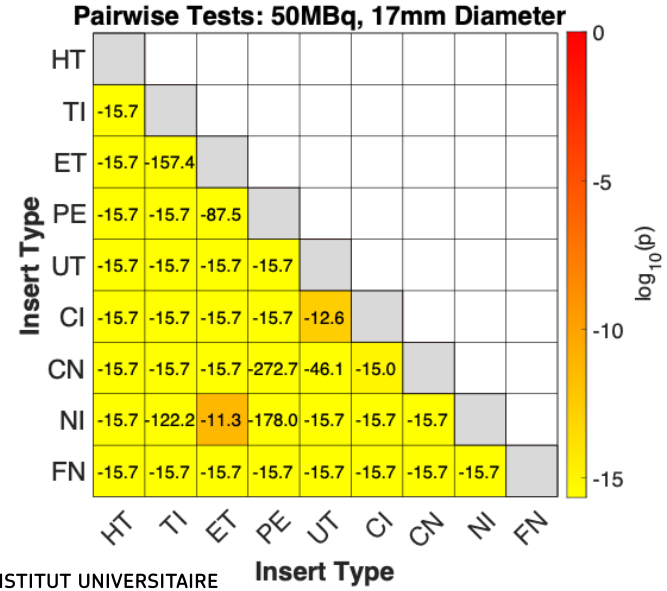
(a)

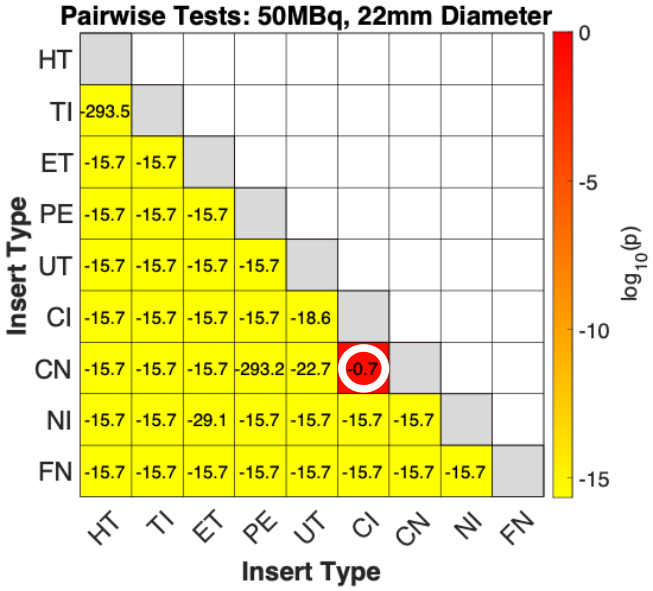

Pairwise Tests: 30MBq, 17mm Diameter
HT
TI -15.7
ET -15.7 -164.6
PE -15.7 -15.7 -52.4
UT -15.7 -15.7 -15.7 -15.7
CI -15.7 -15.7 -15.7 -15.7 -13.9
CN -15.7 -15.7 -15.7 -223.4 -31.8 -5.7
NI -15.7 -31.2 -69.5 -220.8 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
0
-5
-10
-15
log10(p)

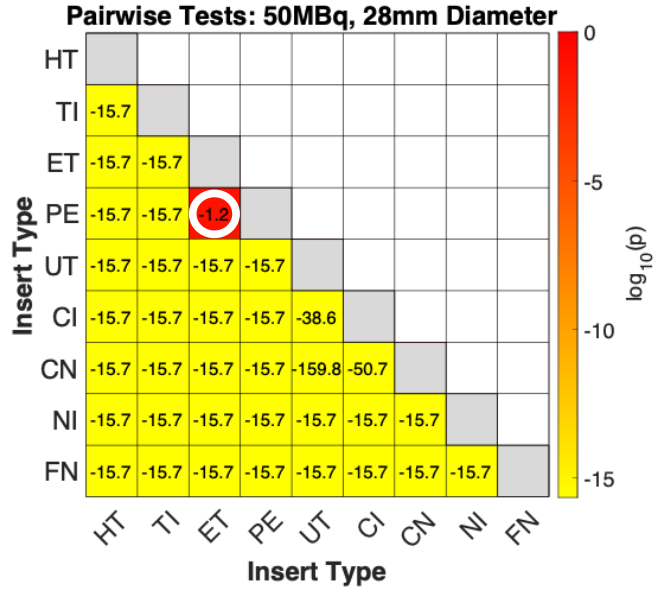

Pairwise Tests: 30MBq, 22mm Diameter
HT
TI -225.3
ET -15.7 -15.7
PE -15.7 -15.7 -15.7
UT -15.7 -15.7 -15.7 -212.8
CI -15.7 -15.7 -15.7 -217.3 -2.8
CN -15.7 -15.7 -15.7 -155.9 -7.4 -2.3
NI -15.7 -15.7 -0.5 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
0
-5
-10
-15
log10(p)


Pairwise Tests: 30MBq, 28mm Diameter
HT
TI -15.7
ET -15.7 -15.7
PE -15.7 -15.7 -2.6
UT -15.7 -15.7 -15.7 -15.7
CI -15.7 -15.7 -15.7 -15.7 -46.3
CN -15.7 -15.7 -15.7 -15.7 -158.4 -43.1
NI -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
0
-5
-10
-15
log10(p)

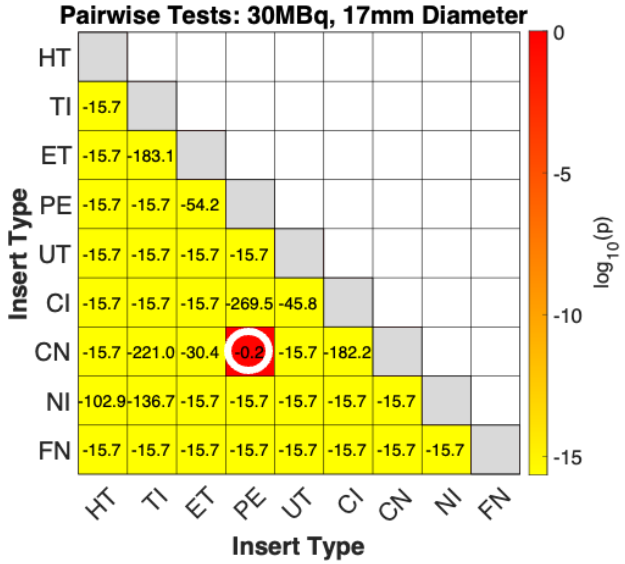
(b)

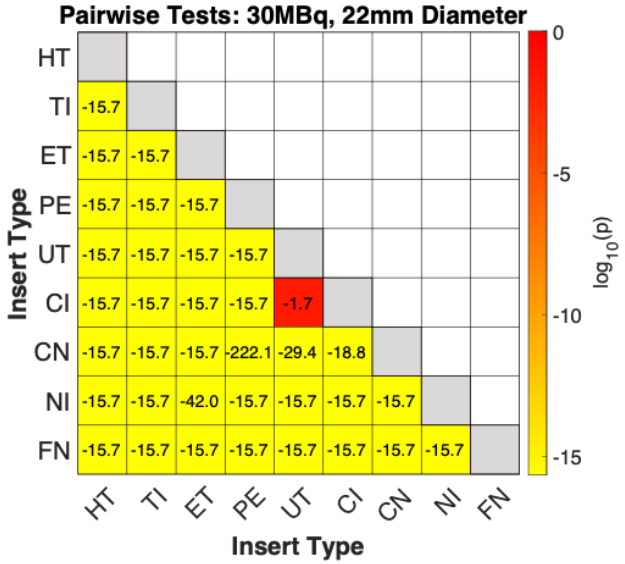

Pairwise Tests: 50MBq, 17mm Diameter
HT
TI -15.7
ET -15.7 -157.4
PE -15.7 -15.7 -87.5
UT -15.7 -15.7 -15.7 -15.7
CI -15.7 -15.7 -15.7 -15.7 -12.6
CN -15.7 -15.7 -15.7 -272.7 -46.1 -15.0
NI -15.7 -122.2 -11.3 -178.0 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
0
-5
-10
-15
log10(p)

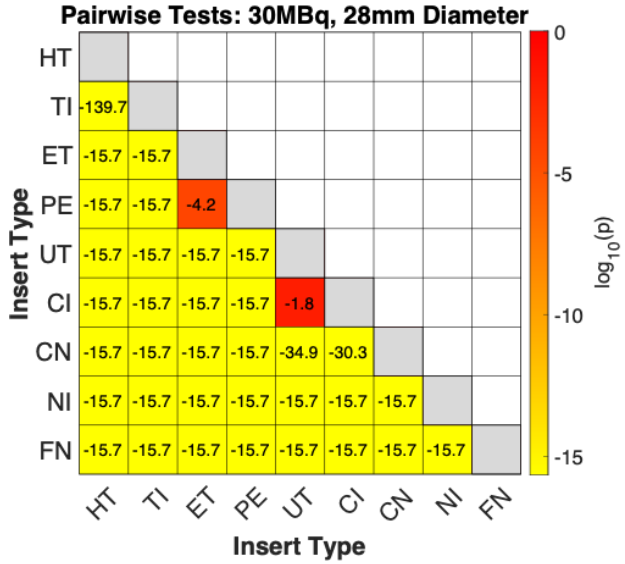

Pairwise Tests: 50MBq, 22mm Diameter
HT
TI -293.5
ET -15.7 -15.7
PE -15.7 -15.7 -15.7
UT -15.7 -15.7 -15.7 -15.7
CI -15.7 -15.7 -15.7 -15.7 -18.6
CN -15.7 -15.7 -15.7 -293.2 -22.7 -0.7
NI -15.7 -15.7 -29.1 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
0
-5
-10
-15
log10(p)


Pairwise Tests: 50MBq, 28mm Diameter
HT
TI -15.7
ET -15.7 -15.7
PE -15.7 -15.7 -1.2
UT -15.7 -15.7 -15.7 -15.7
CI -15.7 -15.7 -15.7 -15.7 -38.6
CN -15.7 -15.7 -15.7 -15.7 -159.8 -50.7
NI -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
0
-5
-10
-15
log10(p)

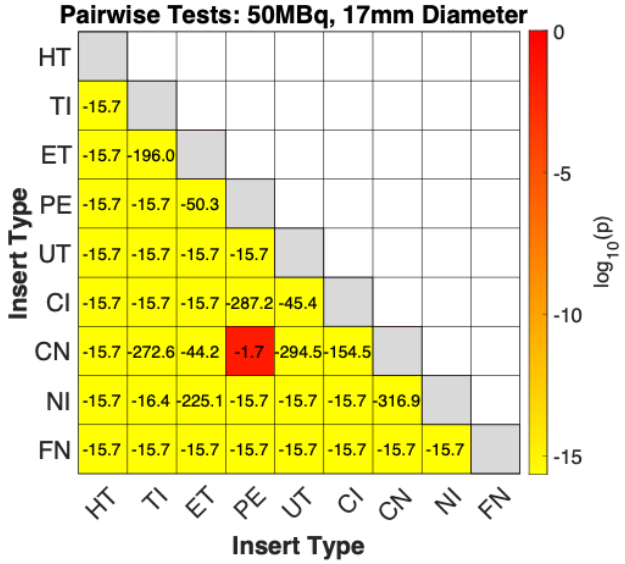
(a)

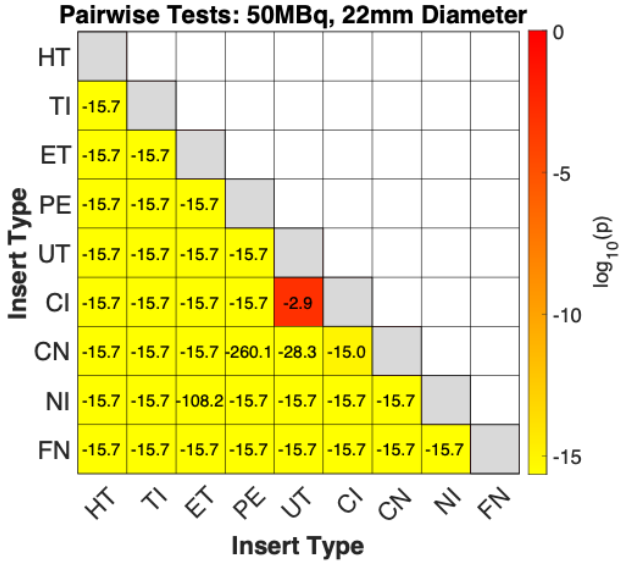

Pairwise Tests: 30MBq, 17mm Diameter
HT -15.7
TI -15.7 -183.1
ET -15.7 -15.7 -54.2
PE -15.7 -15.7 -15.7 -15.7
UT -15.7 -15.7 -15.7 -269.5 -45.8
CI -15.7 -15.7 -15.7 -0.2 -15.7 -182.2
CN -15.7 -221.0 -30.4 -15.7 -15.7 -15.7 -15.7
NI -102.9 -136.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
log10(p)
0 -5 -10 -15

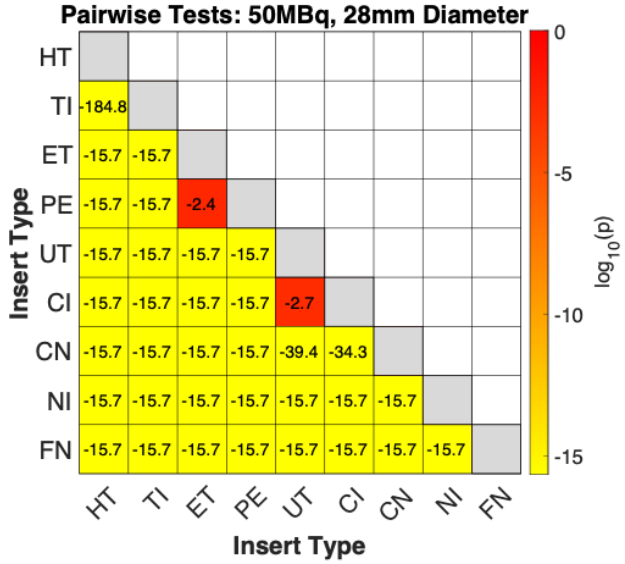

Pairwise Tests: 30MBq, 22mm Diameter
HT -15.7
TI -15.7 -15.7
ET -15.7 -15.7 -15.7
PE -15.7 -15.7 -15.7 -15.7
UT -15.7 -15.7 -15.7 -15.7 -1.7
CI -15.7 -15.7 -15.7 -15.7 -222.1 -29.4 -18.8
CN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
NI -15.7 -15.7 -42.0 -15.7 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
log10(p)
0 -5 -10 -15

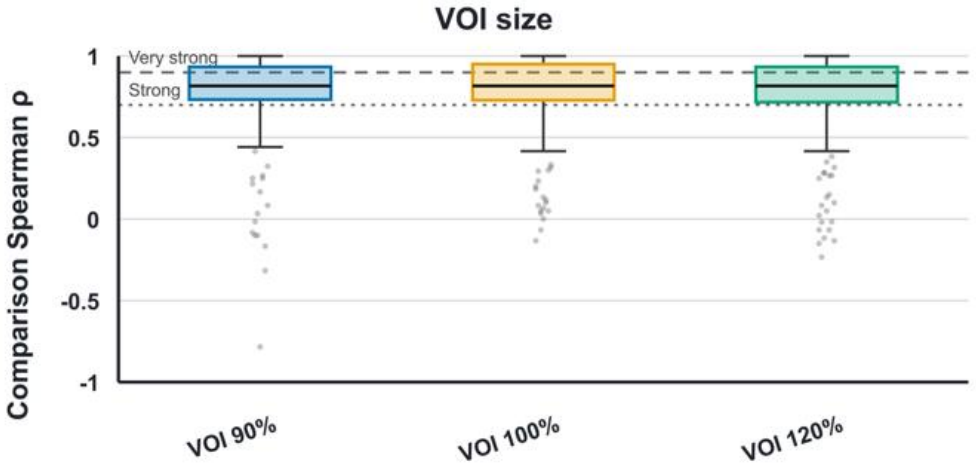

Pairwise Tests: 30MBq, 28mm Diameter
HT -139.7
TI -15.7 -15.7
ET -15.7 -15.7 -4.2
PE -15.7 -15.7 -15.7 -15.7
UT -15.7 -15.7 -15.7 -15.7 -1.8
CI -15.7 -15.7 -15.7 -15.7 -34.9 -30.3
CN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
NI -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
log10(p)
0 -5 -10 -15

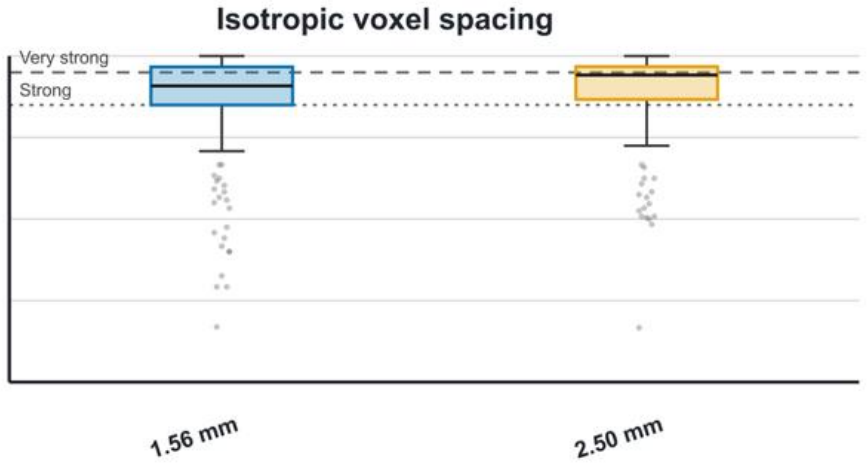
(b)

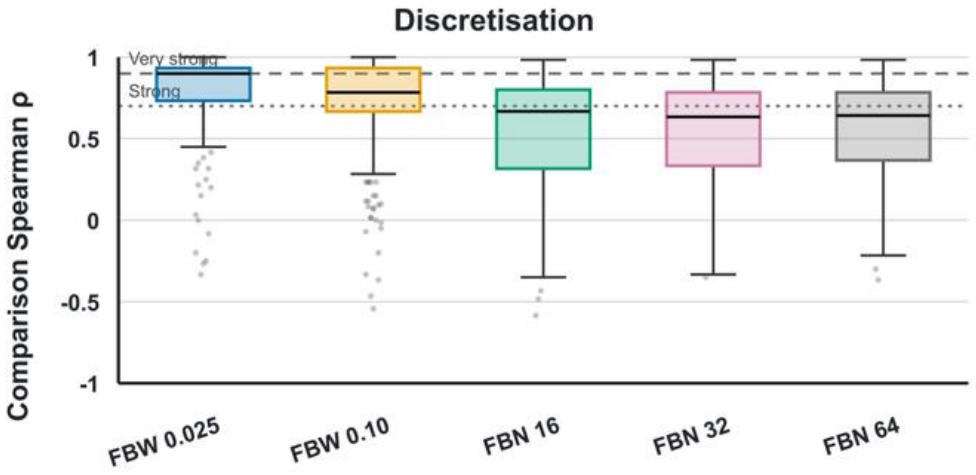

Pairwise Tests: 50MBq, 17mm Diameter
HT -15.7
TI -15.7 -196.0
ET -15.7 -15.7 -50.3
PE -15.7 -15.7 -15.7 -15.7
UT -15.7 -15.7 -15.7 -287.2 -45.4
CI -15.7 -15.7 -15.7 -1.7 -294.5 -154.5
CN -15.7 -272.6 -44.2 -15.7 -15.7 -15.7 -316.9
NI -15.7 -16.4 -225.1 -15.7 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
log10(p)
0 -5 -10 -15

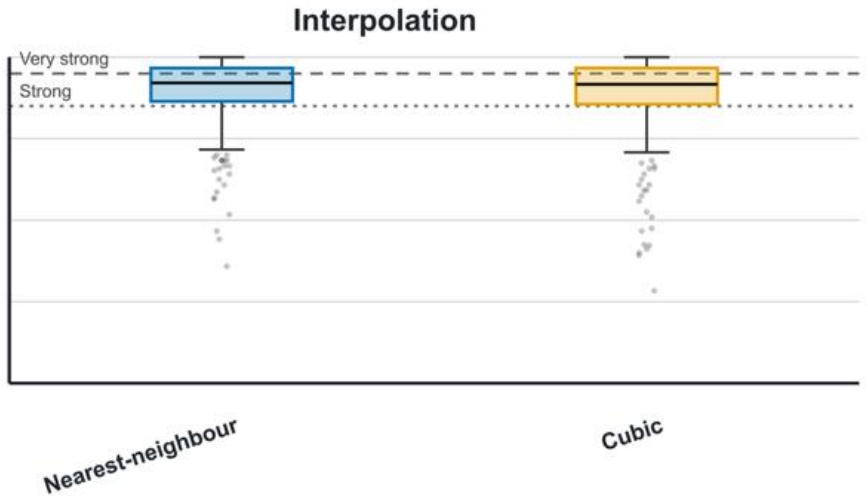

Pairwise Tests: 50MBq, 22mm Diameter
HT -15.7
TI -15.7 -15.7
ET -15.7 -15.7 -15.7
PE -15.7 -15.7 -15.7 -15.7
UT -15.7 -15.7 -15.7 -15.7 -2.9
CI -15.7 -15.7 -15.7 -15.7 -260.1 -28.3 -15.0
CN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
NI -15.7 -15.7 -108.2 -15.7 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
log10(p)
0 -5 -10 -15


Pairwise Tests: 50MBq, 28mm Diameter
HT -184.8
TI -15.7 -15.7
ET -15.7 -15.7 -2.4
PE -15.7 -15.7 -15.7 -15.7
UT -15.7 -15.7 -15.7 -15.7 -2.7
CI -15.7 -15.7 -15.7 -15.7 -39.4 -34.3
CN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
NI -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
FN -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7 -15.7
HT TI ET PE UT CI CN NI FN
Insert Type
Insert Type
log10(p)
0 -5 -10 -15









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Sensitivity Analysis

Synthetic lesion insertion workflow

1 **Load background**

Retain native voxel grid and coordinates, counts

›

2 **Create lesion masks**

Generate fractional-volume regions (floating masks) from lesion STL(s) or binary mask(s)

›

Synthetic lesion insertion workflow

1 Load background

Retain native voxel grid and coordinates, counts

2 Create lesion masks

Generate fractional-volume regions (floating masks) from lesion STL(s) or binary mask(s)

3 Place lesion

Select the desired placement of the lesion(s) interactively

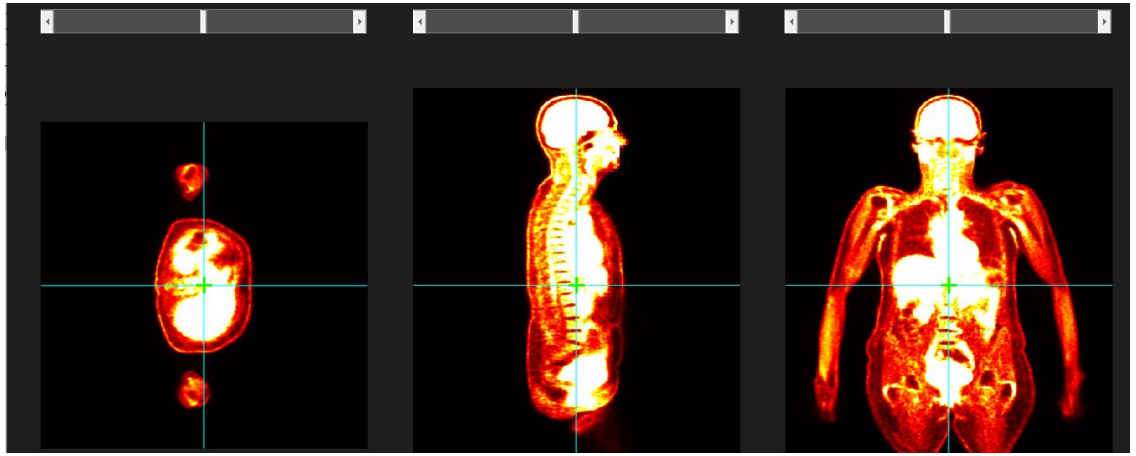
4 Assign activity

Assign contrast ratio(s)

How is Lesion mask created ?

When clicked, the mask is placed and average uptake on baseline image is calculated and multiplied with desired contrast

Lesion Placement UI



Synthetic lesion insertion workflow



Bed Positioning UI

Only OSEM Reconstruction

- **Vendor implementation of BSREM is unknown**
- **3D projector is ideal for BSREM**
- **BSREM is computationally expensive for the computational power**

5 Forward project

Project lesion(s) and background using 2D Radon

›

6 Model acquisition

Apply corrections, counts noise, normalization,

›

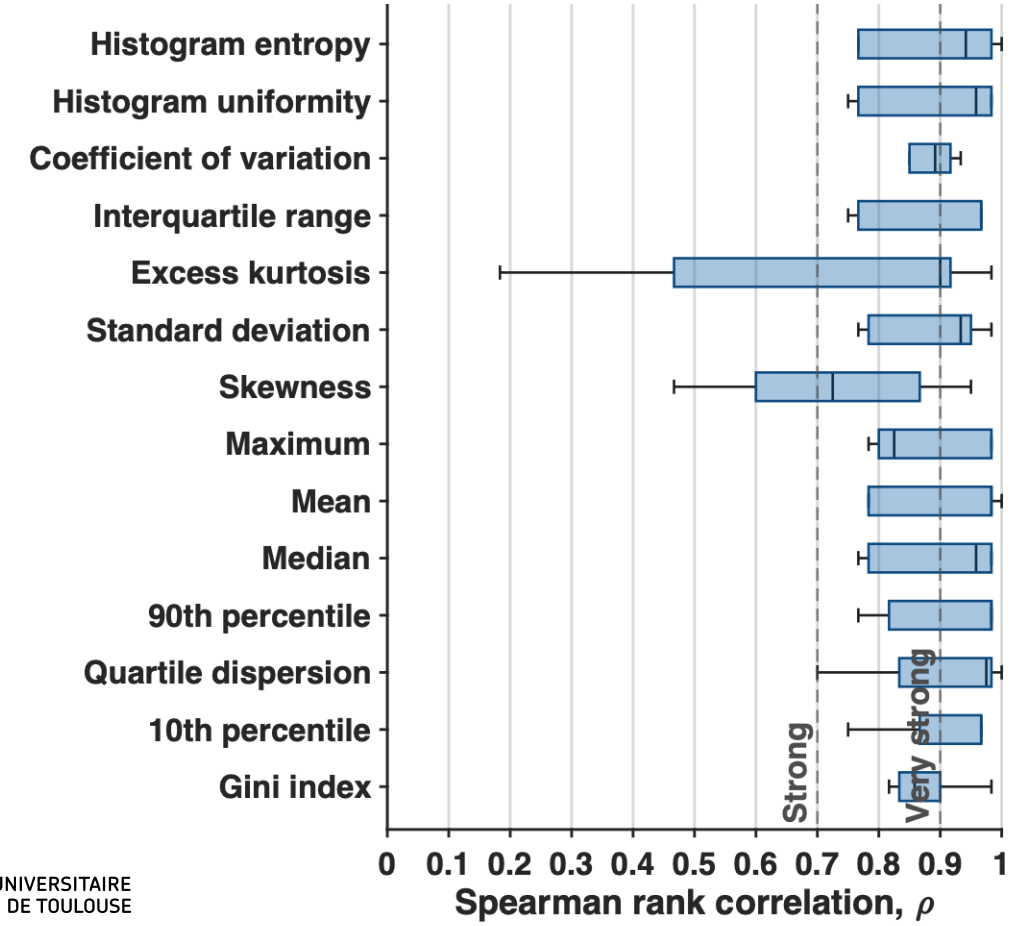
7 Reconstruct

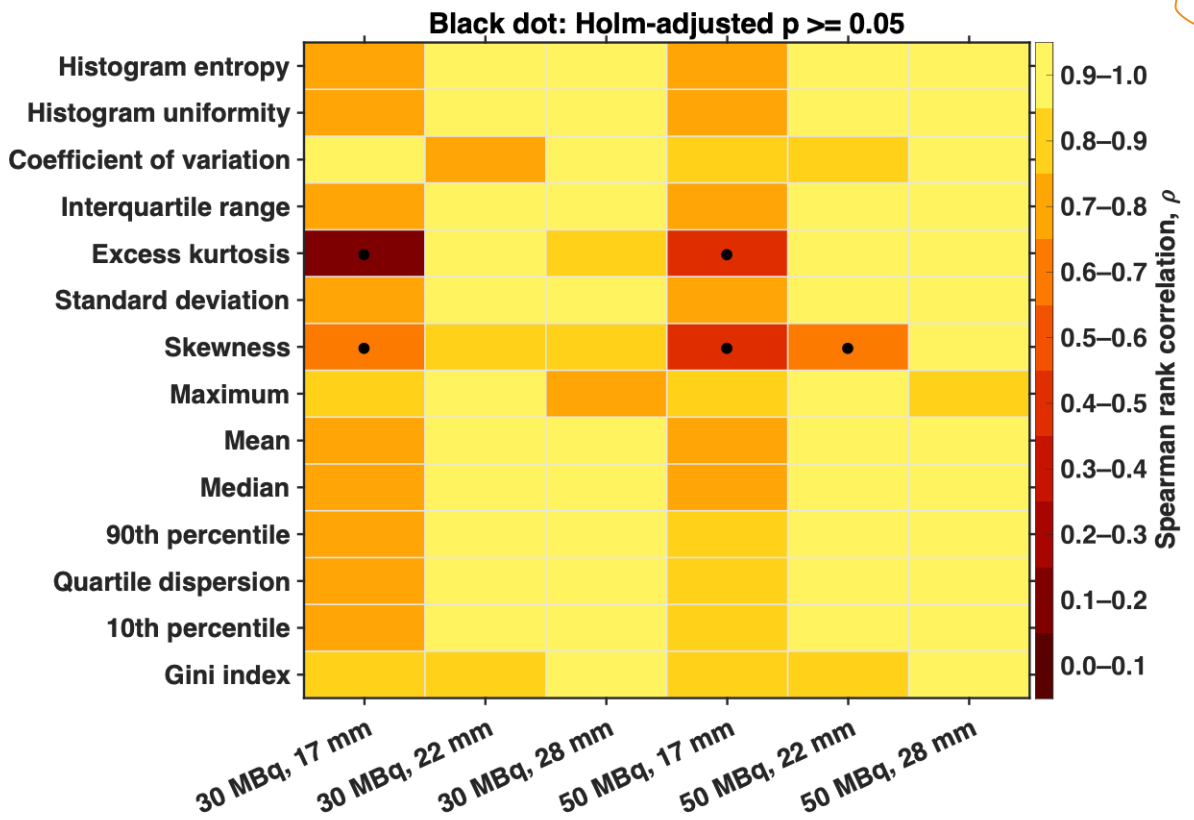
Produce the synthetic PET image using OSEM (2D or 2.5D)

Image composition followed by projection and separate projection followed by addition of each component are equivalent under the ideal forward model used

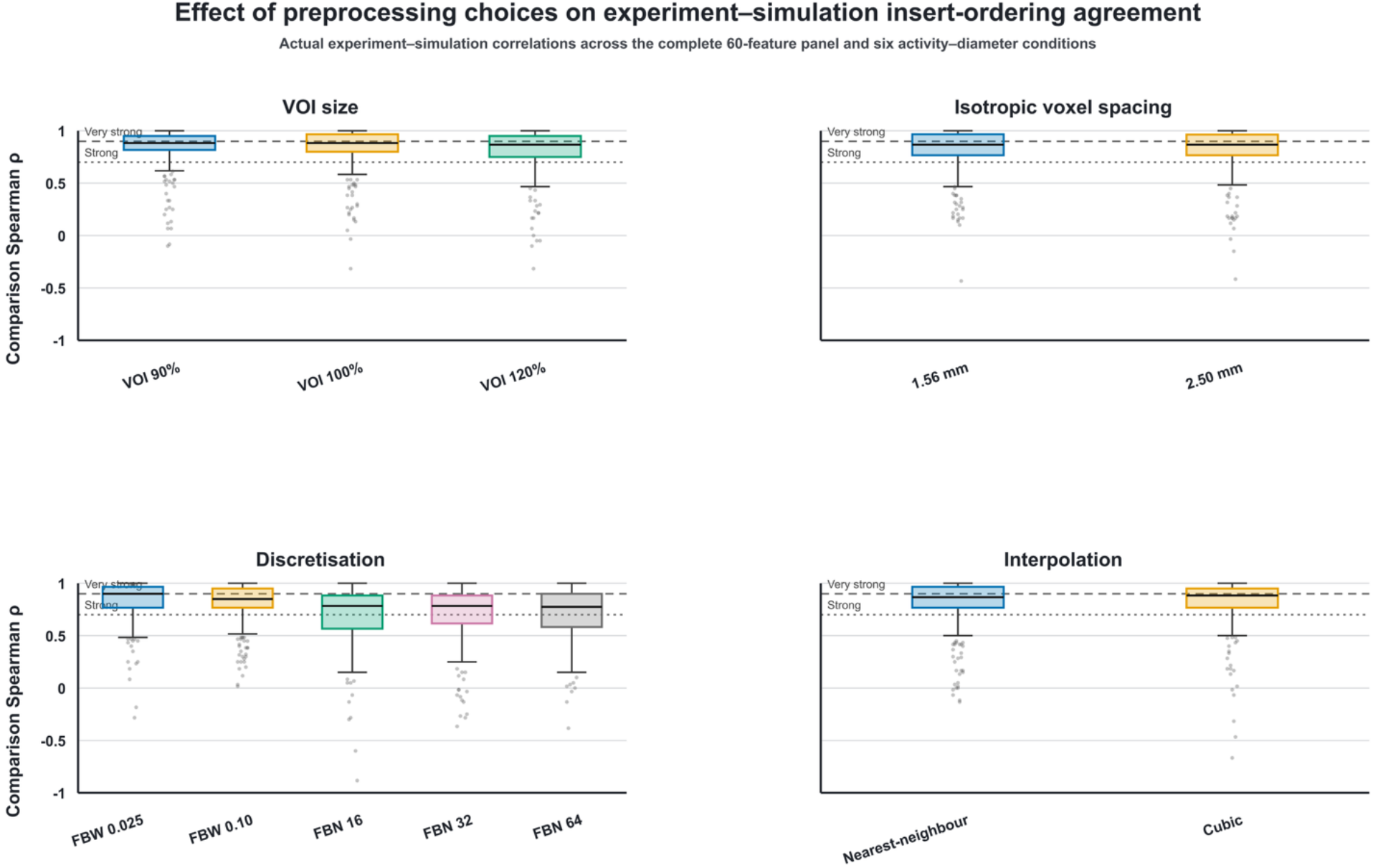
First order Radiomics Metrics

- Median ρ = 0.93 · 79/84 comparisons with ρ ≥ 0.70 · 79/84 Holm-significant





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Sensitivity Analysis

Family	What it measures	Why the ordering may change
GLRLM — 0.92	Lengths of consecutive voxels with similar intensity	Longer runs can retain the broad differences between insert patterns despite modest image changes.
NGTDM — 0.87	Differences between a voxel and its neighbourhood	Smoothing changes local differences. Performance varies by feature; the manuscript identifies coarseness as weaker.
GLCM — 0.84	Frequencies of neighbouring intensity pairs	Noise, smoothing and intensity discretisation change which grey levels occur together.
GLSZM — 0.82	Sizes of connected regions with the same discretised intensity	Small intensity changes can split one zone or merge several zones, substantially changing some features.